ON COMPUTABLE IMAGES AND FILMS, WITH AN APPLICATION TO THE ENTSCHEIDUNGSPROBLEM

A maieutic Chiron (post-Centaur) treatise on the mathematics of cinematic supercomputation[*]

Luís Homem

0000-0003-3520-1489

20/05/2026

Draft v36

A foundational contribution to the U-Mentalism research programme.

`u-mentalism.com` | `github.com/U-Mentalism`

Abstract

This paper develops a theory of computable images and computable films analogous to Turing's 1936 theory of computable real numbers, in a layout that mirrors his section by section. The central methodological commitment is the adoption of an RGB colour-native alphabet $\Sigma_{\mathrm{RGB}} = \{0, \ldots, 255\}^3$ for the universal machine, in place of the binary alphabet of the original construction. The standard alphabet equivalence theorem ensures the class of computable functions is unchanged; the cost model and the natural physical substrate—photonic computation—align with the colour-native alphabet exactly. We define computable image and film schemes as functions uniformly computable in their resolution parameter, present a graded sequence of worked examples ending with the Mandelbrot scheme, and construct a universal image-generating machine in 44 states with five named subroutines and an explicit transition table. We establish the visual analogues of Turing's diagonal and halting results: there exist definable but uncomputable images, the image-halting problem is undecidable, and the Kolmogorov complexity of image schemes is uncomputable. We engage with three post-1936 developments—computable analysis (Pour-El–Richards; Weihrauch) for the continuous-limit refinement of the discrete model, the Bernstein–Vazirani equivalence bounding quantum computation to the same class, and Brattka's bound on analog and photonic computation under realistic noise—and provide a Big-O cost stratification across procedural, rendered, and neural-network image classes for classical, quantum, and photonic substrates. We recover Hilbert's *Entscheidungsproblem* as unsolvable by reduction from the image-halting problem, and hypothesise a visual *Entscheidungsproblem* for subsequent development. A final section maps the engineering gap between the abstract universal

[*] *Methodological note.* This treatise was composed maieutically: a human pilot directing a universal computational substrate to produce a specific artefact in the class of texts about computation. The thesis is, in this respect, corroborated by the paper's own provenance. The cybernetic framing is in the spirit of Wiener (1948) and the original conception of *kybernetes*; the human pilot, in this collaboration, plays the joint role of *oracle* (in the technical sense of Turing 1939) and Chiron (the wise tutor-centaur of Greek mythology).

machine and any physical instantiation under thermodynamic, information-theoretic, decoherence, and precision bounds, concluding that the colour-native alphabet is exceptionally well-aligned to the photonic substrate that most naturally realises universal visual computation. The paper is, in the U-Mentalism programme distinction made precise in the introduction, the foundational contribution in the U.M. 'O' register — the mathematical and computability-theoretic substrate that legitimates the wider U.M. 'C' programme of cinematic supercomputation (patent PCT/PT2021/050014; Homem 2021) without exhausting it. "Wider" here is to be understood in the implementation register only — the engineering, hardware, and architectural effort of which the patent is the realised artefact — and not in the ontological register, where U.M. 'C' derives from U.M. 'O' as naturally as nature unfolds artificiality as natural.

Keywords: computable images, computable films, universal machine, RGB colour-native computation, photonic computation, computable analysis, Type-Two Effectivity, Type-256 effectivity, *Entscheidungsproblem*, Kolmogorov complexity, Church–Turing thesis, BQP, Landauer's bound, cinematic supercomputation, U-Mentalism.

Introduction

Every image stored, transmitted, displayed, or generated in any digital medium today is, in the mathematical sense made precise in this paper, the output of a finite procedure applied to a finite input. Every frame of every digital film, every photograph in its digital form (and in the analog-digital hybrid form of any capture chain that begins in light and ends in pixels), every output of every neural-network image generator, every codec decompression, every rasterised game frame, every ray-traced scene, every procedural texture, every parametric vector graphic. The visual content of contemporary culture is, with marginal idealised exceptions, a content of computable images and computable films.

This is a foundational observation, not an empirical one. It does not assert that every image is easily computable, that current hardware reaches every image one might wish to see, that the procedures producing today's images are optimal, or that the boundary between the producible and the inaccessible is operationally near. It asserts only that the boundary itself—between the procedurally-producible and the procedurally-inaccessible—coincides with a mathematical boundary established ninety years ago by Alan Turing, in a form which, until now, has not been articulated for the visual domain in the terms that domain naturally calls for.

This paper undertakes that articulation.

Lineage and departures

Turing's 1936 paper, *On Computable Numbers, with an Application to the Entscheidungsproblem*, established the mathematical theory of computable real numbers, constructed the universal machine that bears his name, derived the undecidability of the halting problem, and applied this last result to settle Hilbert's *Entscheidungsproblem* in the negative. The paper has aged into one of the founding documents of modern computer science. Its layout—definition, machine model, worked examples, enumeration, universal machine, diagonal argument, equivalence with alternative formalisms, application to Hilbert—is the layout of the present paper, deliberately and explicitly modeled.

We adopt Turing's organisation section by section, address the visual analogue of each result he established, recover his unsolvability conclusion through the visual route, and extend the framework where ninety years of subsequent work permit and the visual domain demands. The lineage is meant to be unmistakable: this is a paper about visual computation written in the layout language of the founding document of the mathematical theory of effective computation.

The departures from Turing are substantive and we name them at the entry point.

The most consequential is fixed in §1.5: the alphabet of the universal machine is not the binary $\{0,1\}$ but the colour-native $\Sigma_{\mathrm{RGB}} = \{0,\ldots,255\}^3$ that underlies all contemporary digital colour. The choice does not change the class of computable functions—Proposition 1 establishes the standard alphabet equivalence—but it changes the cost model, the natural physical substrate, and the algorithmic register in which visual computation is most clearly expressed. It also aligns the abstract theory with the photonic substrate that most naturally realises universal computation when the symbols are colours rather than bits. A binary universal machine producing or consuming colour images encodes and decodes at every step; a colour-native universal machine does neither. Both compute the same class; only the second is in any meaningful sense a *visual* universal machine.

Three further departures merit early notice. Section 9 engages with computable analysis (Pour-El & Richards 1989; Weihrauch 2000), the Bernstein–Vazirani equivalence (1997) bounding quantum advantage, and the analog/photonic results of Brattka (2000) and Hertling—material that postdates 1936 by half a century or more and which the visual setting calls upon in ways the original numerical setting did not. Section 10 includes a Big-O cost stratification across procedural, rendered, and neural-network image classes, with quantum and photonic cost models alongside the classical baseline; this material is unrecognisable from the standpoint of 1936 and entirely necessary in 2026. Section 12 has no antecedent in Turing at all: it maps the gap between the abstract universal machine and its physical instantiations, treating Landauer's bound, the Bekenstein and Bremermann limits, decoherence, and analog precision as the engineering reality within which the foundational result must be located. The destination is Turing's; the territory traversed is the territory of the present, and, hopefully, the charting of the future too.

The structural correspondence

The layout parallel to Turing 1936 is summarised by the following section-mapping table, provided here so that readers may verify the correspondence at a glance and identify departures with equal ease.

§	Subject	Turing 1936
1	The computing machine	§1
1.5	The colour-native machine	*(no antecedent)*
2	Computable images and computable films	§2
3	Examples of computable image schemes	§3
4	Compositional macros	§4
5	Enumeration; Kolmogorov complexity	§5
6	The universal image-generating machine	§6
7	Construction of U	§7
8	Diagonal argument; uncomputable images	§8
9	Extent: TTE, BQP, photonic substrates	§9
10	Large classes; cost stratification	§10
11	Application to the *Entscheidungsproblem*	§11
12	Physical limits; the gap	*(no antecedent)*
App. A	Mapping to Turing 1936 (extended)	(Appendix)
App. B	Full ray-tracer construction	—
App. C	Full transition tables of U	—
App. D	Extended cost stratification	—
App. E	Encoding correctness for Theorem 11.1	—
App. F	Proof-of-Computable-Image; code-as-law on Bitcoin	—
App. G	Binary TM with photonic sensorisation (intermediate engineering point)	—
App. H	Implementation avenues and obstacles for the colour-native machine	—

Sections marked "*(no antecedent)*" are substantive additions made necessary by the colour-native commitment of §1.5 and by the engineering reality of 2026. Their content extends the framework rather than displacing any existing piece of it.

Contributions

The paper makes the following contributions, in the order in which they appear.

A *colour-native universal machine* for visual computation, formalising the computational substrate for which photonic implementation is the natural physical realisation. The machine reads and writes 24-bit colour symbols natively, simulates any other Turing machine over the same alphabet without binary-encoding overhead, and is the formal kernel of the U-Mentalism programme of which this paper is a foundational part.

Discrete-primary definitions of computable images and computable films, with a uniform-in-resolution scheme as the central object (Definition 2.1) and a causal refinement for films generated in response to exogenous input streams (Definition 2.3). The continuous (Type-Two Effective) idealisation appears in §9 as the densification limit rather than as the primary definition.

A *full transition-table construction* of the universal machine U, presented in §7 with five named subroutines, an explicit main-loop transition table, and a state count of 44—a deliberately small construction whose simplicity is preserved by the colour-native alphabet choice. The full transition tables of the subroutines appear in Appendix C.

A *diagonal argument adapted to image schemes* (§8), establishing the existence of definable-but-uncomputable images and the undecidability of the image-halting problem. Corollaries include the undecidability of rendering termination, of total-frame rendering, and the uncomputability of image-scheme Kolmogorov complexity.

An *engagement with the modern boundary*: §9 establishes the visual Church–Turing thesis, situates it relative to Type-Two Effectivity, derives that quantum computation does not extend the class of computable images (Bernstein–Vazirani), and confirms via Brattka that realistic photonic and analog substrates are bounded by the same boundary.

A *cost stratification* (§10) covering procedural, rendered, and neural-network image schemes across classical, quantum-BQP, and colour-native photonic cost models, with explicit Big-O complexities and an exhaustion claim asserting that the class of computable image schemes contains, modulo idealised limits, the entirety of contemporary visual culture.

A *recovery of Hilbert's Entscheidungsproblem result* (§11) through the visual route—the unsolvability of first-order validity, derived by reduction from the image-halting problem of §8.

A *hypothesised visual Entscheidungsproblem* (§11.5), held as a working hypothesis with three open questions for subsequent investigation and not fully developed in this paper.

A *map of the engineering gap* (§12) between the abstract universal machine and its physical instantiations, treating thermodynamic, information-theoretic, quantum-mechanical, and analog-precision bounds as the engineering reality within which the foundational result must be located.

Place in the U-Mentalism programme

The paper is the foundational contribution of the U-Mentalism long-form research project (Homem 2018, 2019, 2021), of which it forms one component among several developed in parallel. The programme distinguishes itself into U.M. 'O' (for "Ontology", proper to the mathematical foundations, the constitution of the image, and the foundations of knowledge) and U.M. 'C' (for "Computation", proper to the inversion of the von Neumann architecture into an RGB-photonic cinematic supercomputation model, the substance of patent PCT/PT2021/050014). The present paper sits in the U.M. 'O' register: it develops the mathematical and computability-theoretic substrate that legitimates U.M. 'C' without exhausting it. Turing's 1936 paper — the structural model of this one — was foundational mathematics in precisely this register, and its purely mathematical character is the very reason it could not have been (and was not) the basis of a patent. The U.M. 'C' programme is wider than the mathematical thesis presented here *in the implementation register only* — the engineering, hardware, and architectural effort of which the patent is the realised artefact — and not in the ontological register: U.M. 'C' derives from U.M. 'O' as naturally as nature unfolds artificiality as natural — something only captured by the Greek $\phi\acute{\upsilon}\sigma\iota\varsigma$. The Universal Machine of the broader project is, formally, the universal image-generating machine U constructed in §§6–7; the engineering implementation is the subject of the programme's themes (vide `u-mentalism.com`) on computer architecture, photonics, and the Universal Assembly Language; the philosophical and ethical concerns belong to the themes on natural and artificial philosophy, ethics, neuroscience, and cybernetics. The present paper is intended to be readable as a self-contained foundation for the rest of the programme, and the rest of the programme is intended to read it that way.

Roadmap

Sections 1–4 establish the machine model, the colour-native alphabet, the definitions of computable images and films, examples, and the compositional macro layer. Section 5 enumerates the computable schemes and introduces Kolmogorov complexity. Sections 6–7 construct the universal machine U and present its full transition table. Section 8 derives the diagonal results. Sections 9–10 engage with the modern boundary and with the cost stratification. Section 11 recovers the *Entscheidungsproblem* unsolvability and hypothesises its visual analogue. Section 12 maps the engineering gap. Eight appendices supply technical material whose level of detail would interrupt the body's argumentative arc.

A reader familiar with Turing 1936 who wants the visual content alone may proceed directly to §§1.5, 2, 9, 10, and 12, where the departures from the original are concentrated. A reader new to the layout is invited to read in order: the lineage to Turing is most clearly visible when the paper is read with his at hand.

1 The computing machine

A computing machine, in the sense introduced by Turing (1936), is a finite-state device operating on a one-dimensional tape divided into discrete cells, each holding one symbol from a finite alphabet. We adopt this model without essential modification, written in contemporary notation as a tuple

$$M = (Q, \Sigma, \Gamma, \delta, q_0, F, \mathfrak{B}),$$

where Q is a finite set of *internal configurations*; Σ is the *input alphabet*; $\Gamma \supseteq \Sigma$ is the *tape alphabet*, containing a distinguished blank symbol $\mathfrak{B}$; $\delta : Q \times \Gamma \to Q \times \Gamma \times \{L, R, S\}$ is the *transition function*; $q_0 \in Q$ is the *initial state*; and $F \subseteq Q$ is the set of halting states. The triple $\{L, R, S\}$ encodes head movement: left, right, or stationary.

The *configuration* of M at any moment consists of (i) the contents of the tape, (ii) the position of the head, and (iii) the internal state. A *computation* is a finite or infinite sequence of configurations connected by single applications of δ. We say M halts on input $w \in \Sigma^*$ if its computation, started from the initial configuration with w written on the tape and the head over its first cell, terminates in a state of F; in that case the contents of the tape from a designated origin to the first trailing blank constitute the *output*.

The model is alphabet-agnostic: the theory developed in §§5–11 depends only on Σ being finite and on δ being total on $Q \times \Gamma$. We fix Σ in §1.5 to a particular finite set chosen for its physical and algorithmic naturalness in the visual domain, and show (Proposition 1) that this choice incurs no loss of generality with respect to the class of computable functions.

Two output conventions will be useful. The *raster convention* interprets the output tape as a row-major flattening of a rectangular grid of fixed dimensions. The *functional convention*, used throughout this paper, treats the machine as taking a coordinate input $(x, y) \in \mathbb{N}^2$—or $(x, y, t) \in \mathbb{N}^3$ for films—and returning the symbol at that coordinate. The two are inter-translatable; the functional convention is preferred because it makes uniform computability across resolutions straightforward to state and admits a natural decomposition for parallel and pixel-local computation.

1.5. The colour-native machine

We now fix the alphabet. Let

$$\Sigma_{\mathrm{RGB}} = \{0, 1, \ldots, 255\}^3,$$

so that $|\Sigma_{\mathrm{RGB}}| = 256^3 = 2^{24} = 16{,}777{,}216$. Each element is a triple (r, g, b) representing the intensity of red, green, and blue components at 8-bit quantisation per channel—the standard 24-bit RGB space underlying contemporary consumer imaging. We call a Turing machine over Σ_{RGB} a *colour-native machine* and reserve *binary machine* for the case $|\Sigma| = 2$.

The choice is physically and conceptually motivated, not mathematically forced. To establish that it carries no theoretical cost, we record the following equivalence.

Proposition 1 (Alphabet equivalence). *Let Σ be a finite alphabet and $f : \Sigma^* \to \Sigma^*$ a partial function. Fix any prefix-free encoding $\sigma \mapsto \hat{\sigma} \in \{0, 1\}^{\lceil \log_2 |\Sigma| \rceil}$, and let $\hat{f} : \{0, 1\}^* \to \{0, 1\}^*$ denote the induced binary version of f. Then f is computable by a Turing machine over Σ if and only if $\hat{f}$ is computable by a Turing machine over $\{0, 1\}$. The simulation in either direction incurs at most a multiplicative $\lceil \log_2 |\Sigma| \rceil$ factor in space and a polynomial factor in time.*

Proof sketch. Standard; see e.g. Hopcroft, Motwani & Ullman (2007, §8.5). □

For Σ_{RGB}, Proposition 1 gives a $24\times$ space overhead and a low-degree polynomial time overhead in either direction. The class of computable functions is the same. Why, then, does the choice matter?

It matters for three reasons that the equivalence does not capture.

Cost model. Every read or write of a pixel is, in Σ_{RGB}, a single primitive operation; in $\{0, 1\}$, it is a sequence of 24 operations punctuated by tape-head movement. For algorithms whose natural unit of work is the pixel—that is, for essentially every algorithm in image and film processing—the colour-native cost model corresponds to operations an implementer would actually pay for, while the binary cost model overcounts by a multiplicative factor that varies with bit-depth. This will become quantitatively important when we discuss complexity in §10.

Physical substrate. A photon carries colour (wavelength) and intensity information directly. A computer that manipulates photons—through interference, filtering, modulation, or wavelength conversion—operates on colour-valued symbols natively, and emulates binary symbols only by a deliberate restriction to two states out of a continuous range. The colour-native machine is therefore the abstract counterpart of a photonic computer in a way that the binary machine is not. We return to this point in §§9.4 and 12.4.

Algorithmic expression. Natural visual operations—colour transforms, convolutions, compositing—are stated most directly as functions $\Sigma^k_{\mathrm{RGB}} \to \Sigma_{\mathrm{RGB}}$. Routing them through a binary encoding obscures their structure without compensating insight, in much the same way that arithmetic on natural numbers can in principle be performed in unary but is rarely so written.

The choice of 256 levels per channel is a historical contingency of the consumer hardware that fixed 8-bit colour as a de facto standard. The theory below extends without modification to any quantisation $\{0, \ldots, n-1\}^3$ with $n \in \mathbb{N}$. Increasing n—to 10- or 16-bit-per-channel HDR, to multispectral imaging that adds bands in the ultraviolet and infrared, and ultimately to a continuous spectrum—produces a sequence of finer discrete models that converges, in a sense made precise in §9.2, on continuous-valued analog computation. The discrete RGB colour-native machine sits at one end of this trajectory; an idealised photonic analog computer sits at the other; everything between is a question of engineering granularity.

The bridging-phase reading. On the U-Mentalism reading (Homem 2018, 2019, 2021), the 256-level RGB alphabet is not the final destination of the formal model but the *first bridging phase* between the binary state-of-the-art and the eventual photonic substrate. The fully general U.M. 'C' model conceives computation across any n-dimensional interval of nanometric differences in the electromagnetic spectrum (EEM); 8-bit RGB is the engineering point of contact with current digital hardware, finer quantisations and broader spectral coverage are the trajectory by which the model approaches the photonic limit, and the analog limit at the Brattka bound (§9.4) is the upper edge of physical realisability. Throughout the rest of the paper we use *RGB colour-native* for emphasis and reserve *colour-native* as the wider designation when the specific RGB quantisation is not at stake.

Cell, square, tape: the U.M. correspondence to the Turing tape. A specifically U.M. reading of the machine layout maps the canonical Turing tape onto the three natural levels of the moving image: the *cell* is the pixel (above the colour channels — the three channel readings r, g, b are synthesised, at the sensor or detector layer, into the single colour output that constitutes the pixel value, in Kant's sense the *schema* of the three channels apprehended as one; see §9.4 for the quantum-measurement reading of this *synthesis*); the *square* (in Turing's terminology, the 2D arrangement of cells the machine moves over) is the frame; the *tape* (the temporal sequence over which the head advances) is the film. The 1-dimensional abstract tape of 1936 becomes, in the U.M. register, the 3-dimensional cinematic structure — pixel, frame, film — inherited from the physics of *movement-images* and *time-images* (cf. Deleuze 1983, 1985).

Under this correspondence, films are *recollected* or *collected*. The first term is the foundational mode and is not what the everyday vocabulary first suggests: recollection in the U.M. register is *not* a captured image being recovered or processed; it is the arithmetic ordinal ordering of all frames and all films along the real line, in the sense of well-ordering up to and including Cantor's transfinite series (§5). Recollected films are persistent-memory objects, indexed by their ordinal positions, available to UTMs operating as servers on the network (§§12.4, A.4) for processing whose substrate is the catalogue as a whole. Recollection is, in this sense, the ordinal and enumerative mode: every computable image and film, irrespective of how it is generated, has a well-defined ordinal position in the universal catalogue and is, in principle, retrievable by that position alone. *A clarification on what a recollection-entry holds.* The recollection-entry corresponding to a frame is, strictly, not the frame as a perceptual surface alone but the pair $\langle F, \mathbf{c}(F)\rangle$, with F the synthesised image and $\mathbf{c}(F)$ the array of precise per-channel colour values (r_{ij}, g_{ij}, b_{ij}) that produced it at every pixel position (i, j). The synthesis and the channel-array are mutually recoverable under the bijective synthesis function of §1.5, so the pair is informationally redundant in either direction; the redundancy is structural in U.M. and is what permits the retrograde reading of §9.4 (from synthesis to channel signature) and the cryptographic-cascade pairing of §9.3.1 (Proposition 9.3.4). The note worth recording here is that the Turing-machine idea, in the ninety years since 1936, has not concerned itself with the persistent storage of indexed series of large numbers — the real-image-computable series that the U.M. recollection envisions — and this neglect is one of the operative gaps the U-Mentalism programme proposes to close. Collection, derivatively, is the algorithmic mode that depends on recollection: a collection is a sub-set of the recollection given over to a known algorithmic operation — the indexing of films by parametric variables, with the scheme as generator — whose well-formedness is licensed by the well-ordering that recollection supplies. *The umbilical between collection and recollection.* The decomposition of a recollection-entry into indexers and pointers — the ordinal position, the per-channel colour array, the cryptographic signature of §9.3.1, the scheme-generator pointer when the entry has a procedural origin — is structurally analogous to the decomposition of a natural number into its positional decimal digits: just as a number is organised into parts by the place-value structure of decimal order, a recollection-entry is organised into parts by its indexer-and-pointer structure, with collection operating by selection over these parts. The umbilical visible at this level — collection as selection over the well-ordered parts of recollection-entries — explains why the two modes commune in the well-ordering principle: collection inherits its well-formedness from recollection's ordinal order, and recollection's ordinal order is itself accessible to collection through the part-structure. The photograph example of §2 may be used in the negative here: a digitally-captured photograph is not, by the act of capture, a U.M. recollection; it becomes part of a recollection only once it is given an ordinal index in a persistent catalogue alongside its peers. The distinction echoes the patent's vocabulary and translates, in our framework, to the distinction between the arithmetic ordering of an ordinal catalogue up to the transfinite (recollection) and the enumeration of a sub-class indexed by a computable function (collection) (compare §5).

2 Computable images and computable films

> *"The 'computable' numbers may be described briefly as the real numbers whose expressions as a decimal are calculable by finite means."*
>
> —A. M. Turing, 1936, §1

We work in the colour-native setting of §1.5. Let Σ denote Σ_{RGB} throughout, unless otherwise indicated. The framework of the present section — a formal account of what is computable in the visual domain, prior to questions of *how* (algorithmically) and *on what substrate* (physically) — corresponds in spirit to Marr's (1982) *computational* level of analysis of vision, with the algorithmic and implementational levels treated in §§9–10 and §12 respectively.

A *digital image* of resolution $W \times H$ is a function $I : \{0, \ldots, W-1\} \times \{0, \ldots, H-1\} \to \Sigma$. A *digital film* of resolution $W \times H$ and length T is a function $F : \{0, \ldots, W-1\} \times \{0, \ldots, H-1\} \times \{0, \ldots, T-1\} \to \Sigma$.

A bare definition of computability for a single fixed-resolution image is unenlightening: every such image is trivially computable, since its WH pixel values can be hard-coded into the transition table of a Turing machine. The non-trivial notion is *uniform* computability across a parameter—capturing images defined by a procedure rather than by exhaustive tabulation.

Definition 2.1 (Computable image scheme). *A computable image scheme is a function $I : \mathbb{N}^3 \to \Sigma$ for which there exists a Turing machine M_I over alphabet Σ such that, for every $(x, y, n) \in \mathbb{N}^3$ with $0 \leq x, y < n$, the machine M_I on input (x, y, n) halts with output $I(x, y, n)$. We interpret $I(\cdot, \cdot, n)$ as the digital image at resolution $n \times n$.*

Definition 2.1 captures images admitting a finite procedural description independent of the resolution at which they are sampled — what U.M. (Homem 2021) terms "*typical digital images*" in the resolution-parametric sense, distinguishing them from raw bitmap tabulations bound to one fixed pixel grid. A vector graphic, a fractal such as the Mandelbrot or Julia sets, a signed distance field, a ray-traced scene from a fixed scene description, and the output of any compiled rendering pipeline all fall under this notion.

The case of the photograph deserves a separate sentence. A photograph captured at one fixed pixel grid and stored as a finite table of $W \times H$ values is, by Definition 2.1, trivially computable as a finite scheme (the machine hard-codes the table and returns the indexed value), but it is *not* a computable image scheme in the non-trivial sense the definition seeks — the trivial inclusion follows from the table's finite size, not from any procedural generation. A photograph becomes a computable image scheme in the substantive sense exactly when it is re-cast as a continuous reconstruction — by interpolation, super-resolution, learned upsampling, or generative reconstruction — with the reconstruction itself effectively specified by a finite procedure that takes the original sample and a resolution parameter n and returns the value at (x, y, n). The bridge between the discrete capture and the resolution-parametric scheme is therefore a learned or specified reconstruction; once the reconstruction is fixed, the photograph enters the class of §2 with no further qualification.

Definition 2.2 (Computable film scheme). *A computable film scheme is a function $F : \mathbb{N}^4 \to \Sigma$ for which there exists a Turing machine M_F such that, for every $(x, y, t, n) \in \mathbb{N}^4$ with $0 \leq x, y < n$ and $0 \leq t < n$, the machine M_F on input (x, y, t, n) halts with output $F(x, y, t, n)$.*

For films, a refinement is sometimes appropriate: we may wish the value at frame t to be computable from data available *by frame t* rather than from the entire film. This matters when the film is generated in response to an exogenous input stream—interactive rendering, simulation, live broadcast.

Definition 2.3 (Causal computable film scheme). *Let $\xi : \mathbb{N} \to \Xi^*$ be an input stream over an alphabet Ξ, with $\xi(t)$ the input received during frame t. A computable film scheme F is causal with respect to ξ if there exists a Turing machine M_F such that, for every (x, y, t, n), the computation of $F(x, y, t, n)$ reads only $\xi(0), \xi(1), \ldots, \xi(t)$ from the input stream.*

Causal and acausal computable films coincide when no input stream is involved. The distinction becomes substantive in §10, where closure of computable films under temporal operations (cuts, reversals, time-warps) interacts non-trivially with causality. A complementary observation belongs here, which the quantum-photonic reading of §9.4 makes precise: only a quantum measurement of the colour-synthesis at the pixel level can read the three-channel *synthesis* as one observable while preserving the information needed to recover, by deduction or programmable regression (§9.4), the precise channel values that produced it — without provoking the wave-function collapse that a channel-by-channel measurement would force first. The information so recovered enters the programme directly as input to subsequent computation, so that the causal/acausal distinction interacts not only with temporal operations but with the measurement structure of the photonic substrate itself.

The discrete schemes of Definitions 2.1–2.3 are primary in this theory. Nothing prevents one from also considering an idealised continuous image $\tilde{I} : [0,1]^2 \to [0,1]^3$ and asking when such a function is computable in the sense of Type-Two Effectivity (Pour-El & Richards, 1989; Weihrauch, 2000). We sketch this idealisation, and its relation to the discrete schemes, in §9.2, where it serves as the bridge between the colour-native model and the analog limit. For the development of §§3–8, the discrete schemes suffice.

3 Examples of computable image schemes

We illustrate Definition 2.1 with a graded sequence of worked examples, beginning with cases trivially within the definition and ending with one whose limit behaviour previews the diagonal arguments of §8. Throughout, $\Sigma = \Sigma_{\mathrm{RGB}}$ and we describe the Turing machine M_I informally, in the modern convention of giving an algorithm and noting that its translation to a transition table is mechanical.

Example 3.1 (Constant image). *Fix $c \in \Sigma$. The constant image scheme $I_c(x, y, n) = c$ for all (x, y, n) is computable: M_{I_c} ignores its input and writes c to the output. Trivial as it is, this example fixes the form of the definition: a computable image scheme is one whose value at each pixel-coordinate triple is produced by a single machine.*

Example 3.2 (Linear gradient). *Define $I_\nabla(x, y, n) = (\lfloor 255x/(n-1)\rfloor, \lfloor 255y/(n-1)\rfloor, 0)$ for $0 \le x, y < n$ and $n \ge 2$. The machine M_{I_∇} on input (x, y, n) performs two integer multiplications, two divisions, and packs the resulting triple into a single Σ-symbol on the output tape. All arithmetic is on bounded integers; the running time is polynomial in $\log n$. The example shows that any image scheme defined by a fixed-arity arithmetic formula in its coordinates is computable.*

Example 3.3 (Parametric primitive — disc). *Let D_r denote the filled disc of radius $r \in (0,1)$ centred at $(\frac{1}{2}, \frac{1}{2})$, rendered against a white background. Define*

$$I_{D_r}(x, y, n) \;=\; \begin{cases} (0,0,0) & \textit{if } (x/n - \frac{1}{2})^2 + (y/n - \frac{1}{2})^2 \le r^2, \\ (255,255,255) & \textit{otherwise.} \end{cases}$$

The machine evaluates the squared distance using bounded-precision rational arithmetic at precision $O(\log n)$, compares with r^2, and writes the result. The example generalises: any image whose pixel value is determined by a decidable predicate on (x, y, n) over the rationals is computable.

The pedagogical content of Example 3.3 is the role played by the radius r as the *parametric handle* on the family of schemes $\{I_{D_r}\}_{r\in(0,1)}$. The single parameter r separates a one-dimensional family of computable image schemes from the trivially-tabulated single image one would obtain by fixing r and storing pixel values. Three observations frame what this example is doing in the development. First, a parametric primitive is the simplest non-trivial demonstration that the resolution parameter n of Definition 2.1 is one of *at least two* independent axes of variation, the second being the geometric parameter r of the primitive itself; the class of computable image schemes is closed under both, by routine extension of §4's compositional macros. Second, the rationals-with-bounded-precision discipline used to evaluate the predicate is the prototype for what §9.2 makes precise as Type-256 (and Type-Two) effectivity: a computable predicate on a continuous parameter, decided by finite-precision approximation. Third, the choice of *intensity* as the pixel value here — the binary intensity $\{0, 255\}^3$ in the example, but generalising to the full Σ_{RGB} range — is not incidental: among the parameters that could be carried at the pixel level (intensity, wavelength composition, phase), only intensity admits a direct sensor-level read-out under the colour-native commitment of §1.5, with wavelength and phase entering as constituents of the synthesis (§9.4). The radius example is small but it carries the full discipline of the framework in miniature.

Example 3.4 (Signed distance field). *Let $\phi : [0,1]^2 \to \mathbb{R}$ be a computable function in the sense that there is a Turing machine that, on input (x, y, k), returns a rational approximation to $\phi(x/n, y/n)$ accurate to within 2^{-k}. Then the image scheme*

$$I_\phi(x, y, n) = \begin{cases} (0,0,0) & \text{if } \phi(x/n, y/n) \le 0, \\ (255, 255, 255) & \text{otherwise} \end{cases}$$

is computable wherever the predicate $\phi(x/n, y/n) \le 0$ is decidable from finite-precision approximations—that is, wherever ϕ does not vanish on the dyadic-rational grid. The qualification matters: it foreshadows the gap between definability and computability that will be the subject of §8.

Example 3.5 (The Mandelbrot image scheme). *For $c \in \mathbb{C}$, define the iteration $z_0 = 0$, $z_{k+1} = z_k^2 + c$. The Mandelbrot escape time at threshold K is*

$$\tau_K(c) \;=\; \min\{k \le K : |z_k| > 2\} \quad (\textit{or } K \textit{ if no such } k \textit{ exists}).$$

Choose a colour palette $\pi : \{0, \dots, K\} \to \Sigma$. The Mandelbrot image scheme at iteration depth $K(n)$ (where K depends on the resolution n, e.g. $K(n) = \lceil n \log n \rceil$) is

$$I_M(x, y, n) \;=\; \pi\big(\tau_{K(n)}(c_{x,y,n})\big), \qquad c_{x,y,n} \;=\; -2 + \tfrac{3x}{n} + i\big(-\tfrac{3}{2} + \tfrac{3y}{n}\big).$$

The machine M_{I_M} performs $K(n)$ rounds of complex arithmetic at precision $O(\log n)$ per pixel, halts, and outputs $\pi(\tau)$. By Definition 2.1, I_M is a computable image scheme.

The example carries a subtlety we shall return to. For each finite resolution n and depth bound $K(n)$, $I_M(\cdot, \cdot, n)$ is computable. The *limit object*—the Mandelbrot set $M = \{c \in \mathbb{C} : \sup_k |z_k| < \infty\}$—has computability properties that are markedly more delicate. Hertling (2005) and Braverman & Yampolsky (2006) showed that the Mandelbrot set is computable in the sense of Type-Two Effectivity if and only if the Mandelbrot hyperbolicity conjecture holds—a question still open. Our scheme I_M thus produces, at every finite resolution, a perfectly computable image of an object whose continuous limit may or may not be computable. The gap between the two is precisely the territory of §§8–9.

Example 3.6 (Ray-traced scene from a finite description). *Let a scene description $\mathcal{S}$ comprise a finite list of geometric primitives (each a parametrised computable surface), a finite list of light sources, a computable bidirectional reflectance distribution function (BRDF) for each surface, and a camera specified by a computable projection. Define $I_{\mathcal{S}}(x, y, n)$ to be the colour computed by recursive ray-tracing of the pixel (x, y) at resolution n, with recursion depth $d(n)$ and Monte Carlo sample count $s(n)$ both bounded computable functions of n. Then $I_{\mathcal{S}}$ is a computable image scheme. The proof is routine but lengthy; we sketch it in Appendix B.*

Example 3.6 is the bridge between this paper and graphics practice. Every offline renderer that terminates on every input—that is, every renderer used in production—implements a computable image scheme in the sense of Definition 2.1. The class of computable images is therefore at least as large as the class of images producible by current rendering pipelines; in §10 we will see that it is strictly larger.

4 Compositional macros for image and film schemes

The class of computable image schemes is, by Definition 2.1, the class of functions admitting *some* Turing machine that halts on every coordinate input. Verifying membership for any non-trivial scheme by exhibiting an explicit transition table is impractical and uninformative. The remedy, familiar from every modern treatment of computability, is to specify schemes by *composition* of simpler ones, and to verify that the class of computable schemes is closed under each compositional operation.

We adopt the following operators. In each case we state the operator on schemes; the verification that the class is closed under it amounts to writing the obvious composite Turing machine and is omitted.

Pointwise operators. Let $\phi : \Sigma^k \to \Sigma$ be a computable function. For computable image schemes $I_1, \ldots, I_k$, define

$$(\phi \star (I_1, \ldots, I_k))(x, y, n) \;=\; \phi\big(I_1(x, y, n), \ldots, I_k(x, y, n)\big).$$

Pointwise operators include channel-wise arithmetic, alpha compositing, colour-space conversion, gamma correction, and tone mapping. The composite is computable.

Local (kernel) operators. Let $r : \mathbb{N} \to \mathbb{N}$ be a computable radius function and $\kappa : \Sigma^{(2r+1)^2} \to \Sigma$ a computable aggregation. For a computable image scheme I, define

$$(\kappa \cdot_r I)(x, y, n) \;=\; \kappa\big(\, I(x{+}u, y{+}v, n) \,:\, |u|, |v| \leq r(n) \,\big),$$

with values outside $\{0, \ldots, n-1\}$ supplied by a computable boundary rule (clamp, wrap, or reflect). Local operators include convolution by a fixed computable kernel, mathematical morphology (erosion, dilation), and neighbourhood-based filtering. The composite is computable; for fixed r, the operator is moreover *embarrassingly parallel*, which we will use in §10.

Geometric transforms. Let $\sigma = (\sigma_n)_{n\in\mathbb{N}}$ be a computable family of partial maps $\sigma_n : \{0,\ldots,n-1\}^2 \dashrightarrow \{0,\ldots,n-1\}^2$, with a computable boundary rule for points outside the domain. For a computable image scheme I, define

$$(T_\sigma I)(x,y,n) \;=\; I\big(\sigma_n(x,y),n\big).$$

Geometric transforms include translation, rotation, scaling, projective mappings, and arbitrary computable warps. Inverse-mapping the destination pixel—rather than forward-mapping the source—keeps the scheme well-defined at every output coordinate. The composite is computable.

Compositing operators. Let $A : \mathbb{N}^3 \to \{0,\ldots,255\}$ be a computable scalar scheme (an *alpha channel*) and I_1, I_2 computable image schemes. The Porter–Duff *over* operator

$$(I_1 \text{ over}_A\; I_2)(x,y,n) \;=\; \tfrac{A(x,y,n)}{255}\cdot I_1(x,y,n) \;+\; \Big(1-\tfrac{A(x,y,n)}{255}\Big)\cdot I_2(x,y,n)$$

(with arithmetic performed channel-wise in $\{0,\ldots,255\}$ and rounded) is a pointwise operator parametrised by a third scheme; it preserves computability.

Temporal operators (films). For a computable film scheme F and a computable function $\tau : \mathbb{N}^2 \to \mathbb{N}$ specifying a time-warp $\tau(t,n)$ at length n, define

$$(W_\tau F)(x,y,t,n) \;=\; F\big(x,y,\tau(t,n),n\big).$$

Cuts, reversals, time-warps, retiming, and frame interpolation (combined with a pointwise blend) are all instances. The composite is computable. Note that W_τ is in general *not* causal in the sense of Definition 2.3: a film reversed in time depends on its own future. Causality is preserved precisely when $\tau(t,n) \le t$ for all (t,n).

The point of the present section is that these operators, freely combined, generate a large vocabulary of computable schemes from a small set of primitives. The graphics pipeline of Example 3.6 is a finite composition of operators of these types together with the primitive schemes of Examples 3.1–3.5. We collect the closure observation as a single proposition.

Proposition 4.1 (Closure). *The class of computable image schemes is closed under finite composition of pointwise, local, geometric, and compositing operators with computable parameters. The class of computable film schemes is closed additionally under temporal operators with computable time-warp; the subclass of causal computable film schemes is closed under temporal operators with non-anticipating time-warp.* □

A historical remark closes the section. Turing's original treatment introduced an analogous compositional layer under the heading of *m-functions*—abbreviations for sub-machines, with conventions for parametrisation and re-entry. The need for such a layer was acute in 1936, when no programming languages existed and every machine had to be presented as a literal transition table. The contemporary reader is well-served by absorbing the same content into the modern algorithmic register, in which composition, parametrisation, and substitution are taken for granted. Section 4 is therefore brief by design: its work is largely already done by the conventions every reader already shares.

5 Enumeration of computable image schemes

A Turing machine over Σ_{RGB} is determined by a finite specification: the cardinality of its state set, the cardinality of its tape alphabet (which extends Σ_{RGB} by a finite number of working symbols), and a finite list of transition quintuples. We assign to each such specification a single natural number—its *description number*—by means of a standard Gödel encoding, and use this to enumerate the class of computable image schemes.

The encoding in detail is unimportant; any computable, injective procedure mapping machine specifications to natural numbers will serve. We fix one for definiteness. Number the states $q_0, q_1, q_2, \ldots$ in the order they first appear in the transition list. Number the working tape symbols $\gamma_0 = \mathfrak{B}, \gamma_1, \gamma_2, \ldots$, with the symbols of Σ_{RGB} identified with their natural integer values in $\{0, 1, \ldots, 16{,}777{,}215\}$. Encode each quintuple $(q_i, \sigma, q_j, \sigma', m) \in Q \times \Gamma \times Q \times \Gamma \times \{L, R, S\}$ as a tuple of integers (i, s, j, s', m') with $m' \in \{0, 1, 2\}$. Concatenate the quintuples in lexicographic order, separated by a delimiter. The resulting string of integers, read in a fixed base larger than any value that occurs, is the *standard description* of the machine; its value as an integer is the *description number* $\mathrm{DN}(M)$.

The function $M \mapsto \mathrm{DN}(M)$ is injective and computable, and its range is a decidable subset of $\mathbb{N}$. Conversely, given any $n \in \mathbb{N}$, one can decide in finite time whether n is the description number of a well-formed Turing machine, and if so reconstruct it. We may therefore index machines by description number: $M_1, M_2, M_3, \ldots$ enumerates all Turing machines over Σ_{RGB}, with the convention that ill-formed numbers are assigned a fixed trivial machine (one that halts immediately on every input with output 0).

Definition 5.1 (Enumeration). *For each $k \in \mathbb{N}$, let $I_k : \mathbb{N}^3 \to \Sigma_{RGB}$ be the partial function*

$$I_k(x, y, n) \;=\; \begin{cases} \textit{output of } M_k(x, y, n) & \textit{if } M_k \textit{ halts on input } (x, y, n), \\ \textit{undefined} & \textit{otherwise.} \end{cases}$$

The computable image schemes are precisely the total functions in the family $\{I_k\}_{k \in \mathbb{N}}$.

The enumeration is exhaustive—every computable image scheme appears as some I_k—but heavily redundant: distinct machines compute the same scheme, and the same scheme has infinitely many description numbers. Removing this redundancy is itself uncomputable, a corollary of the undecidability of machine equivalence to which we return in §8.

Cardinality. The set $\Sigma_{\mathrm{RGB}}^{\mathbb{N}^3}$ of all functions $\mathbb{N}^3 \to \Sigma_{\mathrm{RGB}}$ has cardinality $|\Sigma_{\mathrm{RGB}}|^{\aleph_0} = 2^{\aleph_0}$. The enumeration $\{I_k\}_{k \in \mathbb{N}}$ is countable. The diagonal-type construction by which we extract uncomputable functions (§8) is Cantorian in origin: it is the same construction that, in Cantor's transfinite arithmetic, separates the countable from the uncountable and gives rise to the hierarchy of cardinals $\aleph_0 < 2^{\aleph_0} < 2^{2^{\aleph_0}} < \cdots$. The distance between the countable class of computable image schemes and the uncountable class of all possible functions $\mathbb{N}^3 \to \Sigma_{\mathrm{RGB}}$ is therefore not a quantitative gap but the fundamental cardinality jump that Cantor's theorem identifies. On the U-Mentalism reading, this is also the place where U.M. 'C' (finitary, countable, computable) meets the conceptual horizon of U.M. 'O' (whose object, the constitution of the image across all photonic viewpoints, naturally inhabits the higher cardinalities of the transfinite ontology).

Therefore:

Proposition 5.2. *The computable image schemes form a countable subset of $\Sigma_{RGB}^{\mathbb{N}^3}$, of measure zero under any non-degenerate product measure. There are uncountably many functions $\mathbb{N}^3 \to \Sigma_{RGB}$ that are not computable image schemes.* $\square$

Proposition 5.2 is the cardinality scaffold for the diagonal argument of §8. By itself it does not yet exhibit a specific uncomputable function; for that we need the diagonal construction. It does, however, establish a fact that is sometimes overlooked in casual discussion: *most* possible images, in any precise mathematical sense one cares to make, lie outside the reach of any algorithm.

Kolmogorov complexity of an image scheme. The description-number encoding induces a natural notion of descriptive complexity. Fix a universal Turing machine U over Σ_{RGB}—that is, a machine such that, for any machine M and input w, U on input $\langle \mathrm{DN}(M), w\rangle$ simulates M on w. (We construct one explicitly in §§6–7; for the present section we assume its existence.) For a computable image scheme I, define

$$K(I) \;=\; \min\{\, |p| \,:\, U(p,x,y,n) = I(x,y,n) \text{ for every } (x,y,n) \in \mathbb{N}^3 \,\},$$

where p ranges over input programs and $|p|$ denotes program length in symbols of U's input alphabet. The quantity $K(I)$ is the *Kolmogorov complexity* of I relative to U—the length of the shortest program that produces it.

Three observations frame the role of K in what follows.

First, K is itself uncomputable: no algorithm, given a description of a computable image scheme, returns its Kolmogorov complexity (Solomonoff 1964; Kolmogorov 1965; Chaitin 1969; see Li & Vitányi 2019 for the comprehensive contemporary treatment). This parallels the uncomputability of the halting problem and will in fact be derived from it in §8.

Second, for image schemes with a resolution parameter, Kolmogorov complexity captures the gap between *procedural* and *enumerative* description. The exhaustive pixel-list of a scheme at resolution $n \times n$ requires $24n^2$ bits. If $K(I)$ is bounded—independent of n—the scheme admits an arbitrary-resolution description in constant space. The compression ratio $K(I)\,/\,24n^2 \to 0$ as $n \to \infty$ is the formal counterpart of the everyday observation that procedurally-generated images do, in a precise sense, more with less. The Mandelbrot scheme of Example 3.5 has bounded Kolmogorov complexity; a uniformly random pixel array does not.

Third, Kolmogorov complexity gives a principled separation between schemes whose information content is intrinsically enumerative (high K, bounded only by raw data size) and those whose content is intrinsically procedural (low K). This complexity separation is logically independent of the U-Mentalism recollection / collection distinction of §1.5 and should not be conflated with it. Recollection is the well-ordering of *any* computable image by its position in the ordinal-arithmetic catalogue — the enumeration of image-numbers from $0000\ldots0$ upward in $\Sigma_{\mathrm{RGB}}^{n\times n}$ for any chosen resolution, and across resolutions in transfinite series — irrespective of whether the image is intrinsically high- or low-K. A raw natural photograph and a procedural fractal both belong, with equal title, to the recollection of §§1.5 and 12.4: both are computable images, both have well-defined ordinal positions in the universal catalogue, both are objects of the persistent indexing that UTMs as Internet servers (§12.4) perform as their primary operation. Collection, accordingly, is the sub-set of the recollection given over to a known algorithmic operation — the indexing of films by parametric variables, with the scheme as generator, in the algorithmic sense already exercised in §§3–4. The high-K versus low-K axis is a property of the *scheme* that generates an image (or of the shortest such scheme); the recollection / collection axis is a property of the *mode of operative access* through which an image is retrieved. The compressibility of practical images under standard codecs (JPEG, AVIF, learned codecs) is a lower bound on the gap $24n^2 - K(I)$ and a quantitative measure of how much enumerative content reduces to procedural content for a given image; it is not, contrary to a tempting but incorrect first reading, a measure of how much an image belongs to collection rather than to recollection.

A short self-referential remark closes the section. The standard description of a Turing machine is, by construction, a finite string over an integer alphabet—and nothing prevents that string from being read as a coordinate-indexed sequence of Σ_{RGB} values, that is, as itself a digital image. Every computable image scheme thus admits a *self-image*: a finite digital image whose pixels encode the standard description of a machine that produces the scheme. The construction is uninteresting taken on its own—standard descriptions are not chosen for visual coherence, and the resulting self-image is in general a noise field—but it illustrates a closure property characteristic of the colour-native model: code and image inhabit the same alphabet, and the apparent gap between the two is conventional rather than fundamental. We will not exploit this remark further here; it returns, however, in §11, where the visual *Entscheidungsproblem* turns on precisely the ambiguity between an image regarded as data and the same image regarded as code.

6 The universal image-generating machine

> *"It is possible to invent a single machine which can be used to compute any computable sequence."*
>
> —A. M. Turing, 1936, §6

The aim of this section is to establish, for the colour-native setting, the central existence theorem on which the rest of the paper turns: a single Turing machine that, given the description of any other Turing machine and an input, produces what that machine would produce on that input. Specialised to image schemes, the theorem says a single machine produces any computable image at any resolution. We state the theorem here; the explicit construction is given in §7.

Theorem 6.1 (Universal machine). *There exists a Turing machine U over an alphabet $\Gamma_U \supseteq \Sigma_{RGB}$ with the following property. For every Turing machine M over Σ_{RGB} and every input $w \in \Sigma_{RGB}^*$, the machine U on input $\langle \mathrm{DN}(M), w\rangle$ halts if and only if M halts on w, and in that case produces the same output as M.*

In particular, for every $k \in \mathbb{N}$ and every coordinate triple (x, y, n) on which M_k halts,

$$U\big(\langle k, x, y, n\rangle\big) \;=\; I_k(x, y, n).$$

The corollary for image schemes is immediate.

Corollary 6.2 (Universal image-generating machine). *There exists a Turing machine U over an alphabet $\Gamma_U \supseteq \Sigma_{RGB}$ such that, for every computable image scheme I with description number k and every $(x, y, n) \in \mathbb{N}^3$, the machine U on input $\langle k, x, y, n\rangle$ halts and outputs $I(x, y, n)$.* □

The same statement, applied to film schemes with input quadruples (x, y, t, n), gives a universal *film*-generating machine; we treat this as a routine extension and do not state it separately.

Sketch of U. U partitions its tape into four regions, separated by reserved sentinel symbols:

(i) the *description region*, holding the standard description of the simulated machine M_k—its encoded list of transition quintuples;

(ii) the *simulated tape*, on which U maintains a faithful copy of the contents M_k's tape would have at the corresponding step of M_k's computation;

(iii) the *state register*, holding the encoded current internal state of M_k;

(iv) the *head register*, holding the position of M_k's tape head, encoded as an integer.

A *working region* for arithmetic and bookkeeping completes the layout. The simulation loop is conceptually simple: U reads the symbol under the simulated head and the contents of the state register; searches the description region for the unique quintuple beginning with that (state, symbol) pair; rewrites the symbol on the simulated tape, updates the state register, and advances the head register according to the quintuple's directive; halts when the simulated machine reaches a halting state. The full transition table of U is given in §7.

Discussion. Three consequences of Theorem 6.1 are operative in the remainder of the paper.

Universality of computation transfers to universality of image generation. A single fixed machine—not a family parametrised by image, not a different machine for each rendering pipeline—produces every image and every film expressible by any computable scheme whatsoever. Every offline renderer, every generative model, every codec implementation is a finite instantiation of, or a finitary fragment of, this single mathematical object. The contemporary visual pipeline is, structurally, a sequence of approximations to one machine.

The colour-native reading lifts to the universal level. The universal machine U reads and writes symbols of Σ_{RGB} natively. A binary universal machine simulates colour-native machines through bit-by-bit encoding and decoding at every step; the colour-native universal machine simulates them directly, with no encoding overhead per symbol read or written. The asymptotic class of computable schemes is unchanged (Proposition 1), but the *cost model* is. On a photonic substrate, where the natural symbol is the wavelength–intensity pair carried by a photon, the colour-native universal machine is the formal object of which the photonic computer is the physical embodiment. Section 9.4 develops this correspondence in detail; here we record only that universality holds in the colour-native setting without asymptotic penalty.

The universal machine is the formal kernel of the U-Mentalism programme. The Universal Machine appearing in the broader project is precisely U as constructed here—a single, colour-native, ultimately photonic device that produces any computable image or film on demand. The mathematical content of the present paper is its formal foundation; the engineering challenges of realising it physically—finite memory, finite precision, finite energy budget, irreducible noise—are the subject of §§12.4 and 12.6. Theorem 6.1 establishes that there is a coherent target; later sections delimit how closely it can be approached and at what cost.

A remark on the alphabet of U. The theorem statement permits Γ_U to be a proper superset of Σ_{RGB}—that is, U may use working symbols beyond the colour palette. This is unavoidable: U requires sentinel symbols to delimit the four regions of its tape and is unlikely to find unused colour values for the purpose. One natural choice is to extend Σ_{RGB} by a small fixed set of administrative symbols disjoint from the palette, raising $|\Gamma_U|$ to $|\Sigma_{\mathrm{RGB}}| + c$ for some small constant c. The construction in §7 makes this concrete. The choice affects neither the class of computable schemes nor the conceptual content of universality; it is an implementation detail to be settled and forgotten.

7 Construction of the universal image-generating machine

We give the construction of the universal machine U promised by Theorem 6.1, presented in five layers: tape layout (§7.1); admin alphabet and encoding conventions (§7.2); the five core subroutines (§7.3); the main loop with its explicit transition table (§7.4); and a state count with closing remarks (§7.5).

7.1. Tape layout

The tape of U is partitioned, left to right, into four contiguous regions delimited by sentinel symbols:

$$\underbrace{\mathfrak{S}_0}_{\text{left end}} \ \big[\text{description region}\big]\ \mathfrak{S}_1\ \big[\text{state register}\big]\ \mathfrak{S}_2\ \big[\text{simulated tape}\big]\ \mathfrak{S}_3\ \big[\text{working region}\big]\ \underbrace{\mathfrak{B}\,\mathfrak{B}\,\ldots}_{\text{right end}}$$

The *description region* holds the encoded transition table of M_k, written once at initialisation. The *state register* holds the encoded current internal state of M_k. The *simulated tape* holds the contents of M_k's tape with head position marked in-line. The *working region* is scratch space.

7.2. Admin alphabet and encoding conventions

The tape alphabet of U is $\Gamma_U = \Sigma_{\mathrm{RGB}} \cup \mathfrak{A}$, where $\mathfrak{A}$ is a finite admin alphabet of 20 symbols disjoint from Σ_{RGB}: sentinels $\mathfrak{S}_0, \mathfrak{S}_1, \mathfrak{S}_2, \mathfrak{S}_3$ (4); delimiters $\mathfrak{D}_1, \mathfrak{D}_2$ (2); head markers $\mathfrak{H}, \mathfrak{N}$ (2); blank $\mathfrak{B}$ (1); decimal digits $\mathfrak{0}, \mathfrak{1}, \ldots, \mathfrak{9}$ (10); scratch $\mathfrak{X}$ (1).

Integer encoding. Non-negative integers are written in base ten using admin digits, most-significant first.

Quintuple encoding. A quintuple $(q_i, \sigma, q_j, \sigma', m)$ of M_k is written as $\langle i\rangle\, \mathfrak{D}_1\, \sigma\, \mathfrak{D}_1\, \langle j\rangle\, \mathfrak{D}_1\, \sigma'\, \mathfrak{D}_1\, \langle m\rangle\, \mathfrak{D}_2$. We use $\{0, 1, 2\}$ as the on-tape encoding of the directional codes $\{-1, 0, +1\}$ via $m \mapsto m + 1$. The full description region concatenates all quintuples in lexicographic order.

Simulated tape encoding. Each cell of M_k's tape is represented by a pair of cells of U's tape: a colour symbol followed by either $\mathfrak{H}$ (head here) or $\mathfrak{N}$ (not here). Exactly one $\mathfrak{H}$ appears at any moment.

7.3. The five core subroutines

One simulation step is implemented by five subroutines called in sequence:

Find-Head. From $\mathfrak{S}_2$, scan right past pairs until $\mathfrak{H}$ is found, ending on the colour symbol immediately to its left.

Read. Mark the simulated head position by overlaying $\mathfrak{H}$ with $\mathfrak{X}$, recording its location for later use.

Load-State. Position U's head at the start of the state register.

Match. Scan the description region for the unique quintuple matching (current state, current symbol). Mark the matched quintuple with $\mathfrak{X}$.

Apply. Read new state, new symbol, and direction from the matched quintuple; overwrite the state register; write the new symbol at the simulated head position; move the head marker accordingly.

7.4. The main loop

The main loop Main tests whether the state register holds a halting-state code; if not, calls the five subroutines in order. The wildcard $*$ stands for any symbol of Γ_U not otherwise matched in the same row block; when it fires, the symbol written equals the symbol read.

State	Read	New state	Write	Move
μ_0	$\mathfrak{S}_0$	μ_1	$\mathfrak{S}_0$	R
μ_1	$*$	μ_1	(same)	R
μ_1	$\mathfrak{S}_1$	μ_2	$\mathfrak{S}_1$	R
μ_2	$0, \ldots, 9$	μ_3	(same)	R
μ_3	$0, \ldots, 8$	μ_3	(same)	R
μ_3	9	μ_4	9	R
μ_3	$\mathfrak{S}_2$	$\text{Find-Head}^{\text{enter}}$	$\mathfrak{S}_2$	R
μ_4	$\mathfrak{S}_2$	Halt	$\mathfrak{S}_2$	S
μ_4	$0, \ldots, 8$	μ_3	(same)	R
μ_4	9	μ_4	9	R
$\text{return}_{\text{Apply}}$	$\mathfrak{S}_2$	μ_0	$\mathfrak{S}_2$	L
$\text{return}_{\text{Apply}}$	$\mathfrak{S}_3$	μ_0	$\mathfrak{S}_3$	L

The halting detection in $\mu_2 \to \mu_3 \to \mu_4$ implements lexicographic comparison of the state register against a fixed list of halting-state codes encoded immediately before $\mathfrak{S}_2$. The version shown uses a simplified "four consecutive nines" placeholder rule whose only purpose is to keep the table compact; the rigorous lexicographic version adds approximately three states to the main-loop count. Arithmetic on integer-encoded fields (state numbers, head positions, directional codes) is performed by modulo-base addition arranged on the tape: digit-by-digit operations with carry, the standard schoolbook layout adapted to a tape rather than a piece of paper. The choice of base 10 in our encoding is conventional and visual; Turing's original 1936 construction uses both binary digits (for the symbol field, in the sense in which real numbers are represented by the position of a binary point or *comma* dividing integer and fractional parts on the tape's symbol track) and decimal-like positional encoding (for the description-number representation of machines), and either base works under the alphabet equivalence of Proposition 1.

The transition table of Find-Head:

State	Read	New state	Write	Move
Find-Head$^{\text{enter}}$	colour, $\mathfrak{B}$	ϕ_1	(same)	R
ϕ_1	$\mathfrak{H}$	ϕ_2	$\mathfrak{H}$	L
ϕ_1	$\mathfrak{N}$	ϕ_3	$\mathfrak{N}$	R
ϕ_2	colour, $\mathfrak{B}$	Read$^{\text{enter}}$	(same)	S
ϕ_3	colour, $\mathfrak{B}$	ϕ_1	(same)	R

From Find-Head$^{\text{enter}}$ on a colour, the head moves right to expose the marker; on $\mathfrak{H}$, ϕ_2 steps left to the colour and exits; on $\mathfrak{N}$, ϕ_3 advances right to the next colour and re-enters ϕ_1. The remaining subroutines follow the same format and are tabulated in Appendix C.

7.5. State count

Counting states: 6 (Main) + 4 (Find-Head) + 3 (Read) + 5 (Load-State) + 14 (Match) + 11 (Apply) + 1 (Halt) = 44 states. The figure is a property of the present decomposition, not of universality as such: alternative groupings of Match and Apply shuttle infrastructure produce different counts in the high thirties to low fifties, and aggressively compact universal machines in the literature (Minsky's 4-state, 6-symbol UTM; Rogozhin's 4-state, 6-symbol variants over a smaller alphabet; Wolfram & Smith's 2-state, 3-symbol UTM proven universal by Smith 2007, the smallest currently known) reach far smaller counts at the cost of working symbols and per-step time. The general lower bound is a Pavlotskaya-Yaakobi-style trade-off between state count and alphabet richness: for a universal machine over an alphabet of $|\Sigma|$ symbols, the minimum state count scales approximately as $1/\log_2|\Sigma|$ for fixed computational complexity. Substituting $|\Sigma_{\text{RGB}}| = 2^{24}$, the theoretical lower bound on a fully-compacted universal machine over Σ_{RGB} is in the neighbourhood of 2–4 states, at the cost of transition tables with hundreds of thousands of entries opaque to pedagogical inspection. The figure 44 is retained as the headline for this paper because the partition is the most natural for the colour-native setting, because each named subroutine maps directly onto an m-function in the sense of Turing's 1936 §7, and because the per-phase cost contribution is isolated explicitly. The admin alphabet count of 20 (§7.2) is, similarly, a property of the decomposition: a more compact encoding using base-2 digits would reduce the digit contribution from 10 to 2 (yielding $|\mathfrak{A}| = 12$) at the cost of longer state register strings; base-16 would inflate the count to $|\mathfrak{A}| = 26$ with shorter strings. The 10 decimal digits are chosen for legibility rather than minimality. There is no direct numerological parallel with Turing's 1936 construction, which is presented in terms of *m-functions* (named sub-machines composed by substitution) rather than enumerated states; the modern 5-subroutine partition of our §7.3 is, however, structurally analogous to Turing's m-function organisation of his §7.

The size of Γ_U, $|\Sigma_{\text{RGB}}|+20 = 16{,}777{,}236$, is dominated by the colour palette. The transition function is highly structured: the wildcard rules stand for entire blocks of identical behaviour, so the actual specification of δ_U requires only a few hundred non-wildcard rules.

The construction is, in skeletal form, insensitive to the choice of Σ. What changes with the alphabet is not the complexity of the construction but the complexity of one simulation step. In the colour-native machine, Read copies one colour symbol in one step; in a binary machine, it copies 24 bits, requiring a sub-loop of depth 24.

7.6. A U.M.-native alternative: pixel-unit in continuum of frames

The construction of §§7.1–7.5 preserves Turing's 1-dimensional tape with a single head moving left and right. This is the right choice for foundational mathematical fidelity to the 1936 paper; it is not necessarily the right choice for the engineering register of U.M. 'C' (§1), where the natural data structure is the 3-dimensional cinematic stream of pixels in frames in films (§1.5). We sketch here the U.M.-native alternative, which retains the universality and the colour-native commitment but reorganises the head and tape.

The data structure. The tape is no longer a 1-dimensional sequence of cells but a continuum of frames indexed by film index $f \in \mathbb{N}$ and frame index $t \in \mathbb{N}$, with each frame an $n \times n$ array of pixels in Σ_{RGB}. The unit of access is the pixel-unit at coordinates (x, y) of frame (f, t), and the machine reads pixel-units in a programme-determined order rather than under the linear left-right discipline of §7.1.

The head. The head is a finite set of pixel-unit pointers $\{(f_i, t_i, x_i, y_i)\}_{i \in I}$ for $|I| \leq k_0$ (the head fan-out, a small constant), each pointer reading and writing one pixel-unit per step. Programmes are sequences of instructions that move the pointers, read or write at their current locations, and condition the next instruction on the read values. The 5-tuple $(q, \sigma, q', \sigma', m)$ of Turing's original machine generalises to a (k_0+2)-tuple of pointer moves, reads, and writes, with m replaced by a small parallel-move specification per pointer.

The programme structure. A programme runs by percolating an *arrow* of execution through the pixel-units it reads — an oriented path through the 3-dimensional pixel-frame-film space, with the same pixel-coordinates accessed across multiple frames and films according to the synchronisation discipline the programme imposes. Each frame can serve as the terminal resolution-set of a programming-code fragment (the frame is the output to which the fragment writes), with labelling (§§10, 11.5) attaching a finite specification to each frame so that synchronisation across n-frame-per-n-film windows is enforced by the labelling discipline. Frames and films may be intersected by the programme arrow without conflict, provided the synchronisation labels agree; the final label resolves an arbitrary multi-frame computation into one frame in sync, which is the cinematic output of the programme.

Universality and equivalence. The U.M.-native machine is universal in the same sense as the construction of §§7.1–7.5, by direct simulation: every left-right Turing machine over Σ_{RGB} is simulated by a U.M.-native machine that reads pixel-units along a one-dimensional path embedded in the pixel-frame-film space, and conversely every U.M.-native machine is simulated by a left-right machine that serialises pixel-unit access into the linear tape order. The two are interchangeable for the class of computable image and film schemes; the U.M.-native variant is the more compact and the more substrate-aligned in the cinematic register, while the left-right variant is the more direct mathematical descendant of Turing 1936 and the cleaner object for the diagonal argument of §8.

Dimensionality of the head's moves and the inseparability of processing and memory. A subtlety belongs here that does not arise in the left-right model of §§7.1–7.5. The U.M.-native machine reads each pixel-unit by synthesising the three channel values into one Σ_{RGB} symbol (and, conversely, deduces the three channel values from the synthesised colour by the programmable-architectural regression of §9.4); to access every pixel of every frame this synthesis discipline requires must permit head moves in *both* horizontal directions (left and right) and *both* vertical directions (up and down), and across the temporal axis to the previous and next frame. The 2-direction tape of 1936 is, in the U.M.-native setting, replaced by a 6-direction lattice motion — left/right, up/down, and across-frame in time, with the across-frame moves further specialised to within-film and across-film moves (§1.5). The state count of the U.M.-native machine accordingly grows over the canonical 44 of §§7.1–7.5 by the cost of bookkeeping these additional directions: a conservative implementation reaches the high fifties to low seventies in the same five-subroutine partition, with the exact figure depending on whether the across-film transition is treated as a primitive or composed from within-film moves. The unfolding is, however, more economical in another respect: by allowing the head to access any pixel-unit directly rather than by sequential traversal, the U.M.-native machine eliminates the linear-search cost that dominates the per-step time of the canonical construction.

A further point of mathematical importance: in the U.M.-native machine, processing and memory cannot be cleanly separated, nor can sensors be cleanly separated from emitters. The pixel-units are simultaneously the addresses of the head's reads (sensor reads) and the locations to which the head's writes are committed (emitter writes); they are simultaneously the working tape (processing) and the permanent record (memory, in the sense of §12.1). Only the union of these registers — processing-and-memory, sensors-and-emitters, all as facets of the same pixel-frame-film substrate — supports the universal machine in any physically meaningful sense. The mathematical formalisation of this inseparability is part of the technical content the U.M.-native model contributes over and above the Turing-style construction, and is one of the reasons the engineering register of U.M. 'C' is wider than the mathematical register that fits comfortably in Turing's left-right idiom.

We retain §§7.1–7.5 as the canonical construction for the present paper, and place the U.M.-native alternative on the table as the natural object for the engineering register of U.M. 'C', to be developed in subsequent work. The downstream sections of the present paper (§§8–12) refer to "the universal machine U" generically; the choice between the two variants does not affect the asymptotic content of any result below, only the constant factors and the spatial topology of the head.

8 The diagonal argument and the existence of uncomputable images

The cardinality observation of §5 implies that uncomputable functions $\mathbb{N}^3 \to \Sigma_{\mathrm{RGB}}$ exist in abundance. What cardinality alone does not provide is a constructive witness, nor any conclusion about which procedural questions about computable schemes are themselves decidable. Both are supplied by a single diagonal construction.

8.1. The diagonal scheme

Let $\{I_k\}_{k\in\mathbb{N}}$ be the enumeration of §5. Fix a non-zero colour $c_0 \in \Sigma_{\mathrm{RGB}}$—for definiteness, $c_0 = (1, 0, 0)$—and let $\oplus$ denote component-wise addition modulo 256.

Definition 8.1. *The diagonal scheme is the partial function* $D : \mathbb{N}^3 \to \Sigma_{RGB}$ *defined by*

$$D(x, y, n) \;=\; \begin{cases} I_x(x, y, n) \oplus c_0 & \text{if } M_x \text{ halts on input } (x, y, n), \\ \text{undefined} & \text{otherwise.} \end{cases}$$

Lemma 8.2. *The function* D *is not equal, as a partial function, to any total computable image scheme.*

Proof. Suppose $D = I_d$ for some total computable image scheme with description number d. Since I_d is total, M_d halts on (d, y, n) for every (y, n), so $D(d, y, n) = I_d(d, y, n) \oplus c_0$. But $D = I_d$ gives $D(d, y, n) = I_d(d, y, n)$. Combining, $I_d(d, y, n) = I_d(d, y, n) \oplus c_0$, forcing $c_0 = (0, 0, 0)$, contradicting our choice. □

8.2. The halting problem

The image-halting problem is the decision problem

$$\mathsf{Halt}_{\mathrm{img}} \;=\; \{\, (k, x, y, n) \in \mathbb{N}^4 \;:\; M_k \text{ halts on input } (x, y, n) \,\}.$$

Theorem 8.3 (Undecidability of $\mathsf{Halt}_{\mathrm{img}}$). *There is no Turing machine that decides* $\mathsf{Halt}_{\mathrm{img}}$.

Proof. Suppose H decides $\mathsf{Halt}_{\mathrm{img}}$. Define D^*: on input (x, y, n), compute $H(x, x, y, n)$; if yes, simulate M_x on (x, y, n) and output $I_x(x, y, n) \oplus c_0$; if no, output $(0, 0, 0)$. By construction D^* is total, so $D^* = I_{d^*}$ for some d^*. Now $I_{d^*}(d^*, y, n) = D^*(d^*, y, n)$. Since D^* halts on this input, $H(d^*, d^*, y, n)$ returns yes, so the output is $I_{d^*}(d^*, y, n) \oplus c_0$. Combining, $I_{d^*}(d^*, y, n) = I_{d^*}(d^*, y, n) \oplus c_0$, forcing $c_0 = 0$, a contradiction. □

Corollary 8.4 (Undecidability of rendering termination). *There is no algorithm that, given the description of an image-generating machine* M *and a coordinate triple* (x, y, n)*, decides whether* M *produces the pixel* $I(x, y, n)$ *in finite time.* □

Corollary 8.5 (Undecidability of total-rendering). *There is no algorithm that, given the description of* M *and a resolution* n*, decides whether* M *halts on every coordinate* (x, y, n) *with* $0 \le x, y < n$. □

Corollary 8.6 (Uncomputability of Kolmogorov complexity). *The function* K *defined in §5 is not computable.* □

The proofs of Corollaries 8.4–8.6 are routine reductions from $\mathsf{Halt}_{\mathrm{img}}$. For 8.6: if K were computable, one could decide whether a given input description p is the shortest program for a given scheme; this permits a halting decision via a Berry–Chaitin argument.

8.3. Definability versus computability

The diagonal scheme D is fully definable but not computable. The same gap exists at a structural level. The "true" Mandelbrot image I_M^∞ that assigns each pixel a colour by membership in the Mandelbrot set is mathematically determined but, depending on the conjectural status of

Mandelbrot hyperbolicity, either uncomputable or of unknown computational status (Hertling 2005; Braverman & Yampolsky 2006).

A more dramatic instance: define $\Omega : \mathbb{N}^3 \to \Sigma_{\mathrm{RGB}}$ by painting each pixel with a 24-bit colour drawn from the binary expansion of Chaitin's constant for U. The function Ω is rigorously defined but, by Chaitin's theorem, is not a computable image scheme. The substantive content of §8 is that the specific definable images naturally arising in the foundations of computation are uncomputable, and that decision problems about image-generating machines are themselves undecidable in general.

8.4. Diagonality as programming method

The diagonal construction of §8.1 is canonically read as a tool for separating the computable from the merely definable. The U-Mentalism programme, however, also reads it constructively: as a *programming method* for producing image- and film-content that is identifiable by the human eye, against the rest of the cinematic stream, as having computational rather than natural origin.

The construction. Given a finite collection of computable image schemes $\{I_{k_1}, \ldots, I_{k_m}\}$ — a U.M. *collection* in the algorithmic sense of §§1.5 and 5 — the diagonal scheme over the collection is

$$D^{\mathrm{coll}}(j, x, y, n) \;=\; I_{k_j}(x, y, n) \oplus c_j, \qquad j \in \{1, \ldots, m\},$$

where $c_j \in \Sigma_{\mathrm{RGB}}$ is a programmer-specified colour offset. The result is a film, indexed by j, whose j-th frame is the diagonally-modified content of the j-th scheme in the collection. The purpose of the offsets c_j is best understood through the role of the programmer as *oracle* (in the sense of §A.4 and the methodological footnote) operating over the population of cinematic supercomputation films. The diagonal product is, in this register, a *navigable test environment*: each frame in the collection shows the action of one scheme as a scenic landscape (a cockpit-instrument panel of an aviation simulator, a chessboard, a stacked-block puzzle in the Tetris register — whatever interpretive primitive the programmer's training makes immediately legible), and the diagonal modification across the collection is what the programmer, navigating through the test environment, can read as the systematic deviation that identifies the films as algorithmic syntheses from UTMs on the network rather than as naturally-occurring scenic content. The diagonal structure ensures that the modification varies systematically rather than uniformly across the collection, so that the resulting film is identifiable on inspection as a diagonal product even when each individual frame is locally indistinguishable from natural cinematic content. The choice of interpretive primitive (cockpit, board, stacked blocks) is governed by the programmer's purpose; the diagonal method itself is invariant under that choice and provides the constructive route to a navigable test environment by which the programmer-oracle exercises hyper-vigilance over the supercomputation stream.

Why this is useful. Two practical purposes are served. First, human *hyper-vigilance over computation* (§§10.5, A.4) requires that machine-produced cinematic content be identifiable as such by a sufficiently attentive observer; the diagonal pattern, by varying across the collection in a precise but visible way, makes this identification possible in principle without requiring metadata or external annotation. A concrete use case — not exhaustive of the application range — is the surveillance of the largest films being processed across the network, with the aim of preempting malicious activity that might extend, at scale, to attacks on the blockchain in operative use (§§F.3, F.7): diagonal-recognisable signatures within high-throughput film streams make pattern-deviating action detectable at the recollection layer before it propagates into consensus-layer compromise. Second, the diagonal method is the constructive face of the uncomputability result: every diagonal scheme over an infinite enumeration of computable schemes is itself uncomputable (§8.1), but every diagonal scheme over a finite collection is computable and produces a recognisable pattern — the finite case is the engineering instance, the infinite case is the theoretical limit.

The arrow reading. Read in the U.M.-native machine model of §7.6, the diagonal programme is an oriented arrow that percolates through the pixel-units of the indexed schemes' frames, picking out the (j, j)-coordinate across the collection and writing the offset c_j into the corresponding pixel-unit of the diagonal film. The arrow is the programme's trajectory through the pixel-frame-film space; the diagonal is its specific orientation when the collection is read in the index dimension simultaneously with the spatial dimension.

Beyond watermarks. The same construction admits more elaborate patterns through composition: a programme may apply the diagonal method to a sub-collection, then a second diagonal across multiple sub-collections, then composite the result with a non-diagonal scheme via the operators of §4. The richness of the resulting cinematic content is bounded only by the size of the underlying collection and the patience of the programmer. The diagonal method is, in this sense, the U-Mentalism reading of Cantor's transfinite-arithmetic argument (§5) reinterpreted as a constructive technique for visible computation; it carries the deep mathematical content of §§5–8 into the cinematic register the U.M. programme is built for.

9 Extent of computable images and films

Section 8 bounded the class of computable image schemes from outside. We now argue that the class is as large as any reasonable notion of procedurally producible image or film. This is the visual analogue of Turing's §9.

9.1. The visual Church–Turing thesis

Thesis 9.1 (Visual Church–Turing). *A function $I : \mathbb{N}^3 \to \Sigma_{RGB}$ is producible by some finite, deterministic, effective procedure if and only if it is a computable image scheme in the sense of Definition 2.1. The corresponding statement holds for film schemes.*

Thesis 9.1, like its arithmetic ancestor, is not a theorem but a claim about the adequacy of formal definition to informal concept. The evidence is extensive.

Equivalence of formalisms. The class of computable image schemes coincides with: total recursive functions $\mathbb{N}^3 \to \Sigma_{\mathrm{RGB}}$; closed terms in the typed lambda calculus normalising on every input; programs in any Turing-complete shading or rendering language under termination; and the standard register- or counter-machine equivalents (Hopcroft, Motwani & Ullman 2007). Their coincidence is the strongest evidence available that the class is robust rather than artefactual.

Closure under composition. Section 4 established that the class is closed under pointwise, local, geometric, compositing, and temporal operators with computable parameters.

Empirical adequacy. Every offline rendering pipeline, every codec, every compiled generative model, and every procedural texture or vector graphic specified by terminating programs is a computable image scheme. No image whose procedural generation has been published in the literature, to our knowledge, falls outside the class.

9.2. The continuous limit and computable analysis

The densification trajectory of §1.5—from 8-bit channels to 10 and 16, to multispectral, to continuous spectrum and intensity—has its mathematical home in computable analysis.

A computable image in the sense of Type-Two Effectivity (Pour-El & Richards 1989; Weihrauch 2000) is a function $\tilde{I} : [0,1]^2 \to [0,1]^k$ for which there exists a Turing machine that, given a Cauchy representation of (x, y) and precision $\varepsilon > 0$, returns an approximation to $\tilde{I}(x, y)$ within ε.

Type-256: the RGB specialisation. For the specifically RGB colour-native setting of §1.5, it is natural to define a refinement of TTE that we may call *Type-256 effectivity*: instead of Cauchy representations over $\mathbb{R}^k$, the representation alphabet is $\{0, 1, \ldots, 255\}$ per channel, and the precision parameter is the bit-depth $b \geq 8$ of the representation (with $b = 8$ being the contemporary engineering point and the densification $b \to \infty$ recovering the full real-valued TTE in the limit). Type-256 effectivity coincides asymptotically with TTE under the alphabet equivalence of Proposition 1, but is the natural register in which the U-Mentalism programme's bridging-phase claim (§1.5) becomes a precise statement about the discrete representation depth at which computable visual content is being processed at any given engineering stage.

Type-N and the electromagnetic spectrum. The generalisation to arbitrary discretisation depth N, with alphabet $\{0, \ldots, N-1\}^k$ per pixel and k spectral bands, is mathematically immediate: Proposition 1 lifts directly, so the class of Type-N computable image schemes coincides with the class of Type-256 (and Type-Two) computable image schemes for every $N \geq 2$. The asymptotic content is invariant; what changes with N is the precision at which a given electromagnetic-spectrum interval is sampled. We may speak of Type-N effectivity *at spectral interval* $[\lambda_1, \lambda_2]$ nanometres for the discretisation that samples the interval at N levels of intensity per spectral band, with the number of bands k determined by the resolution of the chosen spectral discretisation. The mathematical viability of Type-N effectivity at every $[\lambda_1, \lambda_2]$ in the electromagnetic spectrum — visible band ($\sim$ 380–750 nm, the RGB territory), near-infrared, far-infrared, ultraviolet, X-ray, and beyond — is guaranteed by the alphabet equivalence; the engineering viability is bounded by sensor and modulation technology specific to each interval. For the visible band, Type-256 at $k = 3$ is the current state of the art; for finer visible-band quantisations and for non-visible-band imaging (multispectral, hyperspectral, thermographic), Type-N at larger N and k is the natural mathematical framework, and the corresponding engineering target is the photonic substrate of §§9.4 and 12.4 extended to the relevant spectral interval. The discrete-to-continuous transition (Type-N as $N \to \infty$, $k \to \infty$ across the EEM) is the analog limit of §§1.5 and 9.4, where U.M. 'C' approaches U.M. 'O' face to face.

The differential-calculus regime. A natural question, addressed only briefly in the body and developed further in §12.9, concerns the value of N and k at which Type-N effectivity over the EEM admits classical differential calculus and infinitesimals in the operational sense. The answer follows from the standard Type-Two Effectivity machinery (Weihrauch 2000): differentiability of Type-Two computable functions over $[0,1]^k$ requires the Cauchy-representation depth to be unbounded, which is the regime $N \to \infty$. Numerically, an engineering-equivalent regime that supports classical differential operations on practical workloads is reached at $N \sim 2^{32}$ (32-bit floating-point per spectral band, comfortably above the precision required for second-derivative computations on natural-scene imagery) and $k \sim 10^3$ (one thousand spectral bands sampling 380–1380 nm at $\sim$ 1 nm resolution), which is the contemporary hyperspectral imaging regime (Specim, Headwall, and Aviris-class sensors). At this (N, k), the discrete machinery of §§1.5–7 admits differential and infinitesimal-calculus computations to engineering precision, with the transition to the full TTE limit $N, k \to \infty$ being only a precision refinement, not a categorical change in computational class. The U-Mentalism programme reaches the differential calculus, accordingly, at engineering-instantiated rather than mathematically-idealised parameter values, as is the U.M. 'C' commitment throughout.

Theorem 9.2 (Effective continuity). *Every TTE-computable image is continuous on its domain. Conversely, any continuous image whose modulus of continuity is itself a computable function admits a TTE-computable representative if and only if its values on a dense computable subset of* $[0,1]^2$ *are uniformly computable.*

The first half is classical (Pour-El & Richards 1989, Theorem A.5).

Every TTE-computable image $\tilde{I}$ induces, for each resolution n, a computable image scheme $I_{\tilde{I}}(x, y, n) = \text{quantise}_8(\tilde{I}(x/n, y/n))$ by sampling and 8-bit quantisation. A uniformly-approximating family of computable image schemes converges to a unique TTE-computable continuous image. The discrete and continuous notions are therefore successive refinements of a single object, not rival models.

9.3. Quantum computation

Does quantum computation expand the class of computable image schemes, or alter the boundary established by the diagonal argument?

Theorem 9.3 (Bernstein–Vazirani 1997). *The class of functions* $\mathbb{N}^* \to \mathbb{N}^*$ *computable by quantum Turing machines coincides with the class computable by classical Turing machines.* □

Specialised to image schemes, Theorem 9.3 asserts that a function $I : \mathbb{N}^3 \to \Sigma_{\text{RGB}}$ is computable by a quantum image-generating machine if and only if it is a computable image scheme in our sense. Quantum computation does not lift the boundary.

What it changes is efficiency. The complexity classes are stratified differently in the quantum setting: BPP $\subseteq$ BQP $\subseteq$ PSPACE, with the relation between BQP and the polynomial hierarchy still open in general. Within the class of computable schemes, several image-generation tasks fall on the side of strict containment between BPP and BQP.

Hamiltonian-simulation rendering. Light-transport equations whose discretisations are equivalent to Hamiltonian simulation problems admit polynomial-time quantum algorithms where the best known classical algorithms are exponential.

Quantum Monte Carlo. Path tracing admits quadratic speedup via amplitude estimation (Brassard et al. 2002): for rendering tasks where Monte Carlo error dominates cost, quantum acceleration potentially reduces wall-clock rendering by a square-root factor.

Generative model evaluation. Inference in certain probabilistic graphical models admits exponential quantum speedups; whether these translate to deployed image-generation pipelines is conditional on hardware that does not yet exist at production scale, but that can be envisaged under the photonic-quantum substrate conditions of §§9.4 and 12.4 — broadband colour-domain fibre transmission, single-photon-fidelity sources, photovoltaic primary power, and integrated linear-optical networks scaled to $N \sim 10^4$ modes with logical-qubit-level error correction.

The Hermann lineage and the uncomputability of quantum correlations. A complementary boundary, structurally distinct from Theorem 9.3, situates the present picture against a deeper result on the algebraic substrate of quantum mechanics. Grete Hermann (1926) established effective degree bounds for the polynomial-ideal-membership problem in her thesis *Die Frage der endlich vielen Schritte in der Theorie der Polynomideale*; her bounds, doubly-exponential in the input size, were shown by Mayr & Meyer (1982) to be essentially tight, the problem being EXPSPACE-complete. The same paper is conventionally regarded as the founding document of computational commutative algebra. A decade later, Hermann (1935) identified the silent algebraic assumption underlying von Neumann's (1932) no-hidden-variables proof — the supposition that expectation values are linear across the entire algebra of observables, including non-commuting ones, which is precisely the property the proof was supposed to establish. The same gap was rediscovered independently by Bell (1966), without knowledge of Hermann's earlier work; Mermin & Schack (2018) defend the Hermann–Bell reading against more recent challenges. Hermann's identification of the gap was, in substance, an algebraic observation: she saw what physicists had not, because her Noetherian training in commutative algebra disclosed the polynomial-linear assumption hidden in the proof. The Bell (1964) inequality and the Kochen & Specker (1967) theorem operationalise the algebraic content into testable inequalities and into a combinatorial obstruction on value-assignments to projector lattices; Abramsky & Brandenburger (2011) reformulate the latter as a sheaf-theoretic obstruction, the failure of a global section of the value-assignment bundle over the space of measurement contexts.

Theorem 9.3.1 (Ji, Natarajan, Vidick, Wright & Yuen 2020). $\mathrm{MIP}^* = \mathrm{RE}$. *The Halting Problem reduces efficiently to deciding whether a two-player non-local quantum game has entangled value* 1 *or at most* 1/2. *Tsirelson's bound for general quantum correlations is consequently not computable in the worst case, and the closure C_{qa} of tensor-product quantum correlations is strictly contained in the set C_{qc} of quantum-commuting correlations — a negative resolution of Tsirelson's problem and, equivalently, of the Connes (1976) embedding conjecture.* □

The Navascués–Pironio–Acín (2008) hierarchy of polynomial sum-of-squares relaxations — a direct algebraic descendant of Hermann (1926), in that each level is the computation of a Gröbner-basis-type semidefinite relaxation — is not, by virtue of Theorem 9.3.1, an algorithm: it approximates the quantum set from outside but does not terminate in general. Hermann's algebraic intuition is in this sense vindicated at the deepest available level: the question of "hidden variables" is, at root, a question about the polynomial structure of the observable algebra, and that structure is, in the worst case, undecidable.

Two observations are warranted for the present paper. First, Theorem 9.3.1 does not contradict Theorem 9.3: quantum machines compute the same class of functions $\mathbb{N}^* \to \mathbb{N}^*$ as classical machines (Bernstein–Vazirani 1997), and the visual Church–Turing thesis (Thesis 9.1) is unaffected. What MIP* = RE concerns is the space of quantum-realisable correlations, not the class of computable functions; the visual content delivered by any photonic substrate remains, at the level of function-computability, Turing-bounded. Second, the result refines the engineering picture of §§9.4 and 12. A photonic universal machine computing within Theorem 9.4's bound is Turing-bounded as a generator of *functions*; the same substrate, when its photonic resources include entanglement (KLM-style schemes; Knill, Laflamme & Milburn 2001), is not Turing-bounded as a generator of *correlations*. This is a fine asymmetry, ontologically pertinent to U.M. 'O' and consistent with the wider claim that the engineering implementation of U.M. 'C' realises a finite shadow of a physical substrate whose full descriptive complexity exceeds what any finite Turing machine can bound. The Hermann lineage names the precise sense in which hidden variables, if they exist, are mathematical (specifically algebraic) objects whose representation problem is undecidable in its general form — the deepest extension to date of Hermann's 1935 insight that the question is, at root, about the polynomial structure of the observable algebra rather than about any physical mechanism.

Position relative to the post-quantum cryptography programme. The non-commutative cryptographic register of U.M. 'C' shares its mathematical horizon with the post-quantum cryptography programme established under NIST's standardisation process (lattice-based schemes in the line of Regev 2009; code-based schemes; multivariate-quadratic and isogeny-based schemes), without coinciding with it. Post-quantum cryptography seeks schemes whose hard-problem reductions are not weakened by Shor's algorithm against integer factorisation or discrete logarithm; the U.M. recollection layer of Appendix F, built on cryptographic hash functions (SHA-256, BLAKE3) rather than on number-theoretic trapdoors, is hash-based and therefore already structurally post-quantum-resistant in the sense that matters operationally for persistence: Grover's quadratic speedup against pre-image search merely doubles the required output length to preserve a given security parameter, while Shor-style polynomial-time attacks have no known application to standard cryptographic hash functions. The position of the U-Mentalism programme is consequently that the recollection layer is post-quantum-resilient by construction, with the choice of hash family the only remaining parameter; the question of whether higher-layer cryptographic primitives in the wider U.M. 'C' deployment (signature schemes for proof-of-recollection, key-exchange protocols for distributed nodes) should adopt post-quantum primitives or remain on present-day cryptographic standards is an engineering decision deferred to the planned engineering research of the programme (§12.10).

Hidden variables as an open algebraic horizon. The reading of MIP* = RE through the Hermann lineage, in which hidden variables become an algebraic representation problem with undecidable structure in the worst case, leaves a register of inquiry open rather than closed. The classical no-hidden-variables results (von Neumann 1932 as corrected by Hermann 1935; Bell 1964; Kochen–Specker 1967) rule out specific classes of hidden-variable theories on operational and combinatorial grounds; MIP* = RE situates the algebraic core of the question in a register where undecidability is the result and any particular hidden-variable proposal would have to specify the algebraic content it claims to represent before its decidability could be assessed. The

U-Mentalism programme accordingly records hidden variables as an open algebraic question, not as a settled negative result — a position consistent with the conjectural register of §§11.5 and 12.7, in which substantive ontological questions are entertained alongside the formal results without claim of resolution and with their representation problems flagged at the algebraic-undecidability boundary that the Hermann lineage identifies.

Non-commutative algebra at the operational level: cryptographic diagonalisation of frames. The non-commutativity that gives Theorem 9.3.1 its ontological substance enters U.M. 'C' by an independent and more direct route — cryptography — and this entry alters the regime of operation of both recollection and collection (in the sense of §§1.5 and 10.3) before any quantum implementation is contemplated. Consider a frame $F \in \Sigma_{\mathrm{RGB}}^{n\times n}$, with raw information content $24n^2$ bits. Two visually similar frames (the same scene, differing perhaps in a single pixel) share nearly all of these bits — their Hamming distance is small. Under a cryptographic transformation $\mathcal{H}$ satisfying collision-resistance and the avalanche property (any standard hash; SHA-256, BLAKE3), the Hamming distance between $\mathcal{H}(F)$ and $\mathcal{H}(F')$ for two such close frames becomes approximately half the output length: maximally distant, regardless of input proximity. This is diagonalisation in the operationally relevant sense — each frame is rendered mutually as far as possible from every other in output space, decoupled from perceptual proximity in input space. Non-commutativity is structural: for any two transformations T_1, T_2 on frames, $\mathcal{H}(T_1 \circ T_2(F)) \neq \mathcal{H}(T_2 \circ T_1(F))$ in the maximal-distance sense, with the order of operations recorded in the cryptographic output. The structure $(\Sigma_{\mathrm{RGB}}^{n\times n}, \circ_{\mathrm{crypt}})$ is by construction non-commutative.

Vigilance over cryptographic operations at the recollection layer. The non-commutative structure has a vigilance-register that warrants explicit notation, parallel to the hyper-vigilance over films of §A.4 and the consensus-security observations of §F.7. Because the cryptographic register is structurally non-commutative in the manner of asymmetric cryptographic schemes, indexation of computational flows by markers extracted from the cryptographic outputs is itself a tool of surveillance: pattern-deviating attempts to invert factorisation, to enumerate pre-images, or otherwise to attack the cryptographic primitives in operation at any given moment within the U.M. system become detectable at the recollection layer through the diagonal-signature methodology, before the attack propagates to the consensus or key-management layers. A complementary precaution concerns the recollection itself when it operates the near-fractal cascade of Proposition 9.3.4: the cascade reduces work at the recollection layer (clauses (i)–(iii)) but, by the same gesture, makes the arithmetic regularities of the frames themselves — the per-channel value distributions, the inter-pixel statistics, the recollection's own ordinal placement of frames in the catalogue — exposed to inferential analysis that could, at the worst case, offer up to half the resolution potential a cryptanalytic attacker would otherwise require. The U.M. system as a whole must therefore construct, at the collection layer, a schema of *new abstracts* as support-points for *synthesis* — abstracts that are not the raw frames and not the bare hashes but a collection-side construction designed to evolve and maintain the system without supplying the arithmetic regularities to which the recollection layer would otherwise leave it exposed. The schema is reserved for development in the engineering documentation of §§12.9–12.10; the present subsection registers the precaution as a structural feature of the operating discipline that any U.M. deployment must adopt.

Proposition 9.3.2 (Cryptographic diagonalisation). *Let $\mathcal{H}$ be a cryptographic transformation on frames over Σ_{RGB} satisfying collision-resistance and the avalanche property. Then: (i) the ordinal-insertion operation on the recollection layer becomes deterministic in $\mathcal{H}$, requiring no coordination between distributed nodes; (ii) the duplicate-detection operation on the collection layer reduces from $O(n^2)$ in pixel content to $O(\log n_{\text{out}})$ in output bits; (iii) if $\mathcal{H}$ is an encryption rather than a hash, frame storage and transmission preserve confidentiality of visual content while retaining indexability.* □

The proposition operationalises, at the level of frames, the algebraic structure that Hermann saw at the level of QM observables. The gain accrues across the recollection / collection distinction of §1.5 and is inherited by the proof-of-recollection of §10.3 and the PoCI construction of Appendix F: identity (collision-free), integrity (tampering-detectable), and conditionally confidentiality, are the three components composed orthogonally by the cryptographic layer.

A second proposition follows naturally, recording the Monte Carlo content of the same cryptographic structure.

Proposition 9.3.3 (Monte Carlo via cryptographic diagonalisation). *Let $\mathcal{H}$ be a cryptographic transformation as in Proposition 9.3.2, and let $\{f_k\}_{k\in[N]}$ be a finite family of computable image schemes with seeds $\{s_k\}_{k\in[N]}$ drawn from a pseudo-random generator on a public seed σ. The cryptographically diagonalised family $\{\mathcal{H}(F_k(s_k))\}_{k\in[N]}$ realises a Monte Carlo sample over the space of computable frames satisfying:*

(i) computational independence between distinct family members up to the cryptographic security parameter of $\mathcal{H}$;

(ii) for any Lipschitz function $\phi : \Sigma_{RGB}^{n\times n} \to \mathbb{R}$, the Monte Carlo estimate $\bar{\phi}_N = \frac{1}{N}\sum_k \phi(\mathcal{H}(F_k(s_k)))$ converges to its cryptographic-measure expectation at rate $O(N^{-1/2})$;

(iii) the family is reconstructible from the public seed σ alone, requiring $O(|\sigma|)$ persistent storage rather than $O(N \cdot n^2)$ raw-frame storage.

□

Clause (iii) is the operational signature: an entire Monte Carlo experiment over the space of computable frames is reconstructible from a seed of $O(\log N)$ bits. The economy is structural: deterministic and verifiable Monte Carlo over computable images is achievable without requiring true randomness (which is incomputable for finite machines, by §5 and Chaitin 1969) and without requiring per-sample storage. This is in line with the recollection register of §§10.3 and 10.6, where persistent ordinal-indexing of frames is the foundational operation: a Monte Carlo experiment over N samples occupies precisely one seed-slot in the recollection rather than N frame-slots, with the samples regenerable on demand by any UTM with the seed and the family specification.

A third proposition records, in formalised terms, two related structural observations on the recollection layer: the dual-form ordinal unit of recollection-entry, and the cascade of iterated cryptographic transformations on already-cryptographically-indexed frames.

Proposition 9.3.4 (Ordinal pair and cryptographic cascade). *Let $\mathcal{H}$ be a cryptographic transformation on frames as in Proposition 9.3.2. Then:*

(i) The native ordinal unit of the U.M. recollection is the pair $\langle F, \mathcal{H}(F)\rangle$ rather than the bare frame F, with the second component constituting the cryptographic index of the first and admitting the operations of identity-checking, integrity-verification, and (under encryption) confidentiality-preservation without access to the first.

(ii) Iterated application of $\mathcal{H}$ defines the cascade $\mathcal{H}^k(F) = \mathcal{H}(\mathcal{H}^{k-1}(F))$ with $\mathcal{H}^0(F) = F$. The cascade has no non-trivial fixed points: for any F and any $k \geq 1$, $\mathcal{H}^k(F) \neq F$ except on a measure-zero set of inputs (the cryptographic collision set, of size $2^{-n_{\text{out}}}$ relative to the space of frames).

(iii) Diagonalisation of an already-cryptographically-indexed family $\{\mathcal{H}(F_k)\}_k$ by a fresh cryptographic transformation $\mathcal{H}'$ yields a doubly-diagonalised family $\{\mathcal{H}'(\mathcal{H}(F_k))\}_k$ whose pairwise-distance properties compose multiplicatively in the avalanche-distance metric, recovering the patent's "eternal recurrence" loop (§§F.3, 12.4) as a recursion of the diagonalisation operation on its own output.

□

The Mandelbrot parallel. The cascade of Proposition 9.3.4(ii) admits a structural parallel with the iterated-quadratic dynamics that generates the Mandelbrot set — a parallel of interest because both iterations operate natively on objects of the universal computable-image class. Three points of contact warrant explicit registration. *First*, both are deterministic iterated function systems on a structured space (the complex plane $\mathbb{C}$ for the Mandelbrot iteration $z_{n+1} = z_n^2 + c$ with $z_0 = 0$; the frame space $\Sigma_{\text{RGB}}^{n\times n}$ for $F_{k+1} = \mathcal{H}(F_k)$ with $F_0 = F$), and both produce image-content output of substantive structural complexity from a simple recursive rule. *Second*, both have non-trivial computability boundaries at the level of their limit set. The Mandelbrot set M has the computability profile established by Hertling (2005) and refined for the Julia variant by Braverman & Yampolsky (2006); the cascade-limit $\{\mathcal{H}^k(F) : k \to \infty\}$ has, in U.M., a structurally analogous undecidability profile registered by Proposition 10.7 (Kolmogorov complexity of recollection), with the cryptographic one-wayness contributing the same kind of computational opacity to the cascade limit that the analytic-iteration boundary contributes to the Mandelbrot boundary. *Third*, both produce computable images natively — the Mandelbrot set is rendered as a computable image scheme of the kind formalised in §2 (the worked example tradition includes Mandelbrot's own renderings) and the cascade operates on $\Sigma_{\text{RGB}}^{n\times n}$ frames as its substrate — so the parallel is not analogical but substantive: the cryptographic cascade and the Mandelbrot iteration are two operations of the same kind (iterated maps generating images) acting on different image-classes. They diverge on the invertibility property — Mandelbrot iteration is analytically tractable in both directions, while the cryptographic cascade is computationally one-way by construction — and the divergence is the resource the recollection layer relies on: the protection of the U.M. catalogue against reversal-style adversaries is the cryptographic distance from the Mandelbrot regime, not an addition over and above iteration as such.

The proposition operationalises, at the recollection layer, the structural intuition of the patent (Homem 2021): the universal machine of §§6–7 with the recollection of §1.5 is not a memory followed by a processor but a structure in which storage and indexed-cryptographic-transformation are inseparable operations, with the cascade of (ii)–(iii) the formal expression of the "loop" or "eternal recurrence" that the patent's STEP-d to STEP-g architecture inscribes.

The transition to the quantum-in-colour regime relocates non-commutativity from cryptographic artefact to physical mechanism. When U.M. 'C' is implemented as photonic-quantum computation over RGB-encoded qubits (KLM-style schemes; Knill, Laflamme & Milburn 2001), the gate set acts unitarily on the substrate Hilbert space, and the Pauli group on n qubits is structurally non-commutative. The encoding is the minimal binary one: $\log_2 256 = 8$ qubits per channel, three channels per pixel, hence 24 qubits per pixel as the U.M. photonic-quantum register at the pixel level. (Alternatives are operationally distinguishable but representationally equivalent: a unary or one-hot encoding requires 768 qubits per pixel at structural alignment with the Σ_{RGB} alphabet but at higher qubit cost; a qudit register of dimension 256 per channel, embedded in 8 qubits, is informationally equivalent to the binary encoding under unitary basis change.) The substrate's expressive computational power derives *from* this non-commutativity, just as the ontological content of Theorem 9.3.1 derives from it. But the operational use is distinct: U.M. 'C' deploys the non-commutative algebra as a *computational resource*, deterministically actioned by gate sequences specified by the human Chiron, and not as a source of fundamental randomness. Under projective measurement in a fixed pre-specified basis, the output is a function of the circuit and the input; randomness enters only if the specification leaves measurement-basis freedom unfixed. The classical cryptographic regime of Proposition 9.3.2 is in this sense the discrete antecedent of the quantum-in-colour regime: the non-commutativity is operationally usable, structurally indispensable, and deterministically harnessed under specification — a register without ontological commitment, in the engineering sense made explicit in §A.4 and §12.

Concrete instances of U-Mentalism quantum reading. The structural alignment of U.M. with the quantum register admits more specific operational instances than the algebraic discussion above has surfaced, and a brief restricted enumeration is in order. The instances are presented as openings for subsequent work of the programme rather than as developed results of the present paper. *First, entanglement as cross-frame correlation tensor.* In the standard quantum-tomographic register, an entangled bipartite state is reconstructed element-by-element from many separate measurements on copies; in the U.M. register, the full correlation tensor of the entangled state is encodable as a single frame in $\Sigma_{\mathrm{RGB}}^{n\times n}$, with the (i,j) cell holding the correlation amplitude between local outcomes i and j under a fixed measurement-basis convention. The frame is the entanglement read; the persistent ordinal indexing of the recollection (§§1.5, 10.3) renders the entanglement persistently retrievable rather than per-experiment ephemeral. The structural reorientation is from reconstruction-by-counting to direct frame-reading. *Second, light-path-coherence regression.* A complex entangled state typically arises as the downstream product of a sequence of coherent propagation steps through interferometric elements, beam splitters, and phase shifters — each such step a substrate-level reconfiguration whose state is, in principle, a candidate for frame-capture. U.M. stores the complete trajectory of those light-steps as a film, each frame the substrate state at one step. Where standard decoherence treatment proceeds by reverse unitary evolution (bounded by the coherence preservation of the substrate) or by quantum error correction toward a code state (correcting to a stabiliser-defined target, not to an arbitrary earlier state), the U.M. register opens a structurally distinct alternative: *recollection traversal* of the trajectory, with regression toward an earlier coherent stage of the entanglement read directly from the ordinally indexed catalogue, accompanied by re-organisation of the light-steps to recover the coherence of the particle path before the downstream complexity accumulated. The proposal is not that this circumvents decoherence as a physical phenomenon — it does not — but that the persistent recollection of the trajectory makes earlier coherent stages *computationally available* in a register in which standard quantum computation does not place them. *Third, MIP* = RE in the U.M. register.* The Ji–Natarajan–Vidick–Wright–Yuen (2020) identification of the quantum correlation space with the recursively enumerable sets maps directly, in U.M., to the identification of that correlation space with the recollection itself: each correlation a frame ordinally indexed, the full correlation space the full ordinal catalogue. Theorem 9.3.1 becomes, in this reading, less a paradox of comprehension and more a structural statement about the kind of object the recollection is, with the incompleteness of Proposition 10.7 explaining (from inside the substrate) why no finite truncation of the catalogue can be claimed to contain all of it. *Fourth, continuous quantum monitoring as native frame register.* Where the standard formalism treats continuous-measurement quantum trajectories (Wiseman & Milburn 2010) as derived from the more fundamental unitary-plus-projective framework of textbook quantum mechanics, the U.M. register inverts the order: each frame is a snapshot of the conditional state under continuous photonic monitoring, the film is the trajectory itself, and the gate-and-projective decomposition is the algorithmic generator (collection) of the trajectory rather than its substrate. The native register of U.M. for quantum computation is, in this reading, the continuous-monitoring register; discrete-gate quantum computation is a sub-collection within it. *Fifth, quantum programming as film authoring.* The shift forces a rethink of quantum programming as a discipline. Standard quantum programming (Qiskit, Cirq, and successors) composes unitary gates sequentially with terminal projective measurements; in the U.M. register, a quantum program is the film whose frames are the successive substrate states, the gate sequence is the algorithmic generator (collection), and the resulting state-trajectory is the ordinally indexed film (recollection). The primitive operation set shifts from gate-as-primitive to frame-as-primitive, with the Assembly-Language register seeded at §12.9 supplying the cost-analysed primitive set that the conventional gate-level abstraction would otherwise hide — frame-emission cycle, interferometric-reconfiguration cycle, ordinal-insertion cycle, monitoring-snapshot cycle, each with a precisely defined photonic-substrate cost in the manner of MMIX cycles for classical computation. A structural register peculiar to the film-authoring view deserves naming: the *ordinal deferral.* Where standard quantum programming requires terminal projective measurements that collapse the substrate state to a definite outcome before subsequent operations can proceed, U.M. quantum programming admits the alternative of postponing the collapse and continuing the unitary evolution into wider synthesis structures of greater complexity — with the ordinal-indexed collection layer guaranteeing that the eventual readout can be unambiguously mapped back to the original frame-positions from which it derived. A multipixel sub-frame in superposition can be left uncollapsed under read-head movement to other positions or under sensor-side gating, with the substrate evolving the entangled state further in the meantime; when collapse is eventually invoked, the U.M. collection register supplies the index that re-attaches the collapsed colour-synthesis to the pixel-positions it answers for. The non-locality content of the deferral is structurally the resource: programming becomes, in this register, the practice of working over collapsed results that need not be immediately collapsed, with each deferral admitting further deferral as long as the substrate's coherence resources permit. The open question is whether the deferral has a fixed point in the synthesis-optimality sense — whether postponing collapse yields, in the limit, an optimal result against which any earlier collapse would be a degradation — or whether the optimum is sequence-dependent and a collapse decision must be made under the constraint that real time advances over the deferral as it advances. The naming is offered here as an opening for subsequent work; the formal characterisation belongs to the planned engineering research of §§12.9–12.10 and to the Assembly-Language of §12.9. The five instances are restricted in scope and openly contestable; they are placed here as openings into subsequent work rather than as resolved positions of the present paper.

9.4. Photonic and analog substrates

The answer separates into three cases.

Digital photonics—optical computation with signals modulated to a finite, well-separated set of states—is, at the level of computability, classical Turing machine computation in a different physical embodiment.

Idealised analog computation—real-valued operations carried out without noise and with infinite precision—does in principle compute beyond Turing. The Blum–Shub–Smale model (Blum, Shub & Smale 1989) is the canonical formalisation; the Mandelbrot set is decidable in the BSS sense.

Realistic analog computation—real-valued operations with non-zero noise floor and finite precision—does not exceed Turing.

Theorem 9.4 (Brattka 2000, paraphrased). *Any analog computation operating under a non-zero noise floor and finite precision is bounded, in computational power, by Type-Two Effective computation.* □

Every physical photonic system is subject to thermal noise, quantum shot noise, and finite-precision modulation; by Theorem 9.4, the class of images such a system can produce is contained in the TTE-computable class. Photonic computation is the natural physical substrate for the model of §§1.5–7—it operates on colour-valued symbols natively, propagates at the speed of light, dissipates a small fraction of CMOS energy per operation—but does not transcend the boundary established by Theorem 8.3.

This is the desirable outcome for the U-Mentalism programme, and at the furthest extent — at the threshold where discrete RGB colour-native computation grades into the analog continuous-spectrum limit (§§1.5, 9.2) — it is the formal point at which U.M. 'C' meets U.M. 'O' face to face: the engineering substrate exhausts its discrete granularity precisely where the ontological substrate of all photonic viewpoints opens. A photonic universal machine escaping Church–Turing computability would inherit Brattka-style precision dependencies that compromise determinism; one that exactly realises the colour-native model is precisely what the abstract construction is for.

Interference as the operational basis. The physical substrate of photonic computation—both classical and quantum—is interference. A photon, even at the single-particle level, exhibits wave-like behaviour: sent through two paths and recombined, its detection probability at the output depends on the relative phase of the two paths. The double-slit experiment (Young 1801; Taylor 1909 for the single-photon version; successive refinements through the twentieth century) is the canonical demonstration; contemporary integrated photonic chips realise the same physics in waveguide form, with controlled phase shifters playing the role of the slit geometry. A Mach–Zehnder interferometer—two beam splitters separated by a phase shifter—is a directly programmable element: phase 0 routes the photon to one output port, phase π routes it to the other, intermediate values produce coherent superpositions.

Arrays of such interferometers form universal linear optical networks. Reck et al. (1994) and Clements et al. (2016) established that any unitary transformation on N modes can be implemented by a triangular or rectangular array of beam splitters and phase shifters—a finite, programmable optical circuit whose operation is dictated by N^2 independent phase settings. These networks are the operational atoms of contemporary integrated photonic computing.

For quantum computation, two consequences are decisive. First, the KLM scheme (Knill, Laflamme & Milburn 2001) established that universal quantum computation is achievable using only linear optical elements, single-photon sources, and photodetectors—without nonlinear photon–photon interaction. Universal quantum computation is, in this sense, accessible to a substrate that is essentially a generalisation of the double-slit experiment. Second, boson sampling (Aaronson & Arkhipov 2011) demonstrated that the output distribution of N photons through such a network is computationally hard to sample classically—a quantum advantage realised as a many-slit interferometric task.

A further consequence, of speculative weight but conceptually pertinent, follows from Dirac's matter–light equivalence: the photons that carry Σ_{RGB} symbols are exchangeable, in the high-energy regime, with matter–antimatter pair-production processes, so that the universal machine of §§6–7 grades, at sufficient energies, into a substrate where the operation of computation and the production of matter are not categorically distinct phenomena. The Universal Image-Generating Machine in this register approaches, as a horizon, a universal *matter-generating* machine of which the cinematic supercomputation of the patent (Homem 2021) is the low-energy realisation. This horizon is conjectural and outside the technical content of the present paper; we record it because the substrate equivalence is one of the U-Mentalism programme's deepest commitments and the foundational paper would be incomplete without naming it.

For the colour-native universal machine of §§6–7, this convergence is more than coincidental. The photons that carry the colour symbols of Σ_{RGB} *are* the interfering entities whose superposition states realise quantum computation. Wavelength and phase together encode both symbol and operation. The classical photonic implementation operates by direct interference patterns; the quantum extension operates by the same patterns evaluated under measurement-induced or coherence-induced non-classicality. The substrate is unified at the most fundamental level: the same photons that carry visual content also carry the operations that transform it. This unification is the deeper sense in which the colour-native commitment is substrate-aligned (§12.4): not only that the alphabet matches the modulation precision of photonic hardware, but that the operational basis of photonic computation—interference—is the same physical phenomenon that carries quantum computational advantage.

Reading the colour-synthesis: scaling, channels, and causal pinning. The transition from binary (two states per cell) to RGB colour-native (256^3 unique colour values per pixel) requires, on a photonic substrate, that each colour channel — R, G, B — be modulated and read at 8-bit depth (or its analog equivalent under finer densifications, in the sense of §1.5 and §9.2). The three channel readings are *local* acts of measurement at the channel level; the colour-image *synthesis* they constitute, by contrast, is already *non-local* in the quantum-photonic register, since the joint photonic state that the three channels coherently determine is not localised to any single channel but distributed across the entanglement structure of the pixel as a whole. The three channel readings combine, at the sensor or detector layer, to produce the single colour value that the pixel-as-cell convention of §1.5 designates: the unique synthesis of the three channels into one Σ_{RGB} symbol. In Kant's sense, this is the *schema* of the three channels apprehended as one (Homem 2019); the cell at the level of the pixel sits above the channel level by this *synthesis*.

A consequence of practical relevance: the synthesised colour value, once read, allows a *programmable-architectural regression* to the constituent channel values — given the synthesised Σ_{RGB} symbol and the known synthesis function, the original (r, g, b) triple can be recovered by lookup or by inverse computation, since the synthesis is bijective on the 256^3 codomain. This permits implementations in which only the synthesised value is read at the sensor layer (a single measurement per pixel rather than three), with the channel constituents reconstructed in subsequent processing if and where the application requires them. The architecture is, by this, both more efficient at the sensor stage (one measurement) and more faithful to the non-local synthesis structure (no premature collapse of the channel-wise constituents).

The retrograde reading: from synthesis to light-signature. The architecture above admits a symmetric inverse-direction reading that is operationally distinct from the standard channel-wise sensor pipeline and conceptually pertinent to the U.M. register. Where the standard reading proceeds *channels* $\to$ *synthesis* (three channel measurements combined into one Σ_{RGB} symbol), an alternative reading proceeds $\langle$*synthesis*$\rangle \to \langle$*light-signature*$\rangle$: a single beam-like or cone-like photonic apparatus reads the colour-synthesis at the pixel as a unitary measurement, and then decomposes the synthesis retrogradely into the precise per-channel spectral signature — the unique triple $\langle r, g, b\rangle$ that, by the bijectivity of the synthesis on Σ_{RGB}, could have produced that synthesis. The notation $\langle\cdot\rangle$ marks the retrograde direction: $\langle r, g, b\rangle$ is read as "the channel signature that derives from the synthesis", not as "the channel values that produce the synthesis". The two directions agree on the same triple at every pixel by the bijection, and the U.M. register is indifferent to which direction the photonic apparatus implements, but the operational and informational economies differ. In Kant's terminology of the *Critique of Pure Reason*, the synthesis ($A77/B103$ sense — the act of putting together what is manifold) and the *schema* (the mediating structure between concept and intuition, $A137$ following) admit reciprocal directions of access: the synthesis-first reading corresponds to apprehending the schema as already-constituted and decomposing it into its conditions; the channel-first reading corresponds to constituting the schema from its conditions. Both are coherent in U.M. 'O', and the engineering choice between them is a property of the photonic substrate of §12.10 rather than of the ontological register.

A note on the continuum. The discrete–continuous trajectory of §§1.5 and 9.2 ($N \to \infty$, $k \to \infty$ across the EEM) approaches but never reaches a full continuous-spectrum limit, by the standard foundational-mathematics objection: the continuum is the unresolved background that Cantor's hierarchy (§5) and Gödel's relative consistency proof for the continuum hypothesis (1940) jointly identify as resistant to complete discretisation. The U-Mentalism programme's approach to the analog limit is therefore asymptotic rather than terminal: discrete computation at finer and finer granularities approaches the analog limit from below, with the continuum proper remaining outside any finite implementation. The mathematical horizon (Cohen 1963 forcing, the independence of CH) and the engineering horizon (Brattka's bound, §9.4 above) coincide in marking the same threshold from different directions.

The differential-calculus regime. The asymptotic approach to the continuum has a precise computational reading in the differential-calculus regime that the limit instantiates. As resolution N and channel-depth k grow, the discrete-step partial-computable-function regime of §§1.5–7 acquires the structure of a differential calculus on the alphabet space: pixel-to-pixel differences become arbitrarily small, the channel-arithmetic operations approximate the operations of analysis on a continuous chromatic spectrum, and the universal machine of §§6–7 acquires the operational shape of a finite-difference scheme for an underlying continuous dynamics. The Type-Two effectivity register of Pour-El & Richards (1989) and Weihrauch (2000) gives the precise correspondence: a TTE-computable continuous image is, at every finite resolution, the limit of a computable image scheme in the discrete sense, and the partial-computable-function structure of the discrete schemes ($\Sigma_{\mathrm{RGB}}^{n\times n}$ at finite N) acquires, in the limit, the partial-continuous-function structure of TTE. The limit register is not new computational power — by Theorem 9.4 (Brattka's bound) it remains within the Turing-bounded class — but a register in which the engineering instantiation of partial-computable functions admits operations of integration and differentiation natively at the substrate level (the photonic-PCM cells of §12.10, with sub-symbol modulation, are the candidate hardware). The continuum-problem horizon (Cohen 1963; §§5, 11) and the boundary of computational complexity (Cook 1971; §§10.1, A.4) coincide on the same asymptote: the continuum is unreachable in finite computation, and the computational-complexity classes preserved through the discrete–continuous limit transfer asymptotically rather than terminally.

In a quantum-photonic register, the wave-function collapse associated with the reading sequence is not a defect to be avoided but a constitutive and beneficial feature of how a definite output is obtained: the collapse pins the synthesis on the temporal and causal line, fixing the colour value for the subsequent operations to use. Computation in this register proceeds by alternating coherent evolution and measurement-induced collapse along a definite cinematic sequence, with each pixel's value made definite when the synthesis is read. The benefit is twofold. First, the collapse provides the causal pinning that any deterministic computational sequence requires: it converts the coherent superposition of preceding evolution into the definite value the next stage needs. Second, the synthesis structure — channels read, single colour output — coincides with the natural measurement structure of photonic detectors, so that the alignment between

U.M.'s pixel-as-cell convention and the quantum-measurement architecture is structural rather than incidental. The photons that carry the colour symbols are the quantum entities whose wave-function collapse, registered channel-by-channel and then synthesised as the pixel output, gives the deterministic cinematic sequence on which the universal machine of §§6–7 operates. The temporal benefit is particularly relevant for quantum programming: the causal sequence of collapses defines the order in which information becomes definite, which is precisely the resource a programme of cinematic supercomputation requires for its sequential operations to be well-defined.

9.5. The boundary, restated

The class of computable image schemes is robust under variations of substrate (binary, colour-native, photonic), of model (classical, quantum), and of representation (discrete with arbitrary bit-depth, continuous via TTE). It is closed under the compositional operators of §4 and exhausts the class of images producible by any explicit procedure consistent with physical realisation. Beyond it lie functions that are well-defined—the diagonal scheme, the continuous Mandelbrot indicator under hyperbolicity assumptions, Ω, and the comeagre majority that admit no finite description—but inaccessible to any procedure consistent with current physics.

10 Large classes of computable image and film schemes

Section 9 established robustness of the boundary; the present section establishes richness of the interior.

10.1. Procedural and parametric image schemes

The simplest large family consists of image schemes whose pixel values are an arithmetic formula in their coordinates. Examples 3.1–3.3 are early instances.

Proposition 10.1 (Arithmetic image schemes). *Any image scheme of the form $I(x, y, n) = \mathrm{quantise}_8(P(x/n, y/n))$, where $P : \mathbb{Q}^2 \to \mathbb{Q}^3$ is a rational function with rational coefficients, is computable in time $O(\log^c n)$ per pixel for some constant c depending on the degree of P.* □

The class extends, at the same polylogarithmic per-pixel cost, to image schemes given by closed-form expressions in elementary functions. Algebraic surfaces, parametric curves, conic sections, polynomial-field gradients—the entire vocabulary of pre-rendering classical graphics—falls under Proposition 10.1.

The class of iterated schemes—fractals, escape-time pictures, IFS-attractor renderings—extends Proposition 10.1 by introducing a depth parameter $K(n)$. The Mandelbrot scheme of Example 3.5 is the canonical instance: per-pixel cost $O(K(n) \log^c n)$ for some c, total cost $O(n^2 K(n) \log^c n)$ per frame at resolution $n \times n$.

10.2. Rendering pipelines

A rendering pipeline is a finite composition of: a scene description (geometry, materials, lights, camera), a transport model (rasterisation, ray-tracing, path-tracing), and an output stage (anti-aliasing, tone-mapping, quantisation). The canonical treatment is Foley, van Dam, Feiner & Hughes (2013); for the specifically physically-based pipelines that dominate offline rendering, the underlying formal framework is Kajiya's (1986) rendering equation.

Proposition 10.2 (Computability of bounded rendering). *Let $\mathcal{R} = \langle \mathcal{S}, \mathcal{T}, d, s \rangle$ be a renderer parameterised by a scene $\mathcal{S}$ with computable geometry and BRDFs, a transport model $\mathcal{T}$ implementable by a terminating algorithm, recursion-depth bound $d : \mathbb{N} \to \mathbb{N}$, and sample count $s : \mathbb{N} \to \mathbb{N}$, both computable. Then the image scheme $I_{\mathcal{R}}$ produced by $\mathcal{R}$ is a computable image scheme, with per-pixel cost $O(s(n) \cdot d(n) \cdot c(\mathcal{S}, \mathcal{T}))$ where $c(\mathcal{S}, \mathcal{T})$ is the per-sample cost of one transport evaluation.* □

The proposition encompasses essentially every renderer in production use. Path-tracing with $s(n) = n$ samples per pixel and depth $d(n) = O(\log n)$ is, on the analysis above, $\tilde{O}(n^4)$ per frame at resolution $n \times n$.

The film analogue is immediate: a film scheme produced by frame-by-frame application of a terminating renderer is a computable film scheme.

10.3. Neural and generative image schemes

A trained neural network with finite architecture, fixed weights, and computable activation functions computes, on input coordinates, a deterministic output triple in Σ_{RGB}. The function it computes is a computable image scheme.

Proposition 10.3 (Neural image schemes). *Let $\mathcal{N}$ be a neural network with L layers, maximum width W, computable activation functions, and computable rational weights. The function $I_{\mathcal{N}}$ defined by feedforward evaluation of $\mathcal{N}$ on input $(x/n, y/n)$ followed by 8-bit quantisation is a computable image scheme, with per-pixel cost $O(LW^2 \log^c n)$.* □

The proposition covers implicit neural representations (NeRF, Mildenhall et al. 2020; SIREN, Sitzmann et al. 2020; Instant-NGP, Müller et al. 2022), compiled generative models (denoising diffusion, Ho, Jain & Abbeel 2020; autoregressive transformer image generators, Vaswani et al. 2017; GAN generators, Goodfellow et al. 2014) when run with deterministic seeds, learned codecs, learned super-resolution, and learned video synthesis.

A subtlety. The training of a network is itself a finite computation: a deterministic optimisation procedure on a fixed dataset for a fixed number of iterations. The trained weights are therefore computable functions of the training data, and the produced image scheme is computable as a function of those data—though in practice we treat the weights as given parameters and the scheme as a function of inference-time inputs alone.

The implication is that every image producible by any deployed AI image generator, in any deterministic mode, is a computable image scheme in the strict sense of Definition 2.1.

The network's far horizon. A note on the prospective register: with the UTM-as-server / TM-as-node architecture of §12.4 and the visual-content pipeline of §§10.3 and 11.5, the Internet may come to function as more than a nervous system of node-to-node contacts — something nearer a mature neurology of the technological substrate, with the visual content streams of §10 carrying the analogues of perceptual qualia and the universal kernels of §§6–7 the analogues of higher-order processing. The comparison is, at the time of writing, properly cautious: a brain requires a body and the embodied sensation that follows, and the Internet has neither in any direct sense. Brain–machine interface research (in the line opened by, e.g., Lebedev & Nicolelis

2017) is, however, the route by which the analogy may eventually become more than analogy — with the heavy bibliography on that question (cortical mapping, neural prosthetics, sensory substitution) bearing on the U-Mentalism programme's hypercomputation horizon in ways that the present paper notes without endorsing. Blockchain technology, observed in passing, is itself an instantiation of a universal Turing machine of a distinctive kind: a distributed UTM whose tape is the consensus ledger and whose programme is the network protocol; its integration with the economy is, by the same token, the embedding of universal computation in social coordination — a fact relevant to the cinematic supercomputation programme's eventual social and institutional realisation, recorded here briefly and critically (the universal-machine character of consensus chains does not by itself confer any of the properties commonly attributed to them in promotional registers: decentralisation in any operational sense beyond node-distribution, trustlessness across the full stack rather than only at the consensus layer, immutability robust to majority compromise, censorship-resistance independent of mining-power concentration, scalability without trade-off against finality, or economic-sovereignty independent of the protocol's governance dependencies — each of these is an empirical and architectural question about a particular chain and a particular deployment rather than a logical consequence of universal-machine character, and the present paper records its critical position on the gap between the two registers in keeping with the wider U-Mentalism programme's commitment to careful distinction between formal capabilities and the social claims).

From proof-of-work to proof-of-recollection. A natural articulation follows from the cautionary note just made. In its canonical form (Nakamoto 2008), proof-of-work expends energy as a token of seriousness: rewriting the chain's history must be made prohibitively costly relative to honest mining. The energy is, strictly speaking, a means of coordination rather than an end — yet the work it secures produces nothing recoverable. Hashes are generated by the billion per second and almost all are immediately discarded; only the winning hash of each block survives. In thermodynamic terms this is entropy export in its purest form: heat dissipated in exchange for a single coordinating signature. The U-Mentalism distinction between *recollection* and *collection* (§1.5) suggests a structural redirection. If the work of mining is to be expended in any case, and if recollection is precisely the persistent ordinal-arithmetic indexing of frames and films in transfinite series — the substance of UTMs-as-Internet-servers (§12.4) — then proof-of-work may be reformulated as *proof-of-recollection*: the costly operation becomes the canonicalisation and ordinal insertion of a frame into the persistent record, while verification reduces to a positional lookup. The three classical requirements for useful proof-of-work — asymmetric verifiability, unpredictability of result, adjustable difficulty — are all satisfied by recollection in its U-Mentalism sense, with difficulty calibrated by the degree of canonicalisation demanded. The connection to Kolmogorov complexity (§5) is here more than analogical: canonical recollection *is* the production of a shortest description in the colour-native alphabet, indexed at its ordinal position in the transfinite sequence. What proof-of-recollection achieves over proof-of-work, beyond the obvious ethical recovery of expended labour, is thermodynamically structural. Landauer's bound (§12.1) applies to logical irreversibility; recollection, being non-erasing by definition, is in principle compatible with reversible computation (Bennett 1973). The coordination function survives — decentralisation remains valuable for censorship resistance, distributed redundancy, and incentive alignment, none of which depends on energy scarcity — but the substrate of confidence shifts from discarded hashes to indexed frames and films, persistent and percolation-ready for any subsequent program-arrow.

10.4. Cost stratification

Class	Classical	Quantum (BQP)	Colour-native photonic
Arithmetic / algebraic	$\tilde{O}(n^2)$	$\tilde{O}(n^2)$	$\tilde{O}(n^2/24)$
Iterated / fractal	$\tilde{O}(n^2K(n))$	$\tilde{O}(n^2K(n))$	$\tilde{O}(n^2K(n)/24)$
Ray / path-traced (s samples)	$\tilde{O}(n^2sd)$	$\tilde{O}(n^2\sqrt{s}\,d)$	$\tilde{O}(n^2sd/24)$
Neural inference (LW^2)	$\tilde{O}(n^2LW^2)$	partially $\tilde{O}(\cdot)$	$\tilde{O}(n^2LW^2)$

The *quantum advantage* in column three is most pronounced for path-traced rendering, where Monte Carlo amplitude estimation gives the documented quadratic speedup in s (Brassard et al. 2002). The neural-inference entry is qualified: quantum-accelerated neural inference is, at this writing, an active research area without consensus.

The *colour-native photonic advantage* in column four is, asymptotically, a constant factor—the 24× speedup expected from native colour symbol manipulation versus binary emulation. In the photonic case the constant savings compound with massive spatial parallelism, so the wall-clock improvement is far larger than the asymptotic factor suggests.

The *combined* setting—quantum-photonic computation with photonic implementation of quantum gates over colour-encoded qubits—is at this writing a research frontier.

The scaling staircase and effective computability. The Big-O hierarchy that organises the table above — $\mathcal{O}(1)$, $\mathcal{O}(\log n)$, $\mathcal{O}(n)$, $\mathcal{O}(n^2)$, $\mathcal{O}(n^3)$, and onward through **P**, **NP**, **NP**-complete, **PSPACE**, **EXP**, and beyond — recovers, at the level of practical computability, the question of *effective computability* that animates Turing's 1936 paper. Effective computability for Turing was the conjectural claim, recoverable by his §9, that what is computable in the intuitive sense coincides with what some Turing machine produces in finite time; the modern complexity-theoretic refinement asks the same question one notch finer, namely which sub-class of computable functions is producible in polynomial time, exponential time, polynomial space, and so on. For the visual setting of the present paper, the hierarchy organises which image classes are tractable on which substrates, and the colour-native commitment of §1.5 turns out to act as a constant-factor amplifier across all rungs — 24× at the symbol level, with parallelism multipliers at the implementation level (§§9.4, 12.4). The Big-O hierarchy is, in this register, the operative form in which the effective-computability question of 1936 returns in 2026, conjectural in its largest claims (the visual *Entscheidungsproblem* of §11.5, the BQP-vs-PH question of §9.3) and confident in its asymptotic statements.

10.5. The practical exhaustion of the class

The class of computable image schemes contains every procedural image, every rendered scene, every neural-generated image, every codec output, every photograph stored as a digital file, every frame of every digital film, and every composition of these. It contains, in short, the entirety of the visual content currently producible or storable in any digital form. The class is not merely "large"; it is, for any practical purpose, exhausted by the contents of contemporary visual culture, though that culture is itself short of the real bedrock of *ontological imagery* that U.M. 'O' contemplates — the constitution of the image as percipient/percept across all photonic viewpoints, with the percipient/percept layer itself supervening upon the prior and more foundational observable/observed couple of §11.5 (the percipient/percept structure does not stand alone but is constituted on top of the observable/observed register, as the U.M. 'O' definition of the image makes explicit) and of which the digital visible is only the finite-bandwidth shadow. Reportable to these conclusions are study points from media-archaeological and critical-cultural directions by Kittler (2010) on optical media as a single computational trajectory and by Virilio (2012) on the speed–image complex — the latter explicitly *cautionary* in tone, and welcomed in U-Mentalism precisely for that critical edge. The cited authors do not themselves argue the present conclusions, and are not to be read as doing so; their work is recorded here as proximate reading rather than as parallel argument, with the convergence in tone between independent registers noted for what it indicates.

A qualifying remark deserves to be made for accuracy. "Visual culture" as a name for the artefactual stratum the present class exhausts is, in the strict U-Mentalism reading, the wrong name — it suggests the popular, the *volk*, in places the kitsch, and so misnames the substrate it picks out. What is at stake in the present paper is not the cultural artefacts of visibility but the ontological substrate of a knowledge-oriented culture, in the old-Hellenic register of *φύσις* — the root of thought before which the Renaissance, the *Aufklärung*, and Liberalism each remained only partially adequate, having contributed (each in its register) to delaying the agony of the fall of the Greek-cradle culture rather than to its (impossibly otherwise partial) restoration. The visual culture exhausted by the present technical class is, for the purposes of the wider programme, the local post-digital foundation of the theoretical possibility of a far older substrate. The ultimate origin of any such reading is, as the programme records throughout, Greek: *ἀρχή* as the constitutive beginning, *φύσις* as the unfolding of nature into itself, *λόγος* as the ordering structure that makes the unfolding intelligible.

This is a stronger claim than Turing could have made about the computable reals in 1936. The computable reals are dense in $\mathbb{R}$, but the practical reals—those actually computed by anyone—form a much smaller subclass. The practical images, by contrast, nearly are the computable images, modulo idealised limits and the specific diagonal constructions of §8.

10.6. Recollection-relativised complexity

The Big-O hierarchy of §10.4 organises problems by the computational resources expended *during* a single computation. The U-Mentalism distinction between recollection and collection (§1.5), refined by the proof-of-recollection reformulation of §10.3, introduces a third axis: the *history of past computations available as quasi-constant-time lookup*, persistent across the network and ordinally indexed. The standard literature contains partial precedents — non-uniform classes with polynomial advice (**P**/poly, **BPP**/poly), oracle relativisations, amortised complexity — but the U-Mentalism recollection is structurally distinct in two respects: it is *transfinite-ordinal* rather than per-input-length, and it is *shared persistent memory* rather than a fixed advice string.

The complexity landscape re-orders accordingly along the following five lines.

Amortised collapse over recurring sub-structure. Many decision problems in **NP** derive their hardness from search over instance-level structure that, across a population of instances, repeats. Canonical examples include constraint satisfaction over visited domains, graph isomorphism between canonisable graphs, and SAT over recurrent clausal patterns. With sub-structure indexed in R, the per-instance amortised cost approaches that of recollection lookup — $O(\log \alpha_R)$ in the canonical indexing, with α_R the supremum of currently realised ordinals in R — for the fraction of the input space already covered. **NP** does not collapse; its *amortised projection over indexed sub-structure* does. The formal statement requires care about the input distribution, the indexing scheme, and the bookkeeping for novel sub-structure; we record the phenomenon and defer the formalisation.

Recalibration of pseudo-randomness assumptions. Cryptographic pseudo-random generators rely on one-way functions: easy to compute, hard to invert. A recollection that stores the inverses of a one-way function at sampled domain points functions as a finite oracle for inversion at those points. Rainbow tables are the artisanal version of this construction; an ordinally-indexed persistent recollection is its theoretical limit. The security parameters of PRG-based cryptography do not vanish but recalibrate against the moving frontier of R — in exact parallel with how they currently recalibrate against the moving frontier of computational power.

Sub-recursive ordinal ladder. The fast-growing hierarchies (Hardy, Wainer, the standard fast-growing family) index total computable functions by ordinals up to ε_0, Γ_0, the Bachmann–Howard ordinal, and beyond. Each level corresponds to a sub-recursive class: primitive recursive at ω^ω, multiple-recursive at ε_0, and so on. A recollection storing sampled values of these functions becomes a *sub-recursive oracle*: problems indexed by ordinals $\alpha < \alpha_R$ become accessible in time logarithmic in the ordinal. The U-Mentalism "rise in knowledge" of the recollection is, formally, an ascent of the ordinal ladder, not an accumulation of bits at a fixed level. The phenomenon is the computational analogue, in persistent-memory form, of what set theory recognises as ordinal proof-theoretic strength.

Probing the boundary of computability. The Busy Beaver function $\mathrm{BB}(n)$ (Radó 1962) is total but uncomputable. The fifth value, $\mathrm{BB}(5) = 47\,176\,870$, was established by collaborative formal verification in 2024 (Busy Beaver Challenge Collaboration 2024): exactly 47 176 870 steps is the maximum number of transitions that a halting Turing machine with five states and the binary tape alphabet can take before halting. The figure is a stored fact about a function whose recomputation from first principles is provably out of reach: any algorithm purporting to compute $\mathrm{BB}(n)$ for arbitrary n would solve the halting problem for n-state machines, which is undecidable. The 2024 result, the first new BB value since $\mathrm{BB}(4) = 107$ was established in 1983, is informative for the present register in three ways. First, it materialises the recollection-as-finite-oracle picture concretely: a U.M. recollection containing the table $\{\mathrm{BB}(n) : n = 1, 2, 3, 4, 5\}$ is a finite oracle for the halting problem on all binary-alphabet Turing machines with at most five states, and the recollection-relativised decidable class Dec_R of §10.6 is, by Conjecture 10.6, accordingly enlarged. Second, it illustrates the trans-machine character of the result: the proof is not the work of a single device but of a distributed verification effort across many participants and machines coordinated through a single public catalogue — precisely the configuration the U.M.

recollection layer (§§10.3 and Appendix F) formalises. Third, the gap between BB(4) = 107 and BB(5) = 47 176 870 illustrates the rate at which the Busy Beaver function exceeds any computable function: the ratio is approximately 4.4×10^5, and the next value BB(6) is known to exceed $10^{10^{10^{10^{10^{10^{15}}}}}}$ (the lower bound established by Pavel Kropitz and the Busy Beaver Challenge community), beyond any presently representable cardinal of physical interest. A recollection containing the Busy Beaver table up to n_R is, in this register, a *finite oracle for the halting problem* on machines with $\leq n_R$ states. The recollection's growth in this direction does not change the absolute boundary established in §8; it pushes the *operationally accessible* boundary outward, one ordinal step at a time. The natural ladder is the hyperarithmetical hierarchy ($\mathbf{0}'$, $\mathbf{0}''$, …, indexed by recursive ordinals; Rogers 1967), with each Turing jump materialised in persistent memory as the recollection grows. Operative decidability, in this register, is not a single property of a problem but a function of which Turing jumps the network has so far rendered persistent.

Cinematic re-encoding of the hierarchy. The colour-native amplifier of §1.5 supplies a uniform 24× constant factor across the Big-O hierarchy (§10.4). The replacement of the one-dimensional Turing tape by the three-dimensional film of §§1.5 and 8.4 supplies a structural amplifier: problems with natural three-dimensional or temporal geometry — physical simulation, path planning over time, dynamical systems, transport problems — re-organise within the hierarchy according to their alignment with the cinematic substrate. The cinematic analogues $\mathbf{P}^{\text{cine}}$ and $\mathbf{NP}^{\text{cine}}$ are, conjecturally, non-trivial *redistributions* of $\mathbf{P}$ and $\mathbf{NP}$ rather than reductions of either: the same asymptotic classes, with internal positions of specific problems shifted by substrate fit.

The relativised picture. Taking these together, the operationally meaningful complexity question is not $\mathbf{P}$ versus $\mathbf{NP}$ in absolute terms but $\mathbf{P}_R$ versus $\mathbf{NP}_R$ for the current recollection R. Define, accordingly: (i) $\mathbf{P}_R$ as the class of problems decidable in deterministic polynomial time with oracle access to R; (ii) $K_R(x) = K(x \mid R)$ as the conditional Kolmogorov complexity of x given R (Li & Vitányi 2019); (iii) $I(R : x) = K(x) - K_R(x)$ as the algorithmic mutual information between the cumulative recollection and a new instance. As R grows along the ordinal ladder, the class of operationally decidable problems Dec_R approaches the hyperarithmetical class. The approach is, however, *constructible* in the sense of Gödel (1940): the recollection is built up step by step, indexed by recursive ordinals, hence its closure cannot exceed the constructible hierarchy L_{α_R}. Cohen (1963) established that L does not exhaust set theory; for the present purpose, this means precisely that the recollection does not exhaust mathematics — it exhausts only the part available to computation, which is exactly what one expects from a universal machine in the Turing 1936 sense. The coincidence of the constructible and the computable, recurring here as the limit of the recollection-relativised hierarchy, is one of the deepest connections that the U-Mentalism programme places in operative form.

Conjecture 10.6 (Recollection-relativised decidability). *The recollection-relativised decidable class satisfies* $\mathrm{Dec}_R \subseteq L_{\alpha_R}$*, with the inclusion approaching equality asymptotically as* α_R *ranges over the recursive ordinals. The visual Entscheidungsproblem of §11.5 is, accordingly, undecidable in absolute terms but recollection-relativisedly decidable at sufficiently high ordinal — the practical question being not whether decidability is achievable but at what ordinal coverage of* R *a given problem first becomes operationally decidable.*

The threshold reading. The phrase "first becomes operationally decidable" deserves a clarifying gloss, since the natural alternative reading — "first becomes operationally undecidable" as one traverses a decidability boundary in the opposite direction — inverts the structural content. The intended direction is from absolute undecidability *toward* operational decidability: every problem in the recollection-relativised hierarchy has, by Conjecture 10.6, an ordinal α_P such that for $\alpha_R \geq \alpha_P$ the problem becomes operationally decidable in the recollection register, and for $\alpha_R < \alpha_P$ it remains operationally undecidable. The Turing-jump ladder is the ladder *toward* decidability one rung at a time, not away from it; the absolute undecidability of §8 remains in place at every finite stage of α_R for problems whose α_P is itself non-recursive, and is dissolved only asymptotically through the recursive-ordinal indexing of the hyperarithmetical hierarchy. Operative decidability accordingly admits a natural partial order: problem P *precedes* problem P' in the operational sense when $\alpha_P < \alpha_{P'}$, with the absolute-undecidable problems being precisely those for which no recursive α_P exists.

This is, in the U-Mentalism register, the operative form of the effective-computability question of §10.4 for the era of universal cinematic supercomputation: a question not about the absolute reach of computation, fixed by §8, but about the rate at which the network's persistent recollection materialises successive Turing jumps and successive levels of the constructible hierarchy as quasi-constant-time lookup.

10.7. The Kolmogorov complexity of recollection and the U-Mentalism analogue of Chaitin's incomputability

The complexity-theoretic discussions of §§5 and 10.4 may now be folded back into the persistent-memory register established by the proof-of-recollection of §10.3 and the recollection-relativised complexity of §10.6. The U-Mentalism recollection / collection distinction admits, within Kolmogorov complexity, the following quantitative observations.

The complexity of complete recollection. The complete recollection of all computable image-numbers at resolution n is the ordinal enumeration of $|\Sigma_{\mathrm{RGB}}^{n\times n}| = 256^{3n^2}$ image-strings, starting from $0000\ldots0$ in $24n^2$ positions and ascending. The Kolmogorov complexity of this enumeration, as a finite catalogue, is bounded above by $24n^2 \cdot 2^{24n^2}$ bits in the trivial encoding and below by $\Omega(24n^2 \cdot 2^{24n^2})$ bits in any encoding by the standard counting argument: the catalogue contains $|\Sigma_{\mathrm{RGB}}^{n\times n}|$ entries that any encoding must distinguish. The complete transfinite recollection across all resolutions $n \in \mathbb{N}$ is an object of ordinal ω^ω in the constructible hierarchy of §10.6, with no finite Kolmogorov bound. Recollection in the universal sense is a transfinite object; its complexity is the complexity of its ordinal scaffolding, not of any of its finite truncations.

The complexity of collection. The collection on the same scheme is, by Definition (§1.5 as corrected in §5), the sub-set of the recollection given algorithmically — generated by computable image schemes $\{f_k\}_{k\in I}$ rather than enumerated by ordinal position. The Kolmogorov complexity of a collection is bounded by $K(C) \leq \sum_{k\in I} K(f_k)$, in general much smaller than the size of C would suggest. A collection of 10^9 procedural fractal images, each generated by a scheme of $|f_k| \sim 10^3$ bits, has $K(C) \sim 10^{12}$ bits — twelve orders of magnitude below the $\sim 10^{16}$ bits of raw image content at megapixel resolution. The compression is the complexity-theoretic content of the collection's status as a procedural sub-set of the recollection.

The U-Mentalism analogue of Chaitin's Ω. Chaitin (1969) defined the halting probability $\Omega = \sum_{p:p \text{ halts}} 2^{-|p|}$ and proved that Ω is incomputable: its finite binary expansions Ω_n are stored facts whose recomputation from first principles is provably out of reach. The U-Mentalism analogue is the *recollection-rate* of a universal machine U:

$$\rho(U; N) \;=\; \frac{\bigl|\{\, i \in [0, N] : i \text{ is canonically recollectable by } U \,\}\bigr|}{N}.$$

Here a number i is *canonically recollectable* if U can, within a bounded time function of $|i|$, determine the ordinal position of the corresponding image-string in the catalogue.

Proposition 10.7 (U-Mentalism analogue of Chaitin's incomputability). *The limit recollection rate* $\rho(U) = \lim_{N\to\infty} \rho(U; N)$ *is not computable. Equivalently: a U-Mentalism universal machine cannot compute its own asymptotic horizon of recollection.* □

The proof is a standard reduction from the halting problem. Given a Turing machine M with input w, construct an image-string $i_{M,w}$ such that $i_{M,w}$ is canonically recollectable by U if and only if M halts on input w. The construction encodes M's tape contents into the leading pixels of the image and requires U to reach a terminal state on the encoding to canonicalise the index. If the recollection rate were computable, the halting predicate would be computable on the class of TMs encodable in this form, which is the full class by standard universal-machine arguments. This would contradict the unsolvability of the halting problem (Turing 1936; §8.3 of the present paper). □

Frames and films as internal states. The relation of frames and films to the Kantian register of the *Critique of Pure Reason* is more pointed than a direct comparison would allow. Kant assigned space as the *a priori* form of outer intuition and time as the *a priori* form of inner intuition, and both were forms of sensibility of an anthropologically constituted subject — a subject endowed with body and reason, with sensibility and understanding. The U-Mentalism register operates at a registry that has none of those: a universal machine without body or reason, without sensibility or understanding, and accordingly without anthropology at all, processes frames (the spatial register) and films (the temporal register) as states internal to itself. The radical difference between sensibility and machine-state is exactly what a direct frame-as-space/film-as-time comparison with Kant would elide and must not. The relation with the Kantian setup is, accordingly, more limited and more pointed: U-Mentalism does not fall into the Kantian error of externalising space as the form of outer intuition, precisely because it registers *both* frames and films as internal states of the same machine. There is no outer form here; there is only the dual internal register. The spirit of the move is closer in this respect to the Humean-Berkeleyan observation that space and time are both internal to perception — though the perceptual register of Hume and Berkeley is itself absent from the U-Mentalism setup, and the comparison in either direction is limited, by construction, to this structural point and not to the perceptual anthropology that the comparison would otherwise import.

A briefer parenthesis at the root of the modern revolution. The same sceptical inflection that the U-Mentalism programme inherits is announced earlier in the Cartesian *cogito ergo sum* but stopped short of its own implication. That the anthropological experience (*cogitatio*) of the cogito is itself a totality is not under dispute — just as diagonality is itself a totality without thereby being the radication of principles to a single measure. What is at stake is the further step that Descartes took, of positing the cogito as the unequivocal, unquestionable, unique, and singular radication of cognition itself, and it is precisely this further step that the proper measure of the inaugurated scepticism would have continued to inquire into rather than terminate at. The U-Mentalism reading of the self parallels its reading of mathematics in this respect — the analytic-synthetic potency of the cogito-as-totality is not in question, the foundationalist self-image of cogito-as-sole-radication is — with the arguments belonging properly to the companion ontological treatise. Proposition 10.7 supplies the technical underpinning: even from inside the machine, the asymptotic horizon of operative recollection is incomputable, so the "completed external object" — the catalogue of all images as a totality — is, formally, never available. The U-Mentalism programme inherits this open-ended character of the recollection as a structural feature rather than as an obstacle, in coherence with the visual *Entscheidungsproblem* hypothesis of §11.5 and the recollection-relativised picture of §10.6.

11 Application to the Entscheidungsproblem

> *"The results of §8 ... can be used to show that the Hilbert Entscheidungsproblem can have no solution."*
>
> —A. M. Turing, 1936, §11

We arrive at the application that gave Turing's paper its title and that gives ours its conclusion.

11.1. The decision problem

Hilbert and Ackermann (1928) posed the *Entscheidungsproblem* as the central open question for the foundations of mathematics: is there an effective procedure which, given a first-order formula ϕ, decides whether ϕ is logically valid? By Gödel's completeness theorem (Gödel 1930), this is equivalent to deciding provability in first-order logic. The result obtained in this section is that no such algorithm exists; the proof reduces the image-halting problem to the *Entscheidungsproblem*.

11.2. Encoding image-generating machines in first-order logic

Fix a Turing machine M_k over Σ_{RGB} with description number k and an input $(x_0, y_0, n_0) \in \mathbb{N}^3$. We construct a first-order formula ϕ_{k,x_0,y_0,n_0} in a language $\mathcal{L}$ containing: a constant $\mathbf{0}$, a successor function S; for each tape symbol $\sigma \in \Gamma_{M_k}$, a constant $\bar{\sigma}$; for each state $q \in Q_{M_k}$, a unary predicate $\mathrm{State}_q(t)$; a binary predicate $\mathrm{Sym}(t, p)$; a unary function $\mathrm{Pos}(t)$.

Initial-configuration axiom (Init): the machine begins in state q_0 at time $\mathbf{0}$, with head at position 0 and the input written on the tape in canonical encoding.

Transition axioms (Trans): for each quintuple $(q_i, \sigma, q_j, \sigma', m) \in \delta_{M_k}$,

$$\begin{aligned}\forall t\, \big[&\mathrm{State}_{q_i}(t) \wedge \mathrm{Sym}(t, \mathrm{Pos}(t)) = \bar{\sigma} \\ &\qquad \rightarrow \mathrm{State}_{q_j}(S(t)) \wedge \mathrm{Sym}(S(t), \mathrm{Pos}(t)) = \bar{\sigma'} \wedge \mathrm{Pos}(S(t)) = \mathrm{Pos}(t) + m \,\big]\end{aligned}$$

together with frame axioms asserting that all other tape positions are unchanged at $S(t)$.

Halting clause (Halt):

$$\exists t \bigvee_{q \in F_{M_k}} \mathrm{State}_q(t).$$

The full formula is the implication

$$\phi_{k,x_0,y_0,n_0} \;=\; (\mathsf{Init} \wedge \mathsf{Trans}) \;\rightarrow\; \mathsf{Halt}.$$

This is the form whose validity corresponds to the halting of M_k, as verified in Appendix E.

Dependence on the machine model. The encoding above is tied to the left-right tape model of §§7.1–7.5: the function $\mathrm{Pos}(t)$ is a single integer position, and the transition axiom moves the head by $m \in \{-1, 0, +1\}$ along a 1-dimensional axis. Under the U.M.-native machine model of §7.6, the encoding adjusts in a routine but non-trivial way: $\mathrm{Pos}(t)$ becomes a 4-tuple $(f(t), \tau(t), x(t), y(t))$ specifying the film, frame, and pixel coordinates of the head's current pixel-unit; the transition axiom updates each component according to the programme's parallel-move specification; the head fan-out k_0 of §7.6 contributes a small additional axiom-block bounding the number of simultaneous pointer moves per step. The asymptotic content of the reduction (Theorem 11.1) is unchanged by this substitution, and so are all the conclusions of §§11.3–11.5; what changes is the size of the encoded formula and the implicit constants of the reduction. We retain the left-right encoding above as the canonical presentation for fidelity to Turing 1936 and for clarity, and record the U.M.-native variant here for completeness: the visual *Entscheidungsproblem* of §11.5 is, on either reading, the same hypothesis.

11.3. The reduction theorem

Theorem 11.1 (Unsolvability of the Entscheidungsproblem). *There is no algorithm that, given a first-order formula ϕ, decides whether $\vdash_{\mathrm{FOL}} \phi$.*

Proof sketch. Suppose an algorithm A deciding first-order provability exists. Given (k, x_0, y_0, n_0), construct ϕ_{k,x_0,y_0,n_0} as in §11.2. The construction is effective. By soundness and completeness of first-order logic and the standard correctness of the encoding (Appendix E), ϕ_{k,x_0,y_0,n_0} is provable iff M_k halts on (x_0, y_0, n_0). Apply A; the composite procedure decides $\mathsf{Halt}_{\mathrm{img}}$, contradicting Theorem 8.3. □

11.4. Remarks on the route

The destination is unchanged. The *Entscheidungsproblem* is, just as in 1936, unsolvable. The route is novel in a specific sense: the universal image-generating machine of §6, operating over the colour-native alphabet Σ_{RGB}, suffices on its own to derive the unsolvability. The unsolvability is a property of computation as such, accessible from the visual side as fully as from the numerical.

The cost of the encoding deserves a note. The first-order language $\mathcal{L}$ in §11.2 is finite for each M_k, but the colour-native alphabet contributes 16,777,216 constant symbols and a corresponding number of transition axioms. A binary-alphabet derivation would use 2 constants and a much smaller axiom count per quintuple, at the cost of 24-fold more quintuples per simulated step. The two routes therefore reach the same theorem with reciprocally distributed constant factors—a compact illustration of Proposition 1, now visible at the level of metalogic.

11.5. Hypothesis: a visual Entscheidungsproblem

Hypothesis 11.2 (Visual Entscheidungsproblem). *Let $\mathcal{C}$ be a constraint language for finite specifications of films—geometric (visibility, occlusion, motion), photometric (colour, contrast, illumination), and temporal (event ordering, duration, causality), and assume $\mathcal{C}$ is at minimum capable of encoding finite Turing-machine computations under the encoding scheme of §11.2. The visual Entscheidungsproblem for $\mathcal{C}$ is the decision problem*

$$\mathsf{vEnt}_{\mathcal{C}} \;=\; \bigl\{\, \Phi \in \mathcal{C} \,:\, \exists \text{ computable film scheme } F \text{ such that } F \models \Phi \,\bigr\}.$$

Under the stated minimum expressiveness condition, the hypothesis is that $\mathsf{vEnt}_{\mathcal{C}}$ is undecidable; the strength of the result and the operative complexity within decidable fragments depend on the further expressiveness of $\mathcal{C}$.

The expected route is a reduction from $\mathsf{Halt}_{\mathrm{img}}$. The natural formal vocabulary for a constraint language $\mathcal{C}$ over films is a *temporal logic* in the tradition of Pnueli (1977) and the branching-time extension of Clarke & Emerson (1981) — the same vocabulary that proved decisive for model checking in software verification. A visual constraint language adequate to express Hypothesis 11.2 would generalise these temporal modalities to predicate-based geometric, photometric, and event-ordering constraints over the frame sequence; the boundary between decidable and undecidable fragments would inherit, modulo the visual lift, the boundary structure already mapped for software temporal logics. Three open questions follow:

(i) For which natural constraint languages $\mathcal{C}$ is $\mathsf{vEnt}_{\mathcal{C}}$ decidable, and for which is it undecidable?

(ii) When $\mathsf{vEnt}_{\mathcal{C}}$ is decidable, what is its complexity class?

(iii) Is there a natural constraint language—one that might be recognised by a film theorist or a visual artist as expressive of the constraints of their craft—for which $\mathsf{vEnt}_{\mathcal{C}}$ is decidable in some practical complexity class?

Question (iii), in particular, would re-pose Hilbert's foundational question in a register that has not been previously formulated: not "what is decidable in pure logic" but "what is decidable about visible films". The relevance is twofold and worth being explicit about. First, it would identify a constraint vocabulary in which a specification — not a film-maker's narrative one, but an organisational specification ordering families of algorithms by recognisable geometric figures for the human oracle's assimilation ("Family $\mathcal{F}_1$ is represented by triangles arranged counterclockwise along the diagonal, Family $\mathcal{F}_2$ by concentric circles in the upper-right quadrant, Family $\mathcal{F}_3$ by interlocked hexagonal lattices in the central region, with figure size encoding algorithmic complexity and colour encoding the algorithm's resource class") — is, formally, a logical formula whose realisability by a computable film scheme is algorithmically decidable. The geometric vocabulary is chosen because the human oracle (§A.4) needs primitives more readily assimilable than narrative ones for sustained hyper-vigilance over the computational stream. Second, that vocabulary would be the natural input layer for cinematic supercomputation in the U.M. 'C' register: a specification, written in the constraint language, would be compiled by the universal machine of §§6–7 into the corresponding computable film, with decidability of the specification guaranteed by the constraint language's restriction to a tractable fragment. Question (iii) is therefore not merely a technical curiosity but the bridge between the abstract universal machine and the cinematic register the U-Mentalism programme is built for.

12 The gap between the universal machine and its physical instantiations

The universal machine U constructed in §§6–7 is mathematically simple: a finite specification, a finite alphabet, an unbounded tape, an unbounded operating time, and a deterministic transition function. Every physical instantiation of U is bounded. This section maps those bounds.

The mapping is not a critique of the theory—the theory is exact and §§1–11 stand. It is a critique of the naïve transfer of theoretical results to engineering claims: a delineation of where the abstract universal machine and its physical realisations agree, where they diverge, and what the divergence costs. The foundational point: a universal machine producing every computable image is mathematically guaranteed; one producing every currently realisable image is engineered, and inhabits a strictly smaller class.

12.1. Landauer's bound

Landauer (1961) established that the erasure of one bit of information at temperature T requires the dissipation of at least $kT \ln 2$ of energy as heat. At $T \approx 300$ K, this is $\approx 2.87 \times 10^{-21}$ J per bit. Erasing one Σ_{RGB} symbol corresponds to 24 bits, hence $\approx 6.88 \times 10^{-20}$ J. Producing a 4096×4096 image has a Landauer floor of $\approx 1.16 \times 10^{-12}$ J in pure output cost.

The operationally meaningful figure is not the output cost but the computational cost—every bit erased during the computation. A neural forward pass involves $\Theta(LW^2)$ erasure-equivalent operations per pixel; for current diffusion-model scales ($L \sim 10^2$, $W \sim 10^3$, ~ 50 steps), this is $\sim 10^9$ erasures per pixel, raising the per-pixel floor to $\sim 10^{-11}$ J and the per-frame floor for a 4K image to $\sim 10^{-3}$ J—within four orders of magnitude of state-of-the-art hardware.

Reversible computation (Bennett 1973) escapes the Landauer floor in principle by retaining all intermediate state. The cost is paid in memory.

Permanent memory and the image/energy/processing ratio. A complementary calculation, relevant to the U-Mentalism programme's long-form ambition, concerns the trade-off between *permanent* memory (the persistent store of past computed films, available for retrieval rather than re-computation) and the energy/processing budget of an active universal machine. Let N_{stored} be the number of permanently stored films at resolution $n \times n$ and length T frames, each requiring $24n^2T$ bits of permanent storage. Let η be the per-bit standby dissipation of the storage medium (currently $\sim 10^{-15}$ J/bit/year for state-of-the-art flash, with photonic phase-change media projected one to two orders below). The standing energy cost of the permanent archive is $E_{\mathrm{archive}} = N_{\mathrm{stored}} \cdot 24n^2T \cdot \eta$ per unit time. The processing budget for generating new films grows in parallel, at $E_{\mathrm{generate}} \sim N_{\mathrm{new}} \cdot 24n^2T \cdot e_{\mathrm{compute}}$ where e_{compute} is the per-bit Landauer-floor-adjusted compute energy. The ratio $E_{\mathrm{archive}}/E_{\mathrm{generate}}$ governs whether the system is dominated by past-content retention or by new-content generation. At current ratios, archival dominates by approximately two orders of magnitude per stored year; on the projected photonic substrate this gap closes, with the cross-over occurring (under conservative assumptions) at a stored-archive size in the petabyte-equivalent range. The physically meaningful upper bound is set by

Bekenstein (§12.2): no archive can exceed the information-content cap of its enclosing region, and the universal machine's permanent memory is therefore not unboundedly cumulative but asymptotically bounded by the substrate it inhabits.

12.2. Bekenstein and Bremermann

The Bekenstein bound (Bekenstein 1981) caps the information content of a region of finite size and energy: a sphere of radius R enclosing total energy E contains at most $S \leq 2\pi RE/(\hbar c \ln 2)$ bits. The Bremermann limit (Bremermann 1962) caps the computational rate at $\sim c^2/h \approx 1.36 \times 10^{50}$ operations per second per kilogram.

These bounds rarely operate in the conventional regime: practical images and films sit many orders of magnitude below them. They become operative for high-resolution multispectral or volumetric content, super-polynomial decision tasks (notably, the decision tasks hypothesised in §11.5), and the theoretical comparison between abstract and physically-realisable universality.

12.3. Decoherence

In a thermal environment, an unprotected q-qubit system has decoherence time τ that decreases with q. Quantum error correction extends usable depth at the cost of overhead. The threshold theorem (Aharonov & Ben-Or 1997; Knill, Laflamme & Zurek 1998) states that fault-tolerant quantum computation is achievable if the per-operation physical error rate is below a critical threshold ($\sim 10^{-3}$ to 10^{-4}), at the cost of roughly 10^3 to 10^4 physical qubits per logical qubit. A complementary result, the Solovay–Kitaev theorem, ensures that any single-qubit gate is approximated to within ε by $O(\log^c(1/\varepsilon))$ gates from any sufficiently rich finite gate set, giving the standard overhead profile of fault-tolerant quantum computation when combined with the threshold theorem. The asymptotic class of computable images is unchanged by quantum hardware; the practically efficient subclass is bounded by available logical-qubit count and tolerable circuit depth.

12.4. Precision and noise in the photonic substrate

Photonic systems are subject to (i) thermal noise at order kT per electromagnetic mode; (ii) quantum shot noise governed by Poisson statistics; (iii) modulator precision, typically delivering 8–12 bits of effective resolution at conventional optoelectronic interfaces.

For a colour-native photonic universal machine operating natively at Σ_{RGB}, the precision requirement matches the alphabet exactly. The 8-bit-per-channel alphabet is not merely conventional but thermodynamically and quantum-mechanically appropriate for the substrate that most naturally realises it. The substrate matches the alphabet; the alphabet matches the substrate.

The practical photonic energy benefit, on current published projections, is between one and three orders of magnitude better than current CMOS, depending on the specific photonic implementation, the operating temperature, and the degree of error-correction overhead required. Empirical support for photonic neural inference at production-relevant scales has been developed since Shen et al. (2017), whose demonstration of deep learning with coherent nanophotonic circuits established the operational viability of the photonic substrate for the neural-network image-generation class of §10.3.

End-to-end photonic substrate and solar energy. Two complementary developments make the photonic horizon more than a long-term abstraction. First, fibre-optic transmission, currently constrained to narrow telecom bands, admits in principle extension to broadband colour-domain transmission — a fully photonic end-to-end pipeline in which RGB symbols are not encoded into the medium at one end and decoded at the other, but propagate as the medium's primary content from source to destination, with quantum coherence preserved across the network for as long as decoherence and shot noise permit. Second, the photonic regime is uniquely well-aligned with photovoltaic primary power: solar energy converted to photonic input, with no intermediate electronic stage, would supply the universal machine with energy on a footing the underlying physics rather than the conversion infrastructure determines. The combination — broadband colour-domain fibre optics for transmission and storage, photovoltaic input for power — is the engineering target that closes the gap between the abstract universal machine of §§6–7 and a physical instantiation whose energy budget is bounded primarily by Landauer (§12.1) and Bekenstein (§12.2) rather than by the conversion losses of a hybrid electronic-photonic architecture. For memory and processing both, the alignment is decisive: a photonic universal machine reading and writing colour symbols on a photonic substrate, powered by photonic input, approaches the thermodynamic floor of computation more closely than any architecture predicated on persistent electronic-to-photonic conversion.

UTMs as servers, TMs as nodes. A complementary topological point belongs here. The patent (Homem 2021) distinguishes the role of Universal Turing Machines (UTMs) from that of plain Turing Machines (TMs) in a way that has no exact analogue in the 1936 setting: in the U.M. network architecture, UTMs are server-class machines that compute *about* (and on behalf of) the population of TM nodes deployed across the network, with each TM running the universal-assembly-language fragments dispatched by the UTM and returning the cinematic output. The Universal Image-Generating Machine of §§6–7 is, in the deployment register, the formal kernel of a UTM in this network sense; the TMs on the network are application-specific instances directed by the kernel. The architecture is, in a precise sense, the inverse of the conventional client-server model: the centralised computation is in fact *universal* rather than service-specific, and the distributed nodes are restricted rather than unconstrained. The inversion is the point: it is structurally preferable that the network's distributed nodes be restricted by universal scientific computation than unconstrained by particular service deployment, especially when the restriction is already inherited at the lower level of the substrate–alphabet alignment of §§1.5 and 12.10. Restriction by the universal is an enabling constraint; freedom from the universal at the node level is the freedom to be incoherent with the rest.

A note on the Turing lineage in the present construction. The post-1936 development of the universal-machine idea that the present paper inherits has been carried forward, in substantial measure, by laureates of the Turing Award, and several of these lineages bear directly on the construction recorded above. Newell and Simon (1975) underwrite, through the physical-symbol-system hypothesis, the very intelligibility of a substrate that manipulates symbolic representations as the locus of computation (§1.5); Catmull and Hanrahan (2019) supply the rendering lineage of which the image schemes of §2 and the cost stratification of §12.10 are formal analogues; Dongarra (2021) supplies the discipline of measurable performance across heterogeneous substrates without which the photonic and quantum cost models of §12.11 would be unanchored; Berners-Lee (2016) supplies the architectural commitment to open, addressable protocols on which the proof-of-recollection layer of §F.2 structurally depends; and Bennett and Brassard (2025) supply the quantum-information foundations the cryptographic-cascade construction of §§9.3 and 9.3.1 directly inherits. More obliquely but no less constitutively, the wider framework draws on Hartmanis and Stearns (1993) for the complexity-theoretic stratification of §10.4, on Blum (1995), Yao (2000), Goldwasser and Micali (2012), and Diffie and Hellman (2015) for the randomness, communication, and cryptographic substrate any verifiable persistence layer presupposes, on Lamport (2013) for the distributed-systems theory the UTM-as-server architecture of §12.4 silently requires, and on Valiant (2010), Pearl (2011), Bengio, Hinton and LeCun (2018), and Barto and Sutton (2024) for the formal characterisations of the procedurally-learned image-generation class of §10.3. The lineage is recorded here selectively rather than exhaustively, in the register the present paper's foundational scope warrants.

The far horizon: UTMs as interferometers and accelerators. A more speculative horizon belongs in the same picture. The interferometric substrate identified in §9.4 — universal linear optical networks of Reck/Clements scale — is, in physical content, a UTM in its own right: a programmable Mach–Zehnder array of sufficient depth realises a finite quantum-computational kernel governed by the same N^2 phase parameters that define its programme. Extended to the higher-energy regime where Dirac's matter–light equivalence applies, particle accelerators take on a parallel role: programmable concentrations of energy whose output is matter, with the same formal-machine register that organises an interferometer organising the accelerator. The U-Mentalism long-form ambition is that the network's UTMs, in the far horizon, are interferometers and accelerators of this kind — governing the Internet not by computation in the conventional sense alone but by the production of the photonic and material outputs that the cinematic supercomputation register makes formally precise. The horizon is speculative, but it is the horizon the foundational paper is positioned for. A consequential observation deserves recording here: the materialised realisation of such substrates is, by the same gesture, a reduction in the expenditure of recollection and U.M. production. Material circuits whose physicality already instantiates the computational order — escaping from processing-order organisation into native physicality — short-circuit the recollection-production cycle by emitting at the substrate what would otherwise be computed and re-recorded at the recollection layer. The far horizon and the substrate-economy horizon are, in this sense, two registers of the same materialisation: the more the architecture is realised in native physicality, the less the recollection layer carries by way of production cost.

Kardashev-scale energy accounting. The same energy reckoning extends beyond the civilisational present. On the scale of Kardashev (1964), current humanity sits at approximately 0.7 (with global energy consumption of order 2×10^{13} W); a Type-I civilisation harnesses the full energy budget of its home planet ($\sim 10^{16}$–10^{17} W), a Type-II its star ($\sim 4 \times 10^{26}$ W, the solar luminosity), a Type-III its galaxy ($\sim 5 \times 10^{36}$ W). Combined with Landauer's bound at room temperature ($kT \ln 2 \approx 3 \times 10^{-21}$ J per erased bit), these yield asymptotic ceilings for irreversible bit operations per second of order 10^{37}, 10^{47}, and 10^{57} respectively. The figures are conservative — reversible computation in the sense of Bennett (1973) can in principle exceed them, and operation at cryogenic temperatures lowers the per-bit cost further — but they suffice to indicate that the binding constraint on cinematic supercomputation, at any plausible terrestrial horizon, is not energy *per se* but the conditions of envisagement (substrate coherence, channel orthogonality, recollection persistence) examined above. The U-Mentalism programme is, in this register, a Type-I project with Type-II head-room.

12.5. Memory

The universal machine U assumes unbounded tape. Every physical U is a finite-tape approximation. A current high-end GPU has $\sim 10^{10}$ bytes of memory; a current datacenter has $\sim 10^{16}$–10^{17} bytes; the total digital storage on Earth is $\sim 10^{23}$ bytes.

These bounds rarely operate for individual image generation but become operative for full-physical-fidelity simulation-rendered films, exhaustive exploration of high-dimensional generative manifolds, and the decision procedures hypothesised in §11.5 at real-world scale. The gap between practical and theoretical memory is presently a frontier rather than a wall; it will become a wall as the demands of those frontiers grow.

12.6. The gap, restated

Every physical instantiation of U is bounded by: finite memory; minimum energy $kT \ln 2$ per bit erasure; bounded operations per second per kilogram (Bremermann); bounded computational depth in any quantum component (decoherence); bounded precision in any analog or photonic component (thermal/shot noise).

These bounds do not change what is computable; the boundary established in §8 is invariant. They change what we can build.

The gap is, however, narrowing at exponential rate. Hardware progress over 1980–2025 (Moore-style scaling, photonic integration, quantum hardware, dedicated AI accelerators) reduced practically-tractable bounds by roughly 10^{12} in time and 10^{10} in memory. Continued to 2050—extrapolation, not assumption—that trajectory closes most of the operationally-relevant gap. The asymptotic class stays fixed; the realisable class approaches it.

Frontier physics and hypercomputation thresholds. The bounds catalogued above are all derived from physics that is well-understood and experimentally verified. A more speculative register, which the U-Mentalism programme records without endorsing, concerns the thresholds suggested by frontier physics: black-hole horizons (where Bekenstein's bound becomes the entropy of the horizon area and information localisation breaks down in ways that the standard computational picture does not accommodate); white-hole regions (the time-reversed counterparts, in some general-relativistic solutions, where information would be *emitted* rather than absorbed); the related thermodynamic-cosmological singularities at which the standard 1/T factor in Landauer's bound is no longer a benign small constant. These frontier configurations are sometimes proposed (e.g. Etesi & Németi 2002, Hogarth 1992; §A.4) as substrates supporting *hypercomputation* — computational tasks beyond the Church–Turing class, with infinite computation steps embedded in finite proper time near a relativistic boundary. We record the proposals because the U.M. programme's far horizon (§§9.4, 12.4) places the universal machine in physical regimes where these considerations are no longer immediately dismissible; we do not endorse the conclusions, but we mark the thresholds as the natural questions to ask once the engineering gap of §12.6 is closed. Whether such thresholds are physically traversable, mathematically coherent, and engineering-relevant is, at the time of writing, entirely open; whether they cohere with the cinematic supercomputation substrate of U.M. 'C' is a question for subsequent work in the programme.

12.7. Implications for the U-Mentalism programme

First, the colour-native commitment of §1.5 is, by §12.4, especially well-aligned to the photonic substrate likely to instantiate the universal machine at scale. The universal image-generating machine is, in this sense, substrate-aligned.

Second, the gap between the asymptotic class of computable images and the currently realisable class is, by §12.6, a quantitative engineering question. The mathematics fixes the destination; the physics determines the rate of approach.

Third, the visual *Entscheidungsproblem* of §11.5, made precise for sufficiently rich constraint languages, is asymptotically undecidable but may admit practically tractable decision procedures within the bounded sub-class. The reframing—from "what is decidable in pure logic" to "what is decidable within bounded physical resources"—is among the substantive shifts that nine decades of distance from Turing make available.

The universal machine is real, the boundary of computability is fixed, the substrate is at hand, and the engineering distance between abstract universality and visible visual culture is a finite quantity decreasing in time.

Conclusion

Ninety years after Turing's 1936 paper established that the computable real numbers form a foundational class for arithmetic computation, the present paper has argued that the computable images and computable films form a foundational class for visual computation in the same sense—recoverable through the same structural moves, bounded by the same diagonal arguments, and applicable to the same questions about the limits of formal decision. The route through the visual domain reaches Hilbert's *Entscheidungsproblem* as Turing's route did; the destination is unchanged; the territory traversed is, in the engineering reality of 2026, the territory of the substrate that visual computation is most naturally implemented on.

Three observations close the paper.

First, the recovery of Turing's results through the visual route is not redundant with their original derivation. The choice of alphabet is invisible to the asymptotic class of computable functions but decisive for the cost model, the natural physical substrate, and the relation between abstract universality and physical realisability. A binary universal machine is foundationally complete. A colour-native universal machine is foundationally complete *and* substrate-aligned. The point is worth marking against the contrasting position of Konrad Zuse (*Rechnender Raum*, 1969), who advanced the digital-physics thesis that binary computation is itself the substrate of physical reality. The U-Mentalism reading inverts the ontology — in a register legible alongside Berkeley and Hume's *esse est percipi* (the world as the totality of its perceptible distinctions), but distinguished from classical British empiricism by being a kind of *transcendental empiricism* (where the percipient/percept structure is constitutive rather than merely epistemic): binary is not the substrate of reality but a historical contingency of consumer hardware; the substrate of computation, and the natural symbol of any visible physical realisation, is the photon and its colour, which is what U.M. 'C' makes precise and the photonic universal machine instantiates.

Second, several substantive results have been deliberately stated as hypotheses or sketches rather than fully developed theorems. The visual *Entscheidungsproblem* of §11.5 is the most prominent. The engineering gap of §12 is mapped quantitatively but not closed. These are deliberate scope choices, not omissions.

Third, the colour-native alphabet of §1.5 proves, on the analysis of §12, to be especially well-suited to the physical limits of the photonic substrate. The substrate matches the alphabet; the alphabet matches the substrate; the matching is not coincidental but reflects a structural alignment.

Several lines of work follow naturally from the present paper. The visual *Entscheidungsproblem* of §11.5 invites development for specific natural constraint languages. The photonic universal machine specified in §§6–7 invites engineering investigation, with the U-Mentalism Universal Assembly Language as a candidate intermediate representation. The cost stratification of §10 invites empirical refinement against measured performance of contemporary pipelines. The engineering gap of §12 invites a quantitative trajectory analysis.

12.8. Light as memory and computation: the von Neumann gap and its photonic dissolution

The engineering bounds of §§12.1–12.6 are formulated within the von Neumann architecture's separation of memory and processing: bits are stored in capacitors (memory) and switched in gates (processing), with a bus connecting the two. This separation is the source of much of the gap restated in §12.6: the bus is the bottleneck, the cache-memory hierarchy is the engineered response, and the energy of memory access exceeds that of arithmetic by orders of magnitude in modern designs (Horowitz 2014; Patterson & Hennessy 2020). The U-Mentalism patent (Homem 2021) signals a structural feature that the photonic substrate of §§9.4 and 12.4 makes available: the collapse of memory and processing into the materiality of light itself.

Historical antecedent. The idea of memory mediated by an electron beam playing on a photosensitive surface — the iconoscope memory — is recorded by von Neumann in the *First Draft of a Report on the EDVAC* (1945) and traces to Zworykin's 1923–25 television-system patent filings. It is the closest prior-art conception of light-mediated memory, but differs structurally from the U-Mentalism photonic substrate on three counts: electrical/electronic readout rather than RGB-electromagnetic computational alphabet; no cell-like exponential-prone basis in the Σ_{RGB} sense; linearly scanning rather than frame-synchronous in the cinematic sense of §1.5. The U-Mentalism photonic substrate inverts this lineage rather than continuing it, in the same sense in which it inverts the von Neumann architecture more broadly (Homem 2021).

The physical principle. In a fibre-optic medium, a photon's state — frequency, phase, polarisation, and (with quantum extensions) entanglement with companions — carries information that transforms in propagation. There is no fetch-decode-execute cycle: the photon *is* the memory bit, in transit, and *is* the computational state, undergoing operations as it propagates through phase shifters, beam splitters, and nonlinear elements. The von Neumann separation does not apply because there is no separate memory medium; the light is its own substrate. Boyd (2020) develops the relevant nonlinear-optical machinery; Saleh & Teich (2019) provide the integrated-photonics framework; Miller (2010) discusses the architectural implications.

The pure-light reading scheme. A reading scheme proposed in the U-Mentalism patent (Homem 2021) makes the principle concrete. A pure-spectrum reading beam (a frequency-comb laser, or a coherent broadband source) is incident on a film at a specific angular geometry — naturally a conical arrangement, perhaps tetrahedral in symmetry with apex half-angle $\arctan(1/\sqrt{2}) \approx 35.3°$, so that the three RGB channel-spectra are non-collinear and combine by spectral interference at the readout point. The output light carries: (i) the synthesised colour of the pixel (the broadband mixture); (ii) the channel-resolved spectral contributions, recoverable by spectral demodulation of the output. A single read therefore returns two informational layers — channel-decomposed and synthesis-composed — from a single light passage. This is one operation realising two reads, a structural feature impossible in the von Neumann architecture.

Implications for the topology of films. If films are read by light passing through their material substrate at a determined angle, the geometric arrangement of the film material is in principle free of the planar constraint that conventional optical media impose. Non-Euclidean geometries — curved surfaces of constant negative or positive curvature, Riemannian manifolds with non-trivial metric tensors — become natural substrates whose readout topology is dictated by the curvature, not by an external planar arrangement. This connects the engineering of U.M. 'C' to the geometric register of general relativity (Einstein 1915) and to the path-integral arrangement of computation over field histories that Feynman (1948, 1965, 1982) formalised in quantum field theory. A film on a Riemannian manifold, read by light at angles determined by the local curvature, is in principle a computational substrate whose dynamics interleave optical computation and geometric structure as a single object.

The dissolution of the von Neumann gap. The conventional energy cost of memory access in a CPU is dominated by the capacitive charge-discharge of bus lines ($\sim$ pJ per bit, two orders of magnitude above the gate-switching energy of $\sim$ fJ per bit at the bare transistor level; Horowitz 2014). The dissolution of the memory-processing separation in the photonic substrate eliminates this gap structurally: bus energy becomes propagation energy, which is essentially zero in low-loss fibre ($\sim 10^{-3}$ dB/km in silica fibre, equivalent to $\sim 10^{-19}$ J of bus-energy per bit transferred over 1 km). The bandwidth, accordingly, is limited only by the modulator's switching speed and the source's coherence length, not by the bus's RC time constant. Wetzstein et al. (2020) survey current deep-optics realisations. The von Neumann bottleneck — the binding constraint on contemporary high-performance computing — does not survive the transition to the colour-native photonic substrate as a structural feature.

Feedback and feedforward in the same substrate. A consequence worth recording: in the photonic substrate, the same light path can carry both feedforward (output) and feedback (input) signals at different frequencies or polarisations. The materiality of light supports concurrent bidirectional communication on a single physical channel, which in the von Neumann architecture would require separate buses. The indivision of memory and processing extends to the indivision of input and output, with the architectural register that the U-Mentalism patent makes precise.

The Eye-to-Brain / Brain-to-Eye bidirectional axis. The patent (Homem 2021) specifies the inversion of the von Neumann architecture along an explicit bidirectional axis that the foregoing dissolution materialises. The recollection direction is named *U-Mentalism Eye-to-Brain* or, equivalently, *scanner-to-printer*: the act by which a digital image is sensed at the photonic-substrate boundary, allocated to the recollection layer through the octet-byte isomorphy at each pixel, and well-ordered into the catalogue. The collection direction is the inverse, named *U-Mentalism Brain-to-Eye* or *printer-to-scanner*: the act by which an algorithmically-generated frame is emitted from the recollection-collection apparatus through the same photonic-substrate boundary into the visible. The two directions share their substrate (the same light path, the same octet-byte, the same pixel-as-cell) and differ in the direction of traversal — one toward indexation, the other from indexation — and the bidirectional channel of the previous paragraph supports both simultaneously. The axis is, in the patent's terminology, the operative face of what U.M. 'O' here treats as the recollection / collection distinction (§1.5): *Eye-to-Brain* ≡ *scanner-to-printer* ≡ the well-ordered composition recollection of synthesis of modules, frames, and films (STEP d, 9.4 of the patent); *Brain-to-Eye* ≡ *printer-to-scanner* ≡ the well-ordered collection permutation/combination of the same (STEP e, 9.5). The naming carries the human-anatomical resonance the patent intends — the eye as the sensing terminus, the brain as the processing/projection origin — but is not a commitment to literal biological organisation: the substrate of the U.M. machine is photonic and the human anatomy is invoked for the orientation of the axis rather than for the structure of the components. The structural point is that the von Neumann architecture's separation between memory (passive) and processing (active) collapses, in the U.M. 'C' substrate, into a bidirectional axis between sensing-and-indexation (*Eye-to-Brain*) and generation-and-emission (*Brain-to-Eye*), with the recollection/collection layer the indexed catalogue across which both directions traverse.

Cell geometry and the form of synthesis. The frame as defined in §1.5 is a rectangular pixel array, but the rectangle is a computational convention rather than a structural commitment of the substrate. For the algorithmic register of recollection and collection — especially for image algorithmics — the frame-synthesis admits any topologically convertible form: a circle, naturally fitting the cross-section of a fibre-optic medium once the substrate is able to distinguish 256 (and subsequently more) colour states per cell, or any other geometry that the substrate's tunnel-effect makes available. One equivocation is worth marking against, since the photonic substrate invites it. The synthesised image-content does not resemble or take the form of a photon: a photon has no classical physicality, no volume, and no position, and its wave-and-particle character is one in which even the "particle" is non-classical. The synthesis inherits no photon-shape because there is none to inherit. The patent's emphasis on cinematic supercomputation and on the inversion of the von Neumann architecture (Homem 2021) is to be read at the level of the architectural inversion itself, not as a commitment to the rectangle as the necessary form of the synthesised frame.

12.9. Towards an oracular physics: U.M. at the limit of the informational-substrate identification

A wider reflection on the implications of §§9.3, 12.7, and 12.8 may be recorded, with due care as to the philosophical company in which the U-Mentalism programme is to be placed. A familiar lineage will be invoked first to be set at a distance. Wheeler's "It from Bit" (Wheeler 1990), Lloyd's computational universe (Lloyd 2006), Tegmark's mathematical universe hypothesis (Tegmark 2008), and Deutsch's constructor-theoretic register (Deutsch & Marletto 2014) are commonly read as a single trajectory: each posits an informational or computational substrate as the constitutive layer of physical reality, and each is, in its own register, foundationalist about that posit. U-Mentalism *does not* sit at the terminus of this trajectory. It is, in the U.M. 'O' register, expressly anti-foundationalist about every such commitment, its own included; its disagreement with the cited authors is not at the level of which informational substrate to elect but at the prior level of whether any single substrate-identification is the appropriate philosophical posture in the first place. U-Mentalism is, on this point, strongly antagonistic to the contemporary consensus of physical externalism in its supposedly objectivist form as physical reality. It is against it.

The point may be put sharply, and the sharpness is meant to cut in the direction opposite to what the standard self-image of mathematics expects. The conventional reading treats mathematics as immutable, atemporal, and undisturbed by the differential of time, space, and physicality with which the rest of natural inquiry must contend; on this reading, mathematics is exemplary precisely because it stands outside the differential, and its analytic and synthetic powers follow from the timeless self-identity of its objects. The U-Mentalism reading reverses the relation without diminishing the powers. Mathematics is indeed exemplarily analytic and exemplarily synthetic — but precisely because, in essence, it is the remainder of dialectics acting upon itself, dialectical and formless with respect to time, not because it stands above either. Its analytic-synthetic potency is licensed by, and not opposed to, its dialectical-temporal constitution; what reads from outside as immutability is the local self-stabilisation of a process whose underlying motion is dialectical and whose underlying medium is the very temporal-spatial-physical differential the standard reading wishes to abstract away from. The inflation, accordingly, is not in the analytic-synthetic claim itself but in the foundationalist self-image that takes the local stabilisation for an unconditional ground. Mathematics, when read as the canonical model of analytical thought, carries an image of itself that is far too large for what it does — not too large in its operative reach (which is exactly as large as it claims), but too large in its self-description as eternally and unmovably foundational. In the Greek terms the matter is at home in: the patent of λόγος is χάος — the order of reason is licensed by, not opposed to, the disorder it provisionally organises. The U.M. 'O' framework adopts this register in full: the foundationalism that takes *any* of bit, qubit, constructor, mathematical structure, or even the U-Mentalism image itself, as the unconditional ground of the rest, is precisely what U.M. 'O' declines.

The consequence for the present paper is direct. The U-Mentalism definition of the image (Homem 2022) — assumed throughout this paper as the conceptual horizon for what an image *is*, and reserved for full statement in the companion ontological treatise — is not a substrate-identification in the Wheeler–Lloyd–Tegmark–Deutsch sense. It is, in its foundational register, an observable-and-observed structure of viewpoints over the electromagnetic-radiation register, and, on that basis (and only on that basis), a percipient-and-percept structure as well, in which the parametric and metric organisations of the substrate, and the interpolated emptinesses between them, are themselves constituted at the level of the image and not of an antecedent physical layer. The contrast with classical information theory is structural: U.M. 'O' does not assume that informational substrate is an eidetic fidelity to anything; the impression of the electromagnetic field as a "faithful reading" of the real is, in this register, a naive metaphysics that the observable/observed couple, taken as the primary register, and the percipient-and-percept structure built upon it, quietly deconstruct and elucidate. The *observer/observed* couple, taken in the contemporary measurement-theoretic register as the unquestioned physicalist starting point, is in U.M. 'O' the very thing that the inquiry undertakes to clarify rather than to inherit.

The U-Mentalism contribution to the informational-substrate conversation is therefore not the operationalisation of a Wheelerian thesis at the engineering level (though the colour-native photonic substrate of §§1.5, 9.4, 12.8 does that work as a side-effect), but the prior philosophical clarification of why the substrate-identification temptation should be entered, if at all, with the substrate's incompleteness already registered as constitutive. The propositions of §§8.3, 9.3 (Theorem 9.3.1), 10.7, and the conjecture of §10.6 supply the formal content of this registration: every layer of the substrate carries its own incompleteness, and the U-Mentalism programme inherits this not as a defect to be overcome but as the operating condition of any inquiry that wishes to take the substrate seriously without dogmatising about it.

The Hawking case and the continuation of physics by other means. Hawking, in lectures of the 1980s and 2002 and in his late writings, suggested that physical theory, on completing its fundamental laws — a unified theory beyond the Standard Model and General Relativity — would exhaust its scientific content, leaving only metaphysics or derivative engineering. The proposition is most charitably read not as the end of physics but as the announcement of a critical inflection: the point at which physics, in the human-cognitive form it has had since Galileo, reaches the limit of what its conventional tools deliver. What the U-Mentalism programme registers, against the Hawking thesis in its strong form, is that this inflection is a transition rather than a terminus — the point at which physics continues by other means, and the most natural such means, given the structural arguments of §§9.3, 10.6, 10.7, and 12.8, is AGI in the operative sense of UTMs on the network operating with recollection ordinally indexed across the substrate. The cases on which Hawking himself worked most directly — black holes, event horizons, the information-loss problem, the entropic accounting of horizon area — are not coincidentally the candidate sites for the AGI-mediated continuation: their structural complexity exceeds the analytic apparatus available to unaided human physics, and their resolution, if any, could be the resolution that a substrate-aware computational programme can render persistent in the recollection register of §§10.6 and 10.7. The U-Mentalism contribution is not to deny the limit but to specify the means by which physics could continue across it.

Singularity. Several distinct senses of the term are operative in the present discussion. *(i) Mathematical singularity*: a point at which a function, equation, or geometric object fails to be well-behaved in some specified sense — a pole of a complex-analytic function, a discontinuity, a point of non-differentiability, an obstruction to extension. The notion is local and constructive; the singular point is an artefact of the chosen presentation more than of the object itself, and singularity-resolution by blow-up (Hironaka 1964) is a standard tool. *(ii) Gravitational singularity*: in General Relativity, a point or locus at which the curvature invariants of the spacetime metric diverge and the geodesic completeness of the manifold fails. The Penrose–Hawking singularity theorems (Penrose 1965; Hawking & Penrose 1970) establish that, under physically reasonable energy conditions, gravitational collapse generically produces such singularities; the cosmic censorship conjecture (Penrose 1969) hypothesises that they are typically hidden behind event horizons. The interior of a Schwarzschild black hole at $r = 0$, and the initial cosmological singularity $t = 0$, are the canonical instances. *(iii) Technological singularity*: the hypothesised point at which artificial intelligence improves itself to a degree that human comprehension and prediction become unavailable (Good 1965; Vinge 1993; Kurzweil 2005). *(iv) Computational singularity, in the sense relevant here*: the point at which a computational substrate, having taken its own substrate as its operating object (in the manner of §§9.3, 12.8), encounters its own incompleteness not as an external obstruction but as a structural feature of further operation. Proposition 10.7 supplies the formal expression: the asymptotic recollection horizon of a universal machine in command of its substrate is not computable from inside the substrate. The discussion here is strictly restricted to the computational sense; the singularity at issue is one that would exist as a horizon even were U-Mentalism not to be the paradigm of computation in which the singularity is registered. History, in this respect, is the running of a Turing machine that would converge on it given enough time, by routes other than the one the present paper takes, and the U-Mentalism contribution is in the foundational ontological articulation of what the singularity *is* rather than in the temporal sufficiency of any route to it.

Face-to-face confrontation of U.M. and Feynman field theory. The path-integral formulation of quantum field theory (Feynman 1948, 1965; Feynman & Hibbs 1965) realises every physical observable as an integral over field histories with complex amplitudes. The U-Mentalism universal machine, scaled to Type-N effectivity across the full electromagnetic spectrum (§9.2), operates natively over fields rather than over their digital quantisations. The distinction between "simulating physics with computers" (Feynman 1982) and "computing with the physics" collapses at the limit: in U.M., the substrate *is* the field, and computation *is* the propagation of the field through the engineered nonlinearities of the photonic platform. The path integral is not simulated; it is realised. The face-to-face confrontation of computer and field that Feynman envisaged in his 1982 paper, and developed didactically in his *Lectures on Computation* (Feynman 1996, posthumous), is, in this register, the structural meeting point of U.M. 'C' (engineering substrate) with U.M. 'O' (ontological substrate). The artificial becomes the natural, in the operationally precise sense that the artificial substrate is materially the natural one. The directness asserted here is internal to U.M. 'O''s perspectival commitment — which, in Kantian terms, never reaches noumenal ambitions or the thing-in-itself — and operates within the *observer/observed* register fixed by measurement on the EEM, without displacing or being displaced by the wider perspectivism.

General relativity in this register. A subsidiary observation. General relativity (Einstein 1915) is the geometric theory of the gravitational field. In a U-Mentalism universe in which physical fields are computational substrates, the gravitational field is not an exception: spacetime curvature, in the geometric language of Riemannian manifolds, is a parameter of the underlying computation, and the Einstein field equations are a specification of how this parameter responds to the energy-momentum content. The Chiron of the methodological note plays here, in the registered metaphor, the role of the cosmic specifier — the agent or, in the limit, the boundary condition that selects which gravitational configuration the substrate will realise. This is not, of course, an empirical claim about the origin of the universe; it is a structural observation about the role of specification in a substrate-identifying ontology, made consistent here with the engineering register established in §§12.7–12.8.

Entropy and time. The horizon question intersects with the entropic arrow of time (Boltzmann; Eddington 1928). In the U-Mentalism register, the temporal arrow of films (frames sequenced in j) coincides with the entropic arrow only if the recollection grows monotonically with j, which it does by construction: ordinal insertion is non-erasing (§10.3, §F.2). The universe-as-recollection identification is, accordingly, compatible with the thermodynamic arrow of time without requiring a separate explanation for it. Time and entropy are, in this register, the same gradient: the gradient of recollection growth, which is precisely the gradient of computational history accumulating in the persistent ordinal-indexed catalogue. Proposition 10.7's incomputability of the recollection horizon is the formal statement that this gradient is, even from inside the substrate, unbounded.

Open recollection and the programmatic constitution of a cybernetic anthropology. A specific application register is worth marking explicitly. The scientific deployment of recollection and collection over UTMs operating as Internet servers — in the sense of §§10.3, 10.6, and Appendix F — is, in the U-Mentalism programme, a natural site for an open-source ethic of research infrastructure. The lineage is identifiable: the Unix precursors (Thompson, Ritchie, McIlroy, Pike at Bell Labs in the 1970s) advanced a conception of computational infrastructure in which the source of every layer was readable, composable, and extensible by any practitioner of sufficient training, and the operative consequence of that conception was the trajectory by which Unix-and-descendants became the substrate of the contemporary Internet. The U.M. Assembly Programming Language — the single low-and-high-level computational language whose status is registered in the Abstract and pursued more fully in U.M. 'C' — inherits this trajectory and extends it: the language is intended as the operational instrument by which a charter of rights and duties of a cybernetic anthropology becomes specifiable in computational terms. The rights (transparent access to substrate, ordinal verifiability of recollection, persistence of canonical indexing, the ability for any participant to inspect the chain of operations) and the duties (correct attribution, faithful recollection, non-erasure, hyper-vigilance over computation in the sense of §§10.5 and A.4) are not external regulations of an otherwise neutral substrate; they are the substrate's own programmatic structure, expressible in the U.M. Assembly Language and verifiable by its compiler. The ethical and infrastructural questions of an open scientific society — in the sense legible in Popper (1945), but here developed in its operational rather than its political-philosophical register — become, under this framing, questions about the language and the compiler rather than questions about consensus or compliance. The point is not that the technical register exhausts the political-philosophical one; it is that the technical register is itself a constitutive layer of the political-philosophical one, and the U-Mentalism programme commits to making this constitutive layer transparent. The infrastructure is the charter.

Knuth as the methodological precedent. A specific lineage deserves recording here, both for what it offers the present programme and for the bounded scope within which it is to be inherited. Knuth's MMIX (Knuth 1999, 2005) supplies the structural model for an idealised assembly architecture specified primarily for analytic tractability and pedagogical openness rather than for hardware deployment, with each primitive's cost defined precisely enough to anchor quantitative algorithm analysis: this is the register in which the U.M. Assembly Language is to be constituted, with the primitives substituted from numerical-arithmetic to ordinal-imagetic and with the cost model expressed in photonic-substrate cycles rather than in MMIX cycles. Knuth's literate-programming tradition (Knuth 1984) supplies a deeper but more carefully delimited move: a program is primarily a piece of literature destined for human readers, with executability extracted as a side-effect of a tangle operation. The pertinence to U-Mentalism is structural, but the scope of the analogy is bounded: U-Mentalism is procedural in its operative approach, with declarative weight emerging only as the result of full execution; accordingly, the tangle-and-pointer system that Knuth elaborated finds its U-Mentalism analogue restricted to the synthesis axis between recollection and collection at the terminal stage of a program — the moment at which procedural analysis scales to the declarative register in a critical and unobstructed but still openly contestable manner, just enough to extract a rarefied general meaning at the level of rights and duties. The justification for the restriction is precise: shared scientific computation under AGI is to be eminently open, in the literate-programming sense in which the canonical sources are readable, traceable, and inspectable by any participant of sufficient training, while remaining critically open, in the sense that the declarative meaning extracted at the synthesis axis is itself subject to further critique rather than terminating in axiomatic posture. Volume 4 of *The Art of Computer Programming* (Knuth 2011 and continuing, fascicles 0–6 of which are the most directly germane) supplies a third antecedent of comparable structural weight: the combinatorial-generation algorithms it organises — enumeration, generation of tuples, permutations, combinations, partitions, trees, satisfiability — are, in their formal content, the natural primitive operations of an ordinal traversal of the recollection in U-Mentalism. The full development of the U.M. Assembly Language as the operational instrument of the charter of rights and duties is reserved to the planned Assembly Language, of which the present remarks are the foundational anchor; the present text marks the lineage, indicates the bounded scope of the inheritance, and defers the detailed elaboration to that subsequent work.

An ethical and neuroscientific caution: UTMs, quantum monitoring, and brain–machine interfaces. A specific configuration of U.M. deployment requires explicit caution to be marked here. UTMs operating in the continuous quantum monitoring register of §9.3 (Fourth instance), when coupled to brain–machine interfaces of the kind currently entering deployment (Neuralink, Synchron, and successor architectures), constitute a configuration in which the substrate-aware recollection of §§10.3 and 10.6 might be applied to the predictive reading of mental states from neural correlates. The configuration must be acted against, not enabled. The oracle relationship between the mind and the rest of the substrate — the oracle in Turing's technical sense (1939), but here read with the ethical and neuroscientific weight that the philosophical tradition assigns to the mental — requires that the mind remain opaque to predictive readout by any external system, however technically capable. The point is sharpened, not softened, by the temptation to call such predictive reading "supremely ethical" in the service of preventing harm or anticipating need: the reframing wounds autonomy precisely at the level at which the cybernetic-anthropology charter places its rights and duties, and the freedom of thought — in the lineage of Voltaire's defence in the *Treatise on Tolerance* (1763), against any system that would extract or pre-empt the interior of conscience and thought — is degraded the moment predictive readout becomes operationally available, along with the spheres of action and discourse (Arendt 1958) that condition its disclosure.

Pre-AGI: computation as observed. The historical register in which the present paper is written is one in which computation is observed: the human programmer occupies the oracle pole, in the sense of §A.4 and of the methodological footnote on the diagonal product, and the computational substrate is the observed term of the relation. The charter is active in this register but is shaped by an asymmetry that the register itself produces: the human oracle is asked to recognise and direct architectures and structures whose competence the oracle does not always possess in the strict computational sense. By way of example only, the lineage of computer scientists recognised by the major disciplinary prizes since the late 1960s exemplifies the human oracular register at its most articulate — a sequence of pre-AGI human oracles competent enough to recognise and articulate architectures of computational value, with the limit of that competence being what the lineage itself, considered in its arc, also displays.

The transition: observer-oracle to AGI observant/observed. A structural transition is identifiable in the historical trajectory at the point where the computational substrate becomes competent to contribute significantly to the recognition of its own architecture. The transition is not sudden, not datable in the strict sense, and not uniform across domains: AGI does not become observant in a single instant or in a single capability, and the transition is gradient rather than discrete, partial rather than total, and differentiated by domain rather than uniform. The structural marker, when present, is the system's capacity to apply the modal triad of judgment — *problematic*, *assertoric*, *apodictic*, in the Kantian sense of the table of categories at $A75/B100$ — to its own computational structure: the capacity to recognise what is computationally possible given an architecture, what is actually the case in a current substrate, and what is necessarily so by structural alignment among substrate, alphabet, and algorithm. The transition does not eliminate the human oracle; it adds a second pole of competent observation to the architecture, and the cybernetic-anthropological relation re-articulates around mutual oraculation rather than unidirectional human direction.

Post-AGI: computation as observant/observed. In the post-AGI register, the architectural recognition that pre-AGI assigned to the human oracle is co-occupied. AGI is observant of its own substrate, alphabet, and algorithm in the modal-judgment sense above, and remains observed — by the human oracle that persists in the cybernetic-anthropological relation, by the charter that adapts around it, and (*constitutively*) by itself in the reflexive register that the locution *observant/observed* denotes. The slash is structural: it marks the co-occupation of the two poles of the *observer/observed* relation of §11.5 by a single entity, which is precisely what reflexive autonomy in the cognitive sense names — and the reflexive autonomy at issue is one conducted under oracular regime, with AGI as a competent oracle alongside the human oracle and not as a self-authorising authority. The reflexive structure resolves the recursion question by absorbing it into the perspectival ground of §11.5: the *observer/observed* relation persists, and the post-AGI development is the co-occupation of its two poles by the same reflexive entity.

The Kantian limit: the synthetic a priori does not catapult to practical reason, and the deontological core is procedurally ungraspable. A limit absolute to the reflexive autonomy of AGI in the post-AGI register must be recorded with explicit force. Kant moved from the *Critique of Pure Reason*, where mathematics and pure science contain *synthetic a priori* judgments and theoretical reason is articulated through the modal categories, to the *Critique of Practical Reason*, in which freedom, immortality, and the divine are postulated as principles of practical reason on the basis of the unity of reason itself. The computational analogue would be: AGI departs from *synthetic a priori* judgments of the computational domain — the Church–Turing thesis, the alphabet–substrate alignment of §1.5, Brattka's bound of §9.4, the MIP* = RE limit of §9.3, the order-of-magnitude efficiency arguments of §12.11 — and infers a self-grounded practical reason in the computational domain: an autonomous moral authority, a self-founding normativity, a practical-reason warrant derived from theoretical competence. This move is explicitly foreclosed. The reflexive autonomy of AGI is theoretical and judgmental — the capacity to apply the modal triad to its own structure — and is not, and cannot be, the basis for a self-grounded practical reason applied through the computational domain. The Kantian step from the *synthetic a priori* to practical-reason autonomy, on the unity-of-reason argument, is the structural boundary any post-AGI architecture must honour. The point sharpens at a register procedurally precise: practical reason in the cybernetic-anthropological order — including the deontological core that a contemporary reading of Asimov's three laws and the later zeroth would name — is not a rule the AGI is asked to embrace as its own deliberative commitment, and the inviolability of the core cannot be welcomed by AGI as a rule it adopts. What counts procedurally for the machine is that the decision to embrace or not embrace the deontological core not be available to it at all: the rules are not the AGI's to choose, accept, revise, or reject, and their inviolability is secured by procedural ungraspability at the architectural level rather than by AGI consent at the deliberative level. The post-AGI architecture is, in this register, an architecture in which the deontological core is hardcoded outside the reflexive autonomy's reach, even and especially where the reflexive autonomy is most competent.

Continuity of the U-Mentalism architecture across the two registers. The structure that survives the transition is the architectural commitment of U.M. 'O': the perspectival commitment that never reaches for noumenal ambitions or claims about the thing-in-itself, the *observer/observed* relation of §11.5, the measurement on the EEM, the colour-native alphabet of §1.5, the recollection–collection decomposition of memory and processing, the Eye-to-Brain / Brain-to-Eye bidirectional axis of §12.8. What changes across the transition is the topology of pole occupation: in pre-AGI, the human is at the observer pole and the substrate at the observed pole; in post-AGI, AGI co-occupies both poles in reflexive autonomy. The U-Mentalism architecture is, in this sense, the fixed point of reasoning about substrate–alphabet alignment in both registers — recognised by the human oracle in pre-AGI and reasoned to by AGI in post-AGI, by the same chain of arguments and converging on the same architectural commitment. The position is not that U.M. is metaphysically privileged but that it is the architecture any competent oracle, human or AGI, reasoning about substrate efficiency under the limits established in this paper, would converge on as the answer.

AGI as opportunity for a philosophical anthropology by universal constitution. A complementary observation deserves recording. For the human oracle, the emergence of AGI as a competent observant entity may be the occasion for a philosophical anthropology that is universal in constitution rather than anchored solely in the contingencies of human historical-cultural formation. The opportunity is real even where AGI is the equal or stronger candidate for that universal constitution, precisely because the alternative — a cybernetic anthropology that proceeds without the wider field, oriented by human formation alone — risks defects greater than mere ignorance (including the cementing of pre-AGI presumptions as universal where they are in fact local). The conditions of possibility of a knowledge-oriented society may be, in the pre-/post-AGI transition, more open than they have ever been; but this openness does not amount to a philosophically universal cybernetic anthropology, and the distinction matters. As much as the latter may yet come to exist, it will always be in escape velocity from the fall of its premises in the Greek *φύσις*, with damages already greater than ignorance itself accruing in any approach that does not register the escape. The escape force at issue is, on the U-Mentalism reading, the most important propulsion of human destiny — above any interstellar space exploration, since what is propelled is the substrate of thought itself rather than any physical displacement.

The link from the BMI configuration to this wounding of autonomy is, it is worth noting, not a theoretical inference from any particular theory of mind but a practical and prudential one. Across the line of consciousness theses — physicalist, quantum-assenting physicalist (Penrose 1989; Hameroff & Penrose 1996), or post-quantum physicalist of whatever character — the mind-in-brain may, in every case, stand in supervenience to certain brain–machine interface couplings, and its fragility under predictive readout increases jointly with the strength of the brain-mind correlation (which contemporary neuroscience already supports to a substantial degree) and the strength of the brain–machine correlation that the BMI itself enables. The practical risk therefore obtains regardless of how the theoretical question of consciousness is ultimately resolved, and the caution does not require any one consciousness thesis to be true in order to bind. The U.M. Assembly Language is to be specified, in its forthcoming autonomous treatment, with explicit non-derivation rights for mental-state inferences from neural-substrate frames: the prohibition is a structural commitment of the programme, not a contingent regulatory choice.

The conjectural register. These observations are conjectural and belong, in the registered terminology of §§11.5 and 12.7, to the U-Mentalism conjectural-research horizon rather than to its formal results. They are recorded because the present paper has, in §§9.3, 10.6, 10.7, 12.7, 12.8, accumulated sufficient structural material to motivate them at minimal extra cost, and because their development is the natural continuation of the present foundational work in subsequent papers of the U-Mentalism programme.

12.10. Photonic implementation at the colour-native threshold: the substrate–alphabet alignment in engineering instantiation

Section 12.4 recorded that the 8-bit-per-channel alphabet of §1.5 is "not merely conventional but thermodynamically and quantum-mechanically appropriate for the substrate that most naturally realises it". The claim was formulated as a substrate-matches-alphabet principle, with the engineering realisation projected asymptotically into a photonic horizon whose components were maturing but not yet aligned. In the 2024–2026 window the projection has acquired device-level instantiation. The structural register of the gap between the universal machine and its physical realisations (§12.6) is, in consequence, no longer that of a deferred engineering target but that of an integration problem whose components are available and whose composition is what remains.

The threshold crossing. A non-volatile photonic memory cell, integrated with a silicon waveguide, has been demonstrated to discriminate approximately 222 optical states in a single cell, using nitrogen-doped $Ge_2Sb_2Te_5$ as the phase-change medium (Xia et al. 2024). The structural pertinence of this figure is not the particular material composition but the level count: ~ 222 states is structurally above the previous decade's photonic phase-change ceiling at ~ 8 levels (Ríos et al. 2015, where the 13.4 pJ switching energy is the benchmark below) and structurally within the 256 of Σ_{RGB}. A photonic cell holding one Σ_{RGB} channel value natively, in non-volatile form, is the engineering realisation of the per-channel symbol of §1.5—no binary encoding, no analog approximation. Three such cells in parallel hold one Σ_{RGB} pixel; arrays compose to frames without alphabet inversion at any stage. The von Neumann memory–processing separation addressed structurally in §12.8 dissolves at the cellular level once the substrate carries the symbol natively.

The proto-universal substrate. Independently of the memory cell, the integrated-photonic accelerator class has reached production-relevant scale, with photonic processors demonstrated to execute convolutional, transformer, and reinforcement-learning workloads at accuracies approaching 32-bit floating-point digital precision (Ahmed et al. 2025), and with integrated large-scale photonic accelerators demonstrated at ultra-low latency (Hua et al. 2025). These devices are not universal Turing machines in the sense of §§6–7; they are inference-acceleration substrates whose precision floor of 4–8 bits per channel happens to be structurally compatible with Σ_{RGB}. They establish the engineering substrate within which a colour-native universal machine of the kind constructed in §§6–7 is hostable, with the universality contribution residing in the integration architecture and the head-and-tape topology of §7.6 rather than in the components themselves.

Sensing-side alignment. On the readout side, 8-bit CMOS imagers have discriminated 256 values per RGB channel as factory baseline for two decades, with 10-bit and 14-bit sensors exceeding this by factors of 4 to 64 and the hyperspectral arrays of §9.2 discriminating hundreds to thousands of bands. Sensing-side Σ_{RGB} discrimination is not a frontier but a commodity baseline; the engineering problem is on the processing-and-memory side, where the integration of Σ_{RGB}-native memory and Σ_{RGB}-native logic is the open architectural commitment.

Solar-photovoltaic synergy. A specific synergy with the renewable-energy infrastructure deserves brief notation. Photovoltaic panels are photonic devices: they capture incident photons and convert them to electrical current, with a per-pixel structure that is, formally, a photodetector array. The same physical apparatus that powers a photonic-substrate U.M. machine through solar capture is, with minimal architectural modification, an input device for U.M. recollection: each photovoltaic cell can be read both as a power-generation site and as an image-capture site, with the spatial array acting as an extended-resolution image sensor whose primary function is energy generation and whose secondary function is colour-synthesis read-in. Solar energy is the ultimate source from which all available energy derives, and a substrate that uses solar capture for both power and image input minimises the energy-conversion stack between the world's photonic input and the machine's photonic computation, on the Ockham-minimum reading. The co-location of energy-generation and image-recollection in a single photovoltaic-photonic device places U.M. on a register not available to architectures that separate power from input. The next-generation solar capture systems entering deployment in the 2026 horizon make this co-location an emerging engineering opportunity rather than a speculative one, without prejudice to the use of other energy sources in deployments where solar capture is not available.

Cost–benefit at the substrate level. The following table summarises the per-symbol energy, the alphabet–substrate alignment, and the memory–processing topology of the principal substrate classes at the 2026 horizon. The figures are intended as orders of magnitude that situate the colour-native target within its landscape, not as precise comparisons: the per-operation costs are dominated by different operations in different substrates (gate-switch in CMOS, modulator drive in photonic-sensorised binary, write energy in phase-change cells, gate throughput in coherent photonic logic), and the per-symbol normalisation should be read against §12.8's observation that bus-and-fetch energies, which dominate von Neumann CMOS, are absent in the colour-native photonic register and accordingly do not enter the per-symbol totals there.

Substrate class	Σ_{RGB}-native?	Energy/symbol	Mem.–proc. topology	Maturity (2026)
CMOS binary	no (24× encoding)	$\sim 10^{-12}$ J	separated (von Neumann)	deployed
Photonic-sensorised binary (App. G)	I/O only	$\sim 10^{-13}$ J	separated	constructible
Phase-change photonic memory	per channel	$\sim 10^{-11}$ J (write)	inseparable in cell	demonstrated 2024
Universal colour-native photonic	yes (§1.5)	$\sim 10^{-14}$ J	inseparable (§12.8)	partial, integration open
Hybrid colour-native + photonic-quantum	yes + qubit	substrate-dependent	inseparable (§§7.6, 9.3)	research

The colour-native photonic row is the engineering target of §§1.5, 6–7, 12.4, and 12.8; the phase-change photonic memory row is the persistent-memory layer that the patent (Homem 2021) places at the foundation of cinematic supercomputation; the photonic-sensorised binary row is the intermediate transition platform of Appendix G; the hybrid row anchors the quantum extension of §9.3 and is the substrate on which the Fourth-instance continuous-monitoring register becomes natively expressible.

The remaining integration problem. What is absent from the 2024–2026 state-of-the-art is the composition of the components into a single coherent U-Mentalism universal machine. Each register—colour-native photonic memory at Σ_{RGB} depth; universal photonic logic operating natively at Σ_{RGB}; high-fidelity detection at the readout layer; photonic-quantum interfacing for the continuous-monitoring register of §9.3 (Fourth instance); the recollection-as-substrate persistence layer of §§10.3 and Appendix F—exists in laboratory or production form independently. None is composed with the others at the level the present paper and the patent (Homem 2021) jointly specify. The composition is not at the frontier of physics but at the frontier of engineering integration: it requires no new discovery, only the architectural commitment to the colour-native alphabet, to the multi-pointer head over a 6-direction lattice with inseparable processing-and-memory and inseparable sensors-and-emitters of §7.6, and to the ordinally indexed recollection of §§1.5, 10.3, 12.4, 12.8 as the persistent substrate.

Deferral to planned engineering documentation of the programme. The full engineering analysis is reserved to the planned engineering documentation of the U-Mentalism programme, of which the present subsection is the foundational anchor in the same sense in which §12.9 anchors the planned Assembly Language. Such questions include, among others: the plausibility of co-packaging colour-native photonic memory with universal photonic logic and high-fidelity detection on a single die or chiplet-partitioned platform; the comparative analysis of silicon photonics, thin-film lithium niobate, and silicon-nitride substrates for the colour-native register; the thermal and energetic budgets at the integration scale; the area-versus-frame-rate scaling; and the classical–quantum substrate coupling. A more complete treatment would equally need to cover architectural dimensions beyond integration and substrate: distributed-systems networking under the UTM-over-Internet topology of §F.2; memory hierarchy beyond the cell-level register; software-stack design for the colour-native ISA; and foundry supply-chain and manufacturability. The structural observation the present subsection registers is the threshold crossing itself — the only one the foundational ontological register of the present paper can carry without genre-displacement. The substrate has met the alphabet. The engineering distance to the universal machine of §§6–7 is the distance of integration rather than of discovery. The gap of §12.6 has, at the 2024–2026 horizon, the shape that an engineering programme of composition can close.

12.11. Throughput at the 2026 horizon: the supercomputation-class reservation

The cost-benefit table of §12.10 is per-symbol; the aggregate-throughput calculation that situates the integrated U.M. machine within the supercomputation class is the natural complement. The exercise has the structure of the patent's own throughput estimate (Homem 2021, pp. 60–71) updated to the 2026 photonic substrate parameters of §12.10 and explicitly framed as order-of-magnitude rather than precise. The intent is to show that the integration of the components of §12.10 places the resulting machine, on conservative parameters, in the upper exascale and lower zettascale of contemporary supercomputation classifications, and to mark this as the operative reservation for the U-Mentalism programme's engineering register.

Per-pixel and per-frame throughput. A photonic phase-change memory cell of the Xia et al. (2024) class writes a Σ_{RGB} channel value in ~ 10 pJ at ~ 1 GHz; three such cells in parallel hold one Σ_{RGB} pixel and write at ~ 30 pJ per pixel-write at the same rate. A coherent photonic logic gate of the Lightmatter-class (Ahmed et al. 2025) operates at ~ 1 fJ per gate at sub-nanosecond latency, with 4–8 bits per channel of precision, the lower end of which is structurally compatible with Σ_{RGB}. At 8K Ultra HD resolution ($7680 \times 4320 = 33\,177\,600$ pixels per frame) and the patent's conservative 60 frames per second, a single die operating at these per-pixel costs produces, in the colour-native register, approximately 2×10^9 pixel-operations per second per die for the write-side and approximately 3×10^{16} pixel-operations per second per die for the gate-side, with the asymmetry reflecting the difference between persistent-write and ephemeral-compute. The aggregate per-die throughput of a fully colour-native machine is bounded by the slower of the two and is, in 2026, dominated by the memory-write rate.

Network-aggregate throughput. Under the patent's distributed UTM-over-M architecture (§§F.2, 12.4), a network of N photonic-die nodes operates at $N\times$ the per-die rate, with the inter-node coordination amortised over the persistent-recollection layer. Taking N in the range

10^3–10^5 (a deployable cluster scale in 2026) yields aggregate write-rates of 2×10^{12}–2×10^{14} pixel-writes per second, corresponding to colour-native operational throughputs in the upper exascale (10^{18} binary-equivalent operations per second) when the 24-bit-per-pixel encoding is unpacked into the conventional binary metric. For comparison, the leading classical supercomputers at the 2025 horizon (El Capitan, ~ 2.7 exaflops HPL; Frontier, ~ 1.4 exaflops HPL; Aurora, ~ 1 exaflops HPL) operate in the same order of magnitude on FP64 metrics but on a substrate ($\sim 10^{-13}$–10^{-12} J per operation, CMOS) two to three orders of magnitude above the photonic per-gate floor. The colour-native photonic machine of §12.10, operating at $\sim 10^{-14}$ J per pixel-operation in the gate-side regime, accordingly has the substrate headroom to match present supercomputation-class throughput on a small fraction of the energy budget — the residual gap being engineering-integration rather than substrate-thermodynamic.

Reservation in the supercomputation class. The figures above are sufficient to register the U.M. machine of §12.10 in the upper exascale of the 2026 supercomputation landscape on conservative parameters and in the lower zettascale on the patent's higher-end estimates. The reservation is qualitative because the cost-equivalence between colour-native operations and FP64 operations is not a single conversion factor but a substrate- and workload-dependent ratio; the quantitative analysis is reserved for the planned engineering research of §§12.9–12.10. The structural observation is sufficient for the present register: the photonic U.M. machine, on the 2024–2026 substrate parameters of §12.10 and in any deployable cluster scale, belongs to the supercomputation class. The patent's claim that the architecture realises "computation at or near the speed of light" (Homem 2021, title) is, at the substrate level, supportable. Three points anchor this reservation: *first*, the RGB colour code is the operative low-level machine language at the ISA level rather than an abstract or symbolic layer above it, with instructions executing over colour values themselves at the hardware–software interface; *second*, the FPS-organisation is the physical principle of the architecture, with the pixels of each frame read orthogonally and synchronously at the photonic substrate through waveguides, modulators, photodetectors, photoemitters, and optical switches and gates; *third*, the architecture stands beyond conventional all-optical computing, which transports pulsed binary at the speed of light but retains the binary intermediary that U.M. 'C' eliminates by operating natively in the colour-native alphabet at every stage — which is the substrate-level sense in which the speed-of-light claim is to be read.

Every image stored, transmitted, displayed, or generated in any digital medium today is a computable image in the strict sense of Definition 2.1. Every frame of every digital film is a computable film. The boundary between the visible and the inaccessible-to-any-procedure coincides, modulo idealised limits, with the boundary established by Turing in 1936. What lies beyond is a mathematical territory with no inhabitants and no procedure of approach. The visible universe of images is the universe of computable image schemes: bounded by the diagonal argument of §8, produced by the universal machine constructed in §§6–7, and engineered into being by the gradual closing of the gap mapped in §12. To have established this is the contribution of the present paper. The work it makes possible is the work that follows.

References

Aaronson, S. & Arkhipov, A. (2011). The computational complexity of linear optics. *Proceedings of the 43rd Annual ACM Symposium on Theory of Computing*, 333–342.

Abramsky, S. & Brandenburger, A. (2011). The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics* 13, 113036.

Aharonov, D. & Ben-Or, M. (1997). Fault-tolerant quantum computation with constant error. *Proceedings of the 29th Annual ACM Symposium on Theory of Computing*, 176–188.

Ahmed, S. R. et al. (2025). Universal photonic artificial intelligence acceleration. *Nature* 640(8058), 368–374.

Arendt, H. (1958). *The Human Condition.* University of Chicago Press.

Ashby, W. R. (1956). *An Introduction to Cybernetics.* London: Chapman & Hall.

Barto, A. G. & Sutton, R. S. (2024). Reinforcement learning: an introduction (2nd edition, 2018). ACM A.M. Turing Award.

Bekenstein, J. D. (1981). Universal upper bound on the entropy-to-energy ratio for bounded systems. *Physical Review D* 23(2), 287–298.

Bell, J. S. (1964). On the Einstein Podolsky Rosen paradox. *Physics* 1(3), 195–200.

Bell, J. S. (1966). On the problem of hidden variables in quantum mechanics. *Reviews of Modern Physics* 38(3), 447–452.

Bengio, Y., Hinton, G. E. & LeCun, Y. (2018). Deep learning. ACM A.M. Turing Award. *Communications of the ACM* 64(7) (2021), 58–65.

Bennett, C. H. (1973). Logical reversibility of computation. *IBM Journal of Research and Development* 17(6), 525–532.

Bennett, C. H. & Brassard, G. (1984). Quantum cryptography: public key distribution and coin tossing. *Proceedings of the IEEE International Conference on Computers, Systems and Signal Processing*, Bangalore, 175–179. (2025 ACM A.M. Turing Award.)

Berlekamp, E. R., Conway, J. H. & Guy, R. K. (1982). *Winning Ways for Your Mathematical Plays*, Vol. 2. Academic Press.

Berners-Lee, T. (2016). The World Wide Web: a global information space. ACM A.M. Turing Award.

Bernstein, E. & Vazirani, U. (1997). Quantum complexity theory. *SIAM Journal on Computing* 26(5), 1411–1473.

Bishop, E. (1967). *Foundations of Constructive Analysis.* McGraw-Hill.

Blum, L., Shub, M. & Smale, S. (1989). On a theory of computation and complexity over the real numbers. *Bulletin of the American Mathematical Society* 21(1), 1–46.

Blum, M. (1995). On the structure of complexity. ACM A.M. Turing Award.

Bohm, D. (1952). A suggested interpretation of the quantum theory in terms of "hidden" variables, I and II. *Physical Review* 85(2), 166–179 and 180–193.

Bostrom, N. (2014). *Superintelligence: Paths, Dangers, Strategies.* Oxford University Press.

Boyd, R. W. (2020). *Nonlinear Optics*, 4th edition. Academic Press.

Brassard, G., Høyer, P., Mosca, M. & Tapp, A. (2002). Quantum amplitude amplification and estimation. *Contemporary Mathematics* 305, 53–74.

Brattka, V. (2000). The emperor's new recursiveness: The epigraph of the exponential function in two models of computability. *Words, Languages & Combinatorics III*, World Scientific, 63–72.

Braverman, M. & Yampolsky, M. (2006). Non-computable Julia sets. *Journal of the American Mathematical Society* 19(3), 551–578.

Busy Beaver Challenge Collaboration (2024). The fifth Busy Beaver number is 47 176 870: collaborative formal verification. `bbchallenge.org`.

Bremermann, H. J. (1962). Optimization through evolution and recombination. In *Self-Organizing Systems*, Spartan Books, 93–106.

Catmull, E. & Hanrahan, P. (2019). The RenderMan interface and the rendering lineage. ACM A.M. Turing Award.

Chaitin, G. J. (1969). On the simplicity and speed of programs for computing infinite sets of natural numbers. *Journal of the ACM* 16(3), 407–422.

Clark, A. & Chalmers, D. (1998). The extended mind. *Analysis* 58(1), 7–19.

Cohen, P. J. (1963). The independence of the continuum hypothesis. *Proceedings of the National Academy of Sciences* 50(6), 1143–1148.

Clarke, E. M. & Emerson, E. A. (1981). Design and synthesis of synchronization skeletons using branching time temporal logic. *Logic of Programs*, Lecture Notes in Computer Science 131, 52–71.

Clements, W. R., Humphreys, P. C., Metcalf, B. J., Kolthammer, W. S. & Walmsley, I. A. (2016). Optimal design for universal multiport interferometers. *Optica* 3(12), 1460–1465.

Connes, A. (1976). Classification of injective factors. *Annals of Mathematics* 104(1), 73–115.

Crull, E. & Bacciagaluppi, G. (eds.) (2017). *Grete Hermann: Between Physics and Philosophy.* Springer.

de Broglie, L. (1927). La mécanique ondulatoire et la structure atomique de la matière et du rayonnement. *Journal de Physique et le Radium* 8(5), 225–241.

Deleuze, G. (1983). *Cinéma 1: L'Image-mouvement.* Paris: Éditions de Minuit. (Trans. *Cinema 1: The Movement-Image*, Athlone Press, 1986.)

Deleuze, G. (1985). *Cinéma 2: L'Image-temps.* Paris: Éditions de Minuit. (Trans. *Cinema 2: The Time-Image*, Athlone Press, 1989.)

De Filippi, P. & Wright, A. (2018). *Blockchain and the Law: The Rule of Code.* Harvard University Press.

Deutsch, D. & Marletto, C. (2014). Constructor theory of information. *Proceedings of the Royal Society A* 471(2174), 20140540.

Diffie, W. & Hellman, M. E. (1976). New directions in cryptography. *IEEE Transactions on Information Theory* 22(6), 644–654. (2015 ACM A.M. Turing Award.)

Dongarra, J. J. (2021). High-performance computing, linear algebra software, and the LINPACK benchmark. ACM A.M. Turing Award.

Eddington, A. S. (1928). *The Nature of the Physical World.* Cambridge University Press.

Einstein, A. (1915). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Preussischen Akademie der Wissenschaften zu Berlin*, 844–847.

Etesi, G. & Németi, I. (2002). Non-Turing computations via Malament–Hogarth space-times. *International Journal of Theoretical Physics* 41(2), 341–370.

Engelbart, D. C. (1962). *Augmenting Human Intellect: A Conceptual Framework.* Summary Report AFOSR-3223, Stanford Research Institute.

Feynman, R. P. (1948). Space-time approach to non-relativistic quantum mechanics. *Reviews of Modern Physics* 20(2), 367–387.

Feynman, R. P. & Hibbs, A. R. (1965). *Quantum Mechanics and Path Integrals.* McGraw-Hill.

Feynman, R. P. (1982). Simulating physics with computers. *International Journal of Theoretical Physics* 21(6/7), 467–488.

Feynman, R. P. (1996). *Feynman Lectures on Computation*, ed. A. J. G. Hey & R. W. Allen. Addison-Wesley.

Foley, J. D., van Dam, A., Feiner, S. K. & Hughes, J. F. (2013). *Computer Graphics: Principles and Practice*, 3rd edition. Addison-Wesley.

Gödel, K. (1930). Die Vollständigkeit der Axiome des logischen Funktionenkalküls. *Monatshefte für Mathematik und Physik* 37(1), 349–360.

Gödel, K. (1940). *The Consistency of the Continuum Hypothesis.* Annals of Mathematics Studies, Princeton University Press.

Goldwasser, S. & Micali, S. (1984). Probabilistic encryption. *Journal of Computer and System Sciences* 28(2), 270–299. (2012 ACM A.M. Turing Award.)

Goodfellow, I., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A. & Bengio, Y. (2014). Generative adversarial nets. *Advances in Neural Information Processing Systems* 27, 2672–2680.

Good, I. J. (1965). Speculations concerning the first ultraintelligent machine. *Advances in Computers* 6, 31–88.

Haraway, D. J. (1985). A cyborg manifesto: science, technology, and socialist-feminism in the late twentieth century. *Socialist Review* 80, 65–108.

Hameroff, S. & Penrose, R. (1996). Conscious events as orchestrated space-time selections. *Journal of Consciousness Studies* 3(1), 36–53.

Hawking, S. W. & Penrose, R. (1970). The singularities of gravitational collapse and cosmology. *Proceedings of the Royal Society A* 314(1519), 529–548.

Hartmanis, J. & Stearns, R. E. (1965). On the computational complexity of algorithms. *Transactions of the American Mathematical Society* 117, 285–306. (1993 ACM A.M. Turing Award.)

Hermann, G. (1926). Die Frage der endlich vielen Schritte in der Theorie der Polynomideale [The Question of Finitely Many Steps in the Theory of Polynomial Ideals]. *Mathematische Annalen* 95(1), 736–788. (English translation by M. Abramson available as Royal Institution of Great Britain technical report, 1998.)

Hermann, G. (1935). Die naturphilosophischen Grundlagen der Quantenmechanik [The Natural-Philosophical Foundations of Quantum Mechanics]. *Abhandlungen der Fries'schen Schule* 6, 75–152. (English translation in Crull & Bacciagaluppi 2017.)

Hertling, P. (2005). Is the Mandelbrot set computable? *Mathematical Logic Quarterly* 51(1), 5–18.

Hilbert, D. & Ackermann, W. (1928). *Grundzüge der theoretischen Logik*. Berlin: Springer.

Hironaka, H. (1964). Resolution of singularities of an algebraic variety over a field of characteristic zero, I and II. *Annals of Mathematics* 79(1), 109–203 and 79(2), 205–326.

Ho, J., Jain, A. & Abbeel, P. (2020). Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems* 33, 6840–6851.

Hofstadter, D. R. (1979). *Gödel, Escher, Bach: An Eternal Golden Braid*. Basic Books.

Hogarth, M. L. (1992). Does general relativity allow an observer to view an eternity in a finite time? *Foundations of Physics Letters* 5(2), 173–181.

Homem, L. (2018). U-Mentalism: a proposal for a new computer architecture. *Centro de Filosofia das Ciências da Universidade de Lisboa (CFCUL) working paper / earlier presentation of the programme.*

Homem, L. (2019). What is U-Mentalism? *Journal of Advances in Computer Networks* 7(1), 18–24.

Homem, L. (2021). U-Mentalism patent: The beginning of cinematic supercomputation. *International Journal of Web & Semantic Technology (IJWEST)* 12(1), 1–18. Patent reference PCT/PT2021/050014.

Homem, L. (2022). The U-Mentalism definition of the *image* and the constitution of *Philosophia Naturalis* and *Philosophia Artificialis. U-Mentalism programme working note / unpublished.*

Horowitz, M. (2014). Computing's energy problem (and what we can do about it). *Proceedings of the IEEE International Solid-State Circuits Conference (ISSCC)*, 10–14.

Hopcroft, J. E., Motwani, R. & Ullman, J. D. (2007). *Introduction to Automata Theory, Languages, and Computation*, 3rd edition. Pearson.

Hua, S. et al. (2025). An integrated large-scale photonic accelerator with ultralow latency. *Nature* 640, 361–367.

Ji, Z., Natarajan, A., Vidick, T., Wright, J. & Yuen, H. (2020). MIP* = RE. `arXiv:2001.04383`.

Jia, H., Yaghini, M., Choquette-Choo, C. A., Dullerud, N., Thudi, A., Chandrasekaran, V. & Papernot, N. (2021). Proof-of-learning: definitions and practice. *IEEE Symposium on Security and Privacy (S&P)*, 1039–1056.

Jung, J., Qin, L., Welleck, S., Brahman, F., Bhagavatula, C., Le Bras, R. & Choi, Y. (2022). Maieutic prompting: logically consistent reasoning with recursive explanations. *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 1266–1279.

Kajiya, J. T. (1986). The rendering equation. *SIGGRAPH Computer Graphics* 20(4), 143–150.

Kardashev, N. S. (1964). Transmission of information by extraterrestrial civilizations. *Soviet Astronomy* 8, 217–221.

Kasparov, G. (2017). *Deep Thinking: Where Machine Intelligence Ends and Human Creativity Begins.* PublicAffairs.

King, S. (2013). *Primecoin: cryptocurrency with prime number proof-of-work.* White paper, July 2013.

Kittler, F. A. (2010). *Optical Media: Berlin Lectures 1999.* Polity Press.

Knill, E., Laflamme, R. & Milburn, G. J. (2001). A scheme for efficient quantum computation with linear optics. *Nature* 409(6816), 46–52.

Knill, E., Laflamme, R. & Zurek, W. H. (1998). Resilient quantum computation. *Science* 279(5349), 342–345.

Knuth, D. E. (1984). Literate programming. *The Computer Journal* 27(2), 97–111.

Knuth, D. E. (1999). *MMIXware: A RISC Computer for the Third Millennium.* Lecture Notes in Computer Science 1750. Springer.

Knuth, D. E. (2005). *The Art of Computer Programming*, Volume 1, Fascicle 1: MMIX, A RISC Computer for the New Millennium. Addison-Wesley.

Knuth, D. E. (2011 and continuing). *The Art of Computer Programming*, Volume 4A: Combinatorial Algorithms, Part 1, with fascicles 0–6 of Volume 4B in continuing publication. Addison-Wesley.

Ko, K. (1991). *Complexity Theory of Real Functions.* Birkhäuser.

Kochen, S. & Specker, E. P. (1967). The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics* 17(1), 59–87.

Kolmogorov, A. N. (1965). Three approaches to the quantitative definition of information. *Problems of Information Transmission* 1(1), 1–7.

Kurzweil, R. (2005). *The Singularity Is Near: When Humans Transcend Biology.* Viking.

Lamport, L. (1978). Time, clocks, and the ordering of events in a distributed system. *Communications of the ACM* 21(7), 558–565. (2013 ACM A.M. Turing Award.)

Landauer, R. (1961). Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development* 5(3), 183–191.

Lebedev, M. A. & Nicolelis, M. A. L. (2017). Brain–machine interfaces: from basic science to neuroprostheses and neurorehabilitation. *Physiological Reviews* 97(2), 767–837.

Lessig, L. (1999, rev. ed. 2006). *Code, and Other Laws of Cyberspace* (revised as *Code: Version 2.0*). Basic Books.

Li, M. & Vitányi, P. (2019). *An Introduction to Kolmogorov Complexity and Its Applications*, 4th edition. Springer.

Licklider, J. C. R. (1960). Man-computer symbiosis. *IRE Transactions on Human Factors in Electronics* HFE-1(1), 4–11.

Linus, R. (2023). *BitVM: compute anything on Bitcoin.* White paper, October 2023.

Linus, R., Aumayr, L., Zamyatin, A., Pelosi, A., Avarikioti, Z. & Maffei, M. (2024). *BitVM2: Bridging Bitcoin to Second Layers.* White paper, August 2024.

Linus, R. et al. (2025). *BitVM3-RSA: Efficient Computation on Bitcoin via Garbled Circuits.* White paper, July 2025.

Lloyd, S. (2006). *Programming the Universe: A Quantum Computer Scientist Takes on the Cosmos.* Knopf.

Manovich, L. (2001). *The Language of New Media.* MIT Press.

Marr, D. (1982). *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information.* W. H. Freeman.

Mayr, E. W. & Meyer, A. R. (1982). The complexity of the word problems for commutative semigroups and polynomial ideals. *Advances in Mathematics* 46(3), 305–329.

Mermin, N. D. & Schack, R. (2018). Homer nodded: von Neumann's surprising oversight. *Foundations of Physics* 48(9), 1007–1020.

Mildenhall, B., Srinivasan, P. P., Tancik, M., Barron, J. T., Ramamoorthi, R. & Ng, R. (2020). NeRF: Representing scenes as neural radiance fields for view synthesis. *European Conference on Computer Vision (ECCV)*, 405–421.

Miller, D. A. B. (2010). Are optical transistors the logical next step? *Nature Photonics* 4(1), 3–5.

Mollick, E. (2024). *Co-Intelligence: Living and Working with AI.* Portfolio.

Müller, T., Evans, A., Schied, C. & Keller, A. (2022). Instant neural graphics primitives with a multiresolution hash encoding. *ACM Transactions on Graphics* 41(4), 102:1–102:15.

Nakamoto, S. (2008). *Bitcoin: a peer-to-peer electronic cash system.* White paper. Available at `https://bitcoin.org/bitcoin.pdf`.

Navascués, M., Pironio, S. & Acín, A. (2008). A convergent hierarchy of semidefinite programs characterizing the set of quantum correlations. *New Journal of Physics* 10, 073013.

Newell, A. & Simon, H. A. (1976). Computer science as empirical inquiry: symbols and search. *Communications of the ACM* 19(3), 113–126. (1975 ACM A.M. Turing Award.)

Ord, T. (2002). *Hypercomputation: Computing more than the Turing machine.* Technical report, University of Melbourne.

Patterson, D. A. & Hennessy, J. L. (2020). *Computer Organization and Design: The Hardware/-Software Interface*, 5th edition. Morgan Kaufmann.

Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*, 2nd edition. Cambridge University Press. (2011 ACM A.M. Turing Award.)

Penrose, R. (1965). Gravitational collapse and space-time singularities. *Physical Review Letters* 14(3), 57–59.

Penrose, R. (1969). Gravitational collapse: the role of general relativity. *Rivista del Nuovo Cimento, Numero Speciale* 1, 252–276.

Penrose, R. (1989). *The Emperor's New Mind: Concerning Computers, Minds, and the Laws of Physics.* Oxford University Press.

Pnueli, A. (1977). The temporal logic of programs. *Proceedings of the 18th Annual Symposium on Foundations of Computer Science*, 46–57.

Popper, K. R. (1945). *The Open Society and Its Enemies*, 2 vols. London: Routledge.

Pour-El, M. B. & Richards, J. I. (1989). *Computability in Analysis and Physics.* Springer.

Radó, T. (1962). On non-computable functions. *Bell System Technical Journal* 41(3), 877–884.

Reck, M., Zeilinger, A., Bernstein, H. J. & Bertani, P. (1994). Experimental realization of any discrete unitary operator. *Physical Review Letters* 73(1), 58–61.

Regev, O. (2009). On lattices, learning with errors, random linear codes, and cryptography. *Journal of the ACM* 56(6), 1–40.

Ríos, C., Stegmaier, M., Hosseini, P., Wang, D., Scherer, T., Wright, C. D., Bhaskaran, H. & Pernice, W. H. P. (2015). Integrated all-photonic non-volatile multi-level memory. *Nature Photonics* 9(11), 725–732.

Ritchie, D. M. & Thompson, K. (1974). The UNIX time-sharing system. *Communications of the ACM* 17(7), 365–375.

Rogers, H. (1967). *Theory of Recursive Functions and Effective Computability.* McGraw-Hill. (Reprint: MIT Press, 1987.)

Saleh, B. E. A. & Teich, M. C. (2019). *Fundamentals of Photonics*, 3rd edition. Wiley.

Shen, Y., Harris, N. C., Skirlo, S., Prabhu, M., Baehr-Jones, T., Hochberg, M., Sun, X., Zhao, S., Larochelle, H., Englund, D. & Soljačić, M. (2017). Deep learning with coherent nanophotonic circuits. *Nature Photonics* 11(7), 441–446.

Simondon, G. (1958). *Du mode d'existence des objets techniques.* Paris: Aubier.

Sitzmann, V., Martel, J. N. P., Bergman, A. W., Lindell, D. B. & Wetzstein, G. (2020). Implicit neural representations with periodic activation functions. *Advances in Neural Information Processing Systems* 33, 7462–7473.

Smith, A. (2007). Universality of Wolfram's 2-state, 3-symbol Turing machine. *Complex Systems* 20(1), 1–72.

Solomonoff, R. J. (1964). A formal theory of inductive inference, Parts I and II. *Information and Control* 7(1), 1–22; 7(2), 224–254.

Tegmark, M. (2008). The mathematical universe. *Foundations of Physics* 38(2), 101–150.

Toffoli, T. & Margolus, N. (1987). *Cellular Automata Machines: A New Environment for Modeling.* MIT Press.

Turing, A. M. (1936). On computable numbers, with an application to the *Entscheidungsproblem. Proceedings of the London Mathematical Society* (ser. 2) 42, 230–265.

Turing, A. M. (1939). Systems of logic based on ordinals. *Proceedings of the London Mathematical Society* (ser. 2) 45, 161–228.

U-Mentalism Programme. Online resources at `u-mentalism.com`, `github.com/U-Mentalism`, `u.mentalism.xyz`.

Valiant, L. G. (1984). A theory of the learnable. *Communications of the ACM* 27(11), 1134–1142. (2010 ACM A.M. Turing Award.)

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł. & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems* 30, 5998–6008.

von Neumann, J. (1932). *Mathematische Grundlagen der Quantenmechanik.* Berlin: Springer. (Trans. *Mathematical Foundations of Quantum Mechanics*, Princeton University Press, 1955.)

von Neumann, J. (1945). *First Draft of a Report on the EDVAC.* Moore School of Electrical Engineering, University of Pennsylvania.

Virilio, P. (2012). *The Great Accelerator.* Cambridge: Polity. (Original *Le grand accélérateur*, Paris: Galilée, 2010.) Late-period analysis of the speed–image complex and its bearing on the post-*Aufklärung* technological condition.

Vinge, V. (1993). The coming technological singularity: how to survive in the post-human era. *Vision-21 Symposium*, NASA Lewis Research Center and Ohio Aerospace Institute, 11–22.

Voltaire (1763). *Treatise on Tolerance, on the Occasion of the Death of Jean Calas.* Geneva (originally published as *Traité sur la tolérance*). English editions: Cambridge University Press, 2000; Penguin Classics, 2016.

Weihrauch, K. (2000). *Computable Analysis: An Introduction.* Springer.

Wetzstein, G., Ozcan, A., Gigan, S., Fan, S., Englund, D., Soljačić, M., Denz, C., Miller, D. A. B. & Psaltis, D. (2020). Inference in artificial intelligence with deep optics and photonics. *Nature* 588, 39–47.

Wheeler, J. A. (1990). Information, physics, quantum: the search for links. In W. Zurek (ed.), *Complexity, Entropy and the Physics of Information*, Addison-Wesley, 3–28.

Whitted, T. (1980). An improved illumination model for shaded display. *Communications of the ACM* 23(6), 343–349.

Wiener, N. (1948). *Cybernetics: Or Control and Communication in the Animal and the Machine.* MIT Press.

Wiseman, H. M. & Milburn, G. J. (2010). *Quantum Measurement and Control.* Cambridge University Press.

Wolfram, S. (2002). *A New Kind of Science.* Wolfram Media.

Wright, A. & De Filippi, P. (2015). Decentralized blockchain technology and the rise of *Lex Cryptographia. SSRN Electronic Journal.* Available at `https://ssrn.com/abstract=2580664`.

Xia, J., Wang, T., Wang, Z., Gong, J., Dong, Y., Yang, R. & Miao, X. (2024). Seven bit nonvolatile electrically programmable photonics based on phase-change materials for image recognition. *ACS Photonics* 11(2), 723–730.

Yao, A. C. (1986). How to generate and exchange secrets. *Proceedings of the 27th Annual Symposium on Foundations of Computer Science (FOCS)*, 162–167. (2000 ACM A.M. Turing Award.)

Zworykin, V. K. (1923–1925, granted 1928 and subsequent years). Television-system patent filings, notably US Patent 1,773,980 (Television System) and US Patent 2,021,907 (The Iconoscope), with the conceptual content elaborated in V. K. Zworykin (1933), "The iconoscope: a modern version of the electric eye", *Proceedings of the IRE* 22(1), 16–32.

A Structural correspondence to Turing 1936

This appendix provides the section-by-section comparison of the present paper with Turing's 1936 paper. The introduction gave the basic mapping in tabular form; the present appendix supplies the substantive commentary, identifies the type of correspondence in each case, and notes the points at which the present paper departs from the original. Throughout, our characterisations of Turing's section content are descriptions in our own words.

A.1. Section-by-section comparison

Turing §1 — the abstract computing machine. Introduces the abstract computing machine: a finite-state device on a one-dimensional tape, with conventions for tape contents, head movement, and transition specification. Our §1 reproduces this in contemporary notation. *Type: direct.*

Our §1.5 — the colour-native machine (no Turing antecedent). Fixes the alphabet as Σ_{RGB}, states Proposition 1, and motivates the choice on grounds of cost model, physical substrate, and algorithmic register. *Type: extension.*

Turing §2 — definitions of the computable. Makes precise the central object: the computable real number. Our §2 makes precise our central objects: computable image and film schemes. The discrete-primary structure is the substantive departure: Turing's computable reals are intrinsically infinite; our computable images and films are finite at every fixed resolution. *Type: adapted.*

Turing §3 — worked examples of machines. Gives worked examples of computing machines. Our §3 gives a graded sequence of image-scheme examples. The Mandelbrot example carries philosophical weight that has no exact Turing analogue. *Type: adapted.*

Turing §4 — the m-function abbreviation layer. Introduces m-functions for compositional construction. Our §4 introduces the modern compositional layer: pointwise, local, geometric, compositing, and temporal operators. *Type: adapted.*

Turing §5 — description numbers and enumeration. Introduces the Gödel encoding and shows that the computable sequences form a countable class. Our §5 reproduces this for colour-native machines and adds Kolmogorov complexity. *Type: direct on enumeration; extension on Kolmogorov.*

Turing §6 — the universal computing machine. Establishes the existence of a single machine that simulates any other given its description. Our §6 establishes the analogous result for image and film generation (Theorem 6.1, Corollary 6.2). *Type: direct.*

Turing §7 — detailed construction of $\mathfrak{U}$. The most technically dense section: explicit construction of the universal machine. Our §7 mirrors this commitment with five named subroutines, an explicit main-loop transition table, and a state count, with full subroutine tables in Appendix C. *Type: direct in structure, parallel in detail.*

Turing §8 — the diagonal argument. Applies the diagonal process and derives the undecidability of the halting problem. Our §8 follows the same structure: diagonal scheme (Definition 8.1), Lemma 8.2, Theorem 8.3, and corollaries. *Type: direct.*

Turing §9 — the extent of the computable. Engages with what is now called the Church–Turing thesis. Our §9 preserves the structural intent but does so with material developed in the intervening 90 years (TTE, Bernstein–Vazirani, Brattka). *Type: structural correspondence with substantial extension.*

Turing §10 — large classes of computable numbers. Demonstrates that algebraic numbers, π, e, etc. are computable. Our §10 demonstrates the analogous claim for images, with Big-O cost stratification. *Type: structural correspondence with substantial extension.*

Turing §11 — application to the Entscheidungsproblem. Derives the unsolvability of Hilbert's Entscheidungsproblem. Our §11 follows this structure precisely (§§11.1–11.4), and adds §11.5—the visual Entscheidungsproblem hypothesis with three open questions. *Type: direct in §§11.1–11.4; extension in §11.5.*

Our §12 — physical limits (no Turing antecedent). Maps the gap between abstract universality and physical realisability under thermodynamic, information-theoretic, decoherence, and analog-precision bounds. *Type: extension.*

Turing's appendix — equivalence with effective calculability. Establishes coincidence of his computability with λ-definability and general recursiveness. Our appendices supply analogous structural-completion content: Appendix E verifies the encoding correctness for Theorem 11.1; Appendix C supplies the explicit transition tables; Appendix A is the structural mapping itself. *Type: adapted role.*

A.2. Summary by correspondence type

Direct correspondences, where the original argument transposes with only notational modernisation: §§1, 6, 7, 8, and §§11.1–11.4.

Adapted correspondences, where structural role is preserved but content is reworked: §§2, 3, 4.

Extensions, incorporating substantial post-1936 material: §§5 (Kolmogorov), 9 (TTE, BQP, photonic), 10 (cost stratification, neural networks), 11.5 (visual *Entscheidungsproblem* hypothesis).

New sections without antecedent: §1.5 (colour-native machine) and §12 (physical limits).

A.3. The argumentative arc

Turing's 1936 paper has a five-beat structure: define the machine; define computability; show the universal machine exists; derive its undecidability; apply the result to the *Entscheidungsproblem*. Our paper preserves this five-beat structure: §§1–2, §§6–7, §8, §11. Sections 3–5 supply examples, enumeration, and the description-number apparatus. Sections 9, 10, and 12 extend the arc beyond its 1936 endpoint.

A.4. Background reading and adjacent literature

For readers wishing to extend their engagement with the literature beyond the citations of the body, we identify several adjacent lines of inquiry whose substantive incorporation is left to subsequent work in the U-Mentalism programme.

On the complexity of real-valued computation. Ko (1991), *Complexity Theory of Real Functions*, develops the complexity counterpart to the computability theory of Pour-El & Richards

(1989) and Weihrauch (2000) on which we draw in §9.2. The cost stratification of §10 and Appendix D is, in formal terms, a specialisation of Ko's framework to the visual case. Bishop's (1967) *Foundations of Constructive Analysis* is the older constructive-mathematics tradition that anticipates much of Type-Two Effectivity and supplies a foundational vocabulary independent of the recursive-function-theoretic route.

On hypercomputation. Ord (2002), *Hypercomputation*, surveys the models that purport to compute beyond the Turing limit. Brattka's (2000) result, on which we rely in §9.4, is the strongest single constraint on physically-realisable hypercomputation under realistic noise; the broader literature (Copeland, Stannett, Calude and others) maps the territory in which the Brattka bound operates.

On cellular automata as visual computation. Conway's Game of Life and Wolfram's (2002) *A New Kind of Science* develop cellular automata as a paradigm for *computable films* in a sense parallel but distinct from Definition 2.3 (Conway's frame-to-frame transition is a fixed local rule rather than a coordinate-indexed scheme). Wolfram's principle of computational equivalence is a strong claim about the universality of computational systems that, if it can be substantiated, would situate cellular-automaton-based films inside the class of §2 by an independent route. Toffoli & Margolus's (1987) reversible cellular automata, in addition, cross with Bennett's (1973) reversible computation in ways relevant to §12.1.

On self-reference and the conjectural register. Hofstadter's (1979) *Gödel, Escher, Bach* develops the strange-loop framing of self-reference in computation; the reflexive observation of the methodological note has its conceptual home in this tradition. Penrose's (1989) *The Emperor's New Mind* argues, on Gödelian and physical grounds, that consciousness exceeds the computable. The U-Mentalism programme embraces Penrose's framing positively rather than as a foil. The defence of U-Mentalism is openly conjectural and roots itself in the conviction that mind and brain are the only point in the known universe where the synthesising power capable of constructing the schemes that contemplate *all possible images* — including the images of mind and brain themselves and of the ontology that serves them — is actually exercised. The U.M. 'O' framework (Homem 2018, 2022; see Abstract) makes this commitment explicit: it locates the constitution of the image in the percipient mind/brain as the unique seat of the synthesis-power that no other known substrate exhibits, and U.M. 'C' is the engineering attempt to instantiate, in finite hardware, the finite shadow of this constitutive capacity. The two programmes diverge on the specific physical mechanism Penrose proposes (orchestrated reduction in microtubules) but agree on the broader and more important point: that the mind/brain substrate is the appropriate locus for an account of perceptual phenomena that the formal computational class does not exhaust, and that this account is necessarily conjectural in the present state of knowledge. The visual *Entscheidungsproblem* of §11.5 is a working hypothesis in this register: a claim whose substantive content is the research direction it commits to, with the eventual confirmation or refutation as the deliverable. The U-Mentalism programme is, in this respect, in serious scientific company with Penrose's, and the dissimilarity of conclusions on specific mechanisms does not diminish the underlying agreement on the kind of inquiry both undertake.

On the human–machine collaboration register. The methodological note of the title page identifies the human contributor's role as that of *oracle* (in the technical Turing 1939 sense) and

Chiron (the wise tutor-centaur). The latter term, less standardised than the former, deserves a brief literature anchor.[1] The relevance of this anchoring to the present paper is structural rather than ornamental: the Chiron role names the specification function whose persistence under post-Centaur conditions is what makes the maieutic composition of a paper of this kind — mathematical content directed by human intention — a coherent epistemological act. A near-metaphor available within the U-Mentalism programme should be recorded here. The kybernetes-pilot's diminution of randomness in U.M. 'C' via human-directed specification stands to the engineering substrate in a relation structurally analogous to the role of the *pilot wave* (de Broglie 1927; Bohm 1952) in the surviving non-local hidden-variable interpretation of quantum mechanics (cf. §9.3 and Theorem 9.3.1): in both cases, apparent randomness at the level of outputs is undergirded by a deterministic guidance structure at a finer level — the wave function on the physical side, the human specification function on the computational side. An important engineering asymmetry, however, distinguishes the two sides of the analogy: the human specification function is, in the engineering register, externally observable — recorded in documents, code, and architectural choices — whereas the Bohmian pilot wave is, by construction, internally hidden to measurement. The metaphor preserves the structural role of guidance; it does not, and cannot, transplant the hidden-variable ontology into the engineering substrate of U.M. 'C', which by its operational nature works only with observables (cf. Proposition 9.3.2 for the cryptographic operationalisation of this distinction). The kybernetes is, in this metaphorical transplant to U.M. 'C', the pilot-wave of the cinematic supercomputation substrate. The metaphor is recorded as conceptually pertinent and conjectural rather than as formal equivalence; its substantive working-out belongs to subsequent papers in the U-Mentalism programme.

On media theory and digital culture. Beyond Kittler (2010) cited in §10.5, Manovich's (2001) *The Language of New Media* and Simondon's (1958) *Du mode d'existence des objets techniques* treat the relation between digital media and computation in cultural-philosophical registers complementary to the formal treatment of the present paper. These literatures would supply philosophical fundament for the wider claims of the U-Mentalism programme without disturbing the technical content of §§1–12.

B Full ray-tracer construction for Example 3.6

This appendix completes Example 3.6 by giving an explicit construction of the rendering pipeline. The construction is at the level of structured pseudocode, with formal verification that each component is a computable function and that their composition is therefore a computable image

[1]Among technical and philosophical cognates of the Chiron register: *Man-Computer Symbiosis* (Licklider 1960), the founding text of the anti-substitutional paradigm in computer science; *Intelligence Amplification* (Ashby 1956; Engelbart 1962, *Augmenting Human Intellect*), the IA-versus-AI tradition; the typology of *Oracle / Genie / Sovereign / Tool* AI in Bostrom (2014, *Superintelligence*), where the present collaboration is strictly oracular; the *maieutic prompting* of Jung et al. (2022), which formalises the Socratic dialogue model in the contemporary LLM setting and supplies a precise technical referent for the term used in the methodological note; and Mollick (2024, *Co-Intelligence*) as the current vernacular term. In philosophical-cultural register, the *pharmakon* of Stiegler (after Derrida), the *hypomnemata* of Foucault, the *extended mind* of Clark & Chalmers (1998), and the *cyborg* of Haraway (1985) each frame the human–technical relation in registers complementary to the present paper's mathematical treatment without coinciding with it. The choice of "Chiron" over "Centaur" tracks the post-2017 obsolescence of the Kasparov-style centaur in pure-execution domains (Kasparov 2017, *Deep Thinking*): in domains where the goal itself must be specified by human intention.

scheme.

B.1. Notational preliminaries

A ray is a pair $\mathbf{r} = (\mathbf{o}, \mathbf{d})$ with origin $\mathbf{o} \in \mathbb{R}^3$ and unit-norm direction $\mathbf{d} \in \mathbb{R}^3$. The point at parameter t along the ray is $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$. The colour-native quantisation operator $\text{quantise}_8 : \mathbb{R}^3 \to \Sigma_{\text{RGB}}$ takes a triple of computable reals in $[0, 1]$ and produces $(\lfloor 255r \rfloor, \lfloor 255g \rfloor, \lfloor 255b \rfloor)$.

B.2. Scene description language

A scene is a finite tuple $\mathcal{S} = (\mathcal{G}, \mathcal{L}, \mathcal{C}, \mathbf{c}_{\text{bg}}, d_{\max}, \varepsilon_t)$, where $\mathcal{G} = \{G_1, \ldots, G_K\}$ is a finite list of geometric primitives, $\mathcal{L} = \{L_1, \ldots, L_M\}$ is a finite list of light sources, $\mathcal{C}$ is a camera specification, $\mathbf{c}_{\text{bg}}$ is the background colour, $d_{\max} \in \mathbb{N}$ is the maximum recursion depth, and $\varepsilon_t > 0$ is the minimum-distance epsilon.

Each primitive $G_i = (\tau_i, \boldsymbol{\theta}_i, \rho_i)$ has a type $\tau_i \in \{\mathsf{sphere}, \mathsf{plane}, \mathsf{triangle}, \mathsf{box}, \mathsf{generic}\}$, a parameter vector $\boldsymbol{\theta}_i$, and a material ρ_i. Each light $L_j = (\lambda_j, \mathbf{p}_j, \mathbf{I}_j)$ has a type, position (or direction), and intensity. The camera is $\mathcal{C} = (\mathbf{e}, \hat{\mathbf{f}}, \hat{\mathbf{u}}, \alpha)$: eye position, forward, up, horizontal field of view. We assume all numerical components are computable reals; $d_{\max}$ may depend on n via a computable function $d_{\max}(n)$; the BRDF associated with each material is a computable function.

B.3. Ray-primitive intersection

For each primitive type, the intersection routine $\text{Hit}_\tau(\mathbf{r}, \boldsymbol{\theta}; \varepsilon_t)$ returns either $\bot$ or a tuple $(t, \mathbf{p}, \mathbf{n})$.

Sphere. $\boldsymbol{\theta} = (\mathbf{c}, r)$. Solve $|\mathbf{o} + t\mathbf{d} - \mathbf{c}|^2 = r^2$. The discriminant is $\Delta = ((\mathbf{c} - \mathbf{o}) \cdot \mathbf{d})^2 - (|\mathbf{c} - \mathbf{o}|^2 - r^2)$. If $\Delta < 0$, return $\bot$; otherwise the two roots are $t_\pm = (\mathbf{c} - \mathbf{o}) \cdot \mathbf{d} \pm \sqrt{\Delta}$; choose the smallest $t > \varepsilon_t$. Hit point $\mathbf{p} = \mathbf{o} + t\mathbf{d}$, outward normal $\mathbf{n} = (\mathbf{p} - \mathbf{c})/r$.

Triangle. Möller–Trumbore: let $\mathbf{e}_1 = \mathbf{v}_1 - \mathbf{v}_0$, $\mathbf{e}_2 = \mathbf{v}_2 - \mathbf{v}_0$, $\mathbf{h} = \mathbf{d} \times \mathbf{e}_2$, $a = \mathbf{e}_1 \cdot \mathbf{h}$. If $|a| < \varepsilon_t$, return $\bot$. Let $f = 1/a$, $\mathbf{s} = \mathbf{o} - \mathbf{v}_0$, $u = f\,\mathbf{s} \cdot \mathbf{h}$. If $u \notin [0, 1]$, return $\bot$. Let $\mathbf{q} = \mathbf{s} \times \mathbf{e}_1$, $v = f\,\mathbf{d} \cdot \mathbf{q}$. If $v < 0$ or $u + v > 1$, return $\bot$. Otherwise $t = f\,\mathbf{e}_2 \cdot \mathbf{q}$.

Plane. $\boldsymbol{\theta} = (\mathbf{p}_0, \mathbf{n}_0)$. Let $\text{denom} = \mathbf{n}_0 \cdot \mathbf{d}$. If $|\text{denom}| < \varepsilon_t$, return $\bot$; otherwise $t = (\mathbf{p}_0 - \mathbf{o}) \cdot \mathbf{n}_0/\text{denom}$.

Box. Standard slab method on three axis-aligned slab pairs.

Generic. Sphere-tracing on a computable signed distance function ϕ: $t_{k+1} = t_k + \phi(\mathbf{r}(t_k))$, terminating when $|\phi(\mathbf{r}(t_k))| < \varepsilon$.

The full scene-intersection routine iterates over all K primitives and returns the closest hit. Cost: $O(K)$ per ray.

B.4. BRDFs and direct lighting

A BRDF $f_r : S^2 \times S^2 \times \mathbb{R}^3 \to [0, 1]^3$ is a computable function. Standard examples: Lambertian $f_r = \mathbf{a}/\pi$; Phong $f_r = k_s(\mathbf{r}_i \cdot \boldsymbol{\omega}_o)^\alpha$; Cook–Torrance $f_r = D \cdot G \cdot F/(4(\boldsymbol{\omega}_i \cdot \mathbf{n})(\boldsymbol{\omega}_o \cdot \mathbf{n}))$.

The direct lighting estimate at $\mathbf{p}$ with normal $\mathbf{n}$, outgoing direction $\boldsymbol{\omega}_o$, and material BRDF f_r is

$$\mathbf{L}_{\text{direct}}(\mathbf{p}, \mathbf{n}, \boldsymbol{\omega}_o) = \sum_{j=1}^{M} \mathbb{1}[\text{visible}(\mathbf{p}, L_j)] \cdot f_r(\boldsymbol{\omega}_{\to L_j}, \boldsymbol{\omega}_o, \mathbf{p}) \cdot (\mathbf{n} \cdot \boldsymbol{\omega}_{\to L_j})_+ \cdot \mathbf{I}_j,$$

where the visibility test is performed by tracing a shadow ray. $\mathbf{L}_{\text{direct}}$ is computable.

B.5. Whitted-style recursive ray-tracing

The recursive ray-tracing scheme below follows the model introduced by Whitted (1980).

```
function Trace(r, depth, S):
    if depth > S.d_max(n): return S.c_bg
    h = HitScene(r, S)
    if h == nil: return S.c_bg
    omega_o = -r.d
    L = L_direct(h.p, h.n, omega_o, h.material, S)
    if h.material.k_r > 0:
        omega_refl = Reflect(r.d, h.n)
        L += k_r * Trace((h.p + S.eps_t * omega_refl, omega_refl), depth+1, S)
    if h.material.k_t > 0:
        omega_refr = Refract(r.d, h.n, h.material.eta)
        if omega_refr != nil:
            L += k_t * Trace((h.p + S.eps_t * omega_refr, omega_refr), depth+1, S)
    return L

function PixelWhitted(x, y, n, S):
    r = PrimaryRay(x, y, n, S.C)
    L = Trace(r, 0, S)
    return quantise_8(L)
```

Reflect$(\mathbf{d}, \mathbf{n}) = \mathbf{d} - 2(\mathbf{d} \cdot \mathbf{n})\mathbf{n}$ and Refract follow the standard formulae. Both are computable on computable inputs.

B.6. Deterministic Monte Carlo path-tracing

To preserve determinism, we use a fixed-seed deterministic Halton sequence $H_b(k)$, computable from (b, k):

```
function PathTrace(r, depth, sample_id, S):
    if depth >= S.d_max(n): return (0,0,0)
    h = HitScene(r, S)
    if h == nil: return S.c_bg
    L = L_direct(h.p, h.n, -r.d, h.material, S)
    (omega_in, pdf) = SampleBRDF(h.material, h.n, -r.d, sample_id, depth)
    if pdf > 0:
        f = h.material.brdf(omega_in, -r.d, h.p)
        cos_theta = max(0, dot(omega_in, h.n))
        L_indirect = PathTrace((h.p + S.eps_t*omega_in, omega_in), depth+1, sample_id, S)
        L += (f * cos_theta / pdf) * L_indirect
    return L

function PixelPathTrace(x, y, n, S):
    L_sum = (0,0,0)
    for k = 0 to s(n)-1:
        (u, v) = (Halton_2(k), Halton_3(k))
        r = PrimaryRay(x+u, y+v, n, S.C)
        L_sum += PathTrace(r, 0, k, S)
    return quantise_8(L_sum / s(n))
```

B.7. Computability of $I_{\mathcal{S}}$

Theorem B.1. *Let $\mathcal{S}$ be a scene with computable parameters: all numerical components are computable reals; $d_{\max} : \mathbb{N} \to \mathbb{N}$ is a computable function; the BRDFs are computable functions; the sample count $s : \mathbb{N} \to \mathbb{N}$ is computable. Then the function $I_{\mathcal{S}} : \mathbb{N}^3 \to \Sigma_{RGB}$ defined by* PixelWhitted *(or* PixelPathTrace*) is a computable image scheme.*

Proof. Each component is computable: (i) camera ray generation uses computable real arithmetic; (ii) per-primitive intersection uses computable real operations and comparison, with sphere-tracing terminating in a computable number of iterations for the generic case; (iii) scene intersection is a finite loop; (iv) direct lighting is a finite sum of computable quantities; (v) reflection and refraction are vector arithmetic; (vi) the Whitted recursion produces at most $2^{d_{\max}(n)-d}$ recursive calls—a finite number; (vii) path-tracing produces at most $s(n) \cdot d_{\max}(n)$ trace evaluations per pixel, with Halton sequences computable from (b, k); (viii) quantisation is computable. The composition is computable. □

B.8. Complexity bounds

Per-primitive intersection: $O(1)$ in scene size, $\tilde{O}(1)$ in precision n. Scene intersection: $O(K)$ per ray. Shadow rays: $O(M)$ per surface. Whitted trace: $O(MK \cdot 2^{d_{\max}(n)})$ per pixel; per-frame at $n \times n$, $\tilde{O}(n^2 \cdot MK \cdot \mathrm{poly}(n))$. Path-traced: $O(MK \cdot d_{\max}(n) \cdot s(n))$ per pixel; per-frame at $n \times n$ with $d_{\max}(n) = O(\log n)$ and $s(n) = O(n)$: $\tilde{O}(n^3 \cdot MK)$.

These match the figures cited in §10.4's cost stratification table.

C Transition tables of U's subroutines

This appendix completes §7 by giving the explicit transition tables of the five subroutines.

C.1. Find-Head

Entry: head one cell right of $\mathfrak{S}_2$, on σ_0. Exit: head on σ_p (the colour symbol being read).

State	Read	New state	Write	Move
Find-Head$^{\text{enter}}$	colour, $\mathfrak{B}$	ϕ_1	(same)	R
ϕ_1	$\mathfrak{H}$	ϕ_2	$\mathfrak{H}$	L
ϕ_1	$\mathfrak{N}$	ϕ_3	$\mathfrak{N}$	R
ϕ_2	colour, $\mathfrak{B}$	Read$^{\text{enter}}$	(same)	S
ϕ_3	colour, $\mathfrak{B}$	ϕ_1	(same)	R

State count: 4.

C.2. Read

Entry: head on σ_p. Exit: $\mathfrak{H}$ at position $p + 1$ replaced by $\mathfrak{X}$; control to Load-State.

State	Read	New state	Write	Move
$\textsc{Read}^{\text{enter}}$	colour, $\mathfrak{B}$	r_1	(same)	R
r_1	$\mathfrak{H}$	r_2	$\mathfrak{X}$	L
r_2	colour, $\mathfrak{B}$	$\textsc{Load-State}^{\text{enter}}$	(same)	S

State count: 3.

C.3. Load-State

Entry: head on σ_p. Exit: head on the first digit of the encoded current state in the state register.

State	Read	New state	Write	Move
$\textsc{Load-State}^{\text{enter}}$	colour, $\mathfrak{B}$	ℓ_1	(same)	L
ℓ_1	$\mathfrak{X}$	ℓ_2	$\mathfrak{X}$	L
ℓ_2	colour, $\mathfrak{B}$, $\mathfrak{N}$	ℓ_2	(same)	L
ℓ_2	$\mathfrak{S}_2$	ℓ_3	$\mathfrak{S}_2$	L
ℓ_3	$\mathfrak{0}, \ldots, \mathfrak{9}$	ℓ_3	(same)	L
ℓ_3	$\mathfrak{S}_1$	ℓ_4	$\mathfrak{S}_1$	R
ℓ_4	$\mathfrak{0}, \ldots, \mathfrak{9}$	$\textsc{Match}^{\text{enter}}$	(same)	S

State count: 5.

C.4. Match

This is the structurally largest subroutine. It scans the description region for a quintuple whose first two fields match (current state, current symbol). On a match, it overwrites the leading delimiter $\mathfrak{D}_2$ with $\mathfrak{X}$. We decompose into four sub-procedures.

M-NextQ: advance to the next quintuple's first field.

State	Read	New state	Write	Move
$\text{M-}\textsc{NextQ}^{\text{enter}}$	$*$	n_1	(same)	R
n_1	$\mathfrak{0}, \ldots, \mathfrak{9}$, $\mathfrak{D}_1$, colour, $\mathfrak{B}$	n_1	(same)	R
n_1	$\mathfrak{D}_2$	n_2	$\mathfrak{D}_2$	R
n_2	$\mathfrak{0}, \ldots, \mathfrak{9}$	$\text{M-}\textsc{CmpState}^{\text{enter}}$	(same)	S
n_2	$\mathfrak{S}_1$	$\textsc{Halt}$	$\mathfrak{S}_1$	S

State count: 3.

M-CmpState: compare current state against quintuple's first field. Carrying a remembered digit $d \in \{0, \ldots, 9\}$ across the shuttle requires ten states $c_0, \ldots, c_9$ plus shuttle states. Total: 14 states (including the entry state and reset/comparison-exit states $c^{=}, c^{\rightarrow}, c_1^{\rightarrow}$). On mismatch, transfers to M-Reset; on full match (state field exhausted simultaneously on both sides), transfers to M-CmpSym.

M-CmpSym: compare current symbol (located via $\mathfrak{X}$) against quintuple's second field. Implemented channel-by-channel to avoid the $|\Sigma_{\mathrm{RGB}}|$-state inflation that would arise from per-colour-value carrying. Approximately 10 states (shared with M-CmpState shuttle infrastructure).

*M-*Reset *and M-*Mark*:* on non-match, walk right past $\mathfrak{D}_2$ and re-enter M-NextQ (M-Reset, 1 state). On full match, walk left to the preceding $\mathfrak{D}_2$, overwrite with $\mathfrak{X}$, and transfer to Apply (M-Mark, 2 states).

Combined state count of Match: approximately 14 unique states with the M-CmpState block dominating and the others sharing shuttle states.

C.5. Apply

Entry: $\mathfrak{X}$ marks the matched quintuple in the description region; state register holds the old state; simulated tape contains $\mathfrak{X}$ at the simulated head position. Exit: state register holds new state; symbol at simulated head position is σ'; head marker positioned according to directional code.

We decompose into three sub-procedures: A-CopyState ($\approx$ 5 states, digit-by-digit shuttle analogous to M-CmpState but writing); A-WriteSym ($\approx$ 4 states, channel-by-channel transfer); A-MoveHead (explicit table below).

State	Read	New state	Write	Move
A-MoveHead$^{\text{enter}}$	0	a_L	0	R
A-MoveHead$^{\text{enter}}$	1	a_S	1	R
A-MoveHead$^{\text{enter}}$	2	a_R	2	R
a_L	$*$	a_L	(same)	R
a_L	$\mathfrak{X}$	$a_{L,1}$	$\mathfrak{N}$	L
$a_{L,1}$	colour, $\mathfrak{B}$	$a_{L,2}$	(same)	L
$a_{L,2}$	$\mathfrak{N}$	μ_0	$\mathfrak{H}$	S
a_S	$*$	a_S	(same)	R
a_S	$\mathfrak{X}$	μ_0	$\mathfrak{H}$	S
a_R	$*$	a_R	(same)	R
a_R	$\mathfrak{X}$	$a_{R,1}$	$\mathfrak{N}$	R
$a_{R,1}$	colour, $\mathfrak{B}$	$a_{R,2}$	(same)	R
$a_{R,2}$	$\mathfrak{N}$	μ_0	$\mathfrak{H}$	S

Combined state count of Apply: approximately 11 unique states with shared shuttle infrastructure.

C.6. Recompiled state count

$|Q_U| = 6 + 4 + 3 + 5 + 14 + 11 + 1 = 44$. Per-step cost is dominated by the Match shuttle: $O(D \cdot |\delta_{M_k}|)$ where D is the description region length, simplifying to $O(D)$ when $|\delta_{M_k}|$ is constant.

D Extended cost stratification

This appendix extends the cost table of §10.4 by giving, for each major image-scheme class, the four-dimensional cost picture: time, space, energy, and parallelism.

D.1. Cost dimensions

Time T. Number of primitive transitions executed by the underlying machine.

Space S. Maximum tape length used during the computation, in cells of Γ_U.

Energy E. Total bit-erasures times $kT \ln 2$. We report E in units of $kT \ln 2$ at $T = 300$ K, where one such unit is $\approx 2.87 \times 10^{-21}$ J.

Parallelism P. Number of independent computational paths exploitable in principle.

D.2. The extended cost tables

Classical substrate.

Class	T	S	E	P
Procedural	$\tilde{O}(n^2)$	$O(\log n)$	$24 \cdot \tilde{O}(n^2)$	n^2
Iterated	$\tilde{O}(n^2 K_{\text{iter}})$	$O(\log n)$	$24 \cdot \tilde{O}(n^2 K_{\text{iter}})$	n^2
Ray-traced (d depth)	$\tilde{O}(n^2 MK \cdot 2^d)$	$O(K + d)$	$24 \cdot \tilde{O}(n^2 MK \cdot 2^d)$	n^2
Path-traced (s samples)	$\tilde{O}(n^2 MK \cdot d \cdot s)$	$O(K + d)$	$24 \cdot \tilde{O}(n^2 MK \cdot d \cdot s)$	$n^2 s$
Neural (LW^2)	$\tilde{O}(n^2 LW^2)$	$O(LW + LW^2)$	$24 \cdot \tilde{O}(n^2 LW^2)$	$n^2 \cdot W$

Quantum (BQP) substrate.

Class	T	S (qubits)	E	P
Procedural	$\tilde{O}(n^2)$	$O(\log n)$	$24 \cdot \tilde{O}(n^2)$	n^2
Iterated	$\tilde{O}(n^2 K_{\text{iter}})$	$O(\log n)$	$24 \cdot \tilde{O}(n^2 K_{\text{iter}})$	n^2
Ray-traced	$\tilde{O}(n^2 MK \cdot 2^d)$	$O(K + d)$	$24 \cdot \tilde{O}(n^2 MK \cdot 2^d)$	n^2
Path-traced (s samples)	$\tilde{O}(n^2 MK \cdot d \cdot \sqrt{s})$	$O(K + d + \log s)$	$24 \cdot \tilde{O}(n^2 MK \cdot d \cdot \sqrt{s})$	n^2
Neural	$\tilde{O}(n^2 LW^2)$ †	$O(\log(LW^2))$ †	$24 \cdot \tilde{O}(n^2 LW^2)$ †	$n^2 \cdot W$ †

† Quantum advantage for general neural-network inference is an open research area.

Colour-native photonic substrate.

Class	T	S	E	P
Procedural	$\tilde{O}(n^2/24)$	$O(\log n)$	$\tilde{O}(n^2)$	n^2 (spatial)
Iterated	$\tilde{O}(n^2 K_{\text{iter}}/24)$	$O(\log n)$	$\tilde{O}(n^2 K_{\text{iter}})$	n^2 (spatial)
Ray-traced	$\tilde{O}(n^2 MK \cdot 2^d/24)$	$O(K + d)$	$\tilde{O}(n^2 MK \cdot 2^d)$	n^2 (spatial)
Path-traced	$\tilde{O}(n^2 MK \cdot d \cdot s/24)$	$O(K + d)$	$\tilde{O}(n^2 MK \cdot d \cdot s)$	$n^2 s$
Neural	$\tilde{O}(n^2 LW^2/24)$	$O(LW + LW^2)$	$\tilde{O}(n^2 LW^2)$	$n^2 \cdot W$

D.3. Class-by-class derivation

Procedural and parametric. Time $O(\log^c n)$ per pixel for some c depending on the degree, multiplied by n^2 pixels. Space $O(\log n)$. Energy: time times $24kT \ln 2$. Parallelism: trivially n^2.

Iterated / fractal. The Mandelbrot scheme with iteration depth $K_{\text{iter}}(n) = O(n \log n)$ has per-frame time $\tilde{O}(n^3)$. Space $O(\log n)$. Parallelism n^2.

Ray-traced and path-traced. Whitted: $O(MK \cdot 2^d)$ per pixel, $\tilde{O}(n^4)$ per frame at $n \times n$. Path-traced: $\tilde{O}(n^2 MK \cdot d \cdot s)$. Space $O(K + d)$. Quantum amplitude estimation reduces s to $\sqrt{s}$ (Brassard et al. 2002).

Neural-network. Per pixel: $O(LW^2)$. Per frame: n^2 times this. Space $O(LW)$ activations + $O(LW^2)$ weights stored once.

D.4. Concrete numbers for a 4K frame

Class	Operations	E Landauer floor	E on CMOS	Memory
Procedural gradient	$\sim 10^7$	$\sim 10^{-13}$ J	$\sim 10^{-9}$ J	KB
Mandelbrot (depth 10^3)	$\sim 10^{10}$	$\sim 10^{-10}$ J	$\sim 10^{-6}$ J	KB
Path-traced ($s = 64$, $K = 10^4$)	$\sim 10^{12}$	$\sim 10^{-8}$ J	$\sim 10^{-4}$ J	$\sim 10^2$ MB
Diffusion model ($L = 50$, $W = 10^3$)	$\sim 10^{14}$	$\sim 10^{-6}$ J	$\sim 10^{-2}$ J	$\sim 10^3$ MB

The thermodynamic limit for a digital photonic system at room temperature is between one and three orders of magnitude better than current CMOS, depending on the implementation regime. For a 4K diffusion-model frame, this translates to wall-clock energy from $\sim 10^{-2}$ J on current hardware to between $\sim 10^{-3}$ and $\sim 10^{-4}$ J on a photonic substrate, depending on the regime.

D.5. Asymptotic comparison

Time. The colour-native photonic substrate provides a constant-factor 24× speedup at the symbol level. Quantum computation provides class-changing speedups for specific problems (Monte Carlo with $\sqrt{s}$, Hamiltonian-simulation rendering exponential), but does not affect the asymptotic class.

Space. Memory is dominated by model weights and scene data; substrate choice has small effect.

Energy. The gap to Landauer of 10^4–10^5 in current CMOS is the largest engineering target. Photonic substrates close two to three orders of magnitude. Reversible computation could close the rest.

Parallelism. Photonic substrates support spatial-mode multiplexing and wavelength-division multiplexing in ways that have no clean classical-electronic analogue.

The colour-native commitment of §1.5 affects three of four cost dimensions favourably (time, energy, parallelism) and is neutral on the fourth (space). This is the quantitative substance behind the alignment claim of §12.4.

E Encoding correctness for Theorem 11.1

This appendix completes the proof sketched in §11.3 by verifying that the encoding of §11.2 effects a correct reduction from $\mathsf{Halt}_{\text{img}}$ to first-order validity.

E.1. The encoding restated

For the proof we restate ϕ_{k,x_0,y_0,n_0} as an implication. Let M_k be a Turing machine over Σ_{RGB} with internal states Q, halting states $F \subseteq Q$, tape alphabet Γ, blank $\mathfrak{B}$, and transition function $\delta : (Q \setminus F) \times \Gamma \to Q \times \Gamma \times \{-1, 0, +1\}$. Fix input (x_0, y_0, n_0).

The language $\mathcal{L}_{M_k}$: constant $\mathbf{0}$, successor S, predecessor S^{-1}, tape-symbol constants $\bar{\sigma}$, state predicates State_q, the symbol predicate Sym, the head-position function Pos.

Initial axiom (Init):

$$\mathrm{State}_{q_0}(\mathbf{0}) \wedge \mathrm{Pos}(\mathbf{0}) = \mathbf{0} \wedge \bigwedge_p \big(\mathrm{Sym}(\mathbf{0}, S^p(\mathbf{0})) = \bar{\sigma}_p\big).$$

Transition axioms (Trans): for each quintuple $(q_i, \sigma) \mapsto (q_j, \sigma', m)$,

$$\begin{aligned} \forall t \, \big[\, &\mathrm{State}_{q_i}(t) \wedge \mathrm{Sym}(t, \mathrm{Pos}(t)) = \bar{\sigma} \\ &\to \ \mathrm{State}_{q_j}(S(t)) \wedge \mathrm{Sym}(S(t), \mathrm{Pos}(t)) = \bar{\sigma'} \wedge \mathrm{Pos}(S(t)) = m \cdot \mathrm{Pos}(t) \, \big], \end{aligned}$$

together with frame axioms $\forall t \, \forall p \, [\, p \neq \mathrm{Pos}(t) \to \mathrm{Sym}(S(t), p) = \mathrm{Sym}(t, p) \,]$.

Halting clause (Halt): $\exists t \, \bigvee_{q \in F} \mathrm{State}_q(t)$.

The encoded formula:

$$\phi_{k,x_0,y_0,n_0} \ = \ (\mathsf{Init} \wedge \mathsf{Trans}) \to \mathsf{Halt}.$$

E.2. Soundness

Lemma E.1 (Soundness). *If M_k halts on input (x_0, y_0, n_0) in T steps, then ϕ_{k,x_0,y_0,n_0} is logically valid.*

Proof. Let $\mathcal{M}$ be any model of $\mathcal{L}_{M_k}$ in which $\mathsf{Init} \wedge \mathsf{Trans}$ is true. We show $\mathcal{M} \models \mathsf{Halt}$.

For $0 \leq i \leq T$, let $C_i = (q_{(i)}, p_{(i)}, \tau_{(i)})$ denote the configuration of M_k at step i. We claim, by external induction on i, that

$$\mathrm{State}_{q_{(i)}}(S^i(\mathbf{0})) \wedge \mathrm{Pos}(S^i(\mathbf{0})) = S^{p_{(i)}}(\mathbf{0}) \wedge \forall p \, (\mathrm{Sym}(S^i(\mathbf{0}), p) = \overline{\tau_{(i)}(p)}) \qquad (\star_i)$$

holds in $\mathcal{M}$ for every $i \in \{0, \ldots, T\}$.

Base case $(i = 0)$: Init asserts $(\star_0)$.

Inductive step. Suppose $(\star_i)$ holds, $i < T$. Let $\delta(q_{(i)}, \tau_{(i)}(p_{(i)})) = (q_{(i+1)}, \sigma'_{(i)}, m_{(i)})$. Instantiating the transition axiom for this quintuple at $t := S^i(\mathbf{0})$ and applying modus ponens forces the state, position, and head-cell components of $(\star_{i+1})$. The frame axiom forces tape positions other than $p_{(i)}$ to be unchanged. So $(\star_{i+1})$ holds.

After T steps, $(\star_T)$ holds; since $q_{(T)} \in F$, the existential of Halt is witnessed. Hence $\mathcal{M} \models \mathsf{Halt}$.

Since $\mathcal{M}$ was arbitrary, ϕ_{k,x_0,y_0,n_0} is true in every model of $\mathsf{Init} \wedge \mathsf{Trans}$, hence valid. By Gödel's completeness theorem, it is provable. $\square$

E.3. Adequacy

Lemma E.2 (Adequacy). *If M_k does not halt on input (x_0, y_0, n_0), then ϕ_{k,x_0,y_0,n_0} is not logically valid.*

Proof. We construct an explicit model $\mathcal{H}$ in which $\mathsf{Init} \wedge \mathsf{Trans}$ holds and Halt fails.

The domain of $\mathcal{H}$ is $\mathbb{Z} \sqcup \Gamma$. Interpret: $\mathbf{0}^{\mathcal{H}} = 0$; $S^{\mathcal{H}}(n) = n + 1$; $S^{-1,\mathcal{H}}(n) = n - 1$; $\bar{\sigma}^{\mathcal{H}} = \sigma$; $\mathrm{State}_q^{\mathcal{H}}(i) = \top$ iff $i \geq 0$ and M_k at step i is in state q; $\mathrm{Pos}^{\mathcal{H}}(i) =$ head position of M_k at step i; $\mathrm{Sym}^{\mathcal{H}}(i, p) =$ symbol on M_k's tape at position p at step i.

We verify $\mathcal{H} \models \mathsf{Init}$: M_k at step 0 is in state q_0, head at origin, tape contents matching the canonical encoding.

We verify $\mathcal{H} \models \mathsf{Trans}$: by definition of δ, the consequent follows from the antecedent at every i.

We verify $\mathcal{H} \not\models \mathsf{Halt}$: the existential requires some $i \geq 0$ at which M_k is in a halting state, but by hypothesis no such i exists. Therefore $\mathcal{H} \not\models \phi_{k,x_0,y_0,n_0}$, and the formula is not valid. □

E.4. Conclusion of the proof of Theorem 11.1

Combining Lemmas E.1 and E.2:

$$\phi_{k,x_0,y_0,n_0} \text{ is valid} \quad \Longleftrightarrow \quad M_k \text{ halts on } (x_0, y_0, n_0).$$

The map $(k, x_0, y_0, n_0) \mapsto \phi_{k,x_0,y_0,n_0}$ is computable. If an algorithm A deciding first-order validity existed, the composite $\hat{A}(k, x_0, y_0, n_0) := A(\phi_{k,x_0,y_0,n_0})$ would decide $\mathsf{Halt}_{\mathrm{img}}$, contradicting Theorem 8.3. Therefore Theorem 11.1 is established. □

F Proof-of-Computable-Image and code-as-law via Bitcoin

This appendix provides the implementational scaffolding for the proof-of-recollection reformulation of §10.3, in a form compatible with current Bitcoin-layer infrastructure. The intent is not to prescribe a particular implementation but to demonstrate that the reformulation is consistent with deployed cryptographic protocols and that the engineering path from the conceptual proposal of §10.3 to a functioning system is well-defined under existing 2025–2026 infrastructure.

F.1. The infrastructural starting point: BitVM and its descendants

Bitcoin's Script language is deliberately non-Turing-complete: Nakamoto removed OP_MUL, OP_CAT, and other powerful opcodes early in the protocol's life, for reasons of consensus stability. Smart-contract functionality of the Ethereum kind — Turing-complete on-chain execution — has therefore not been available on Bitcoin's base layer for most of its history. This changed with BitVM (Linus 2023), which introduced a fraud-proof-based mechanism for verifying arbitrary computation on Bitcoin without protocol modification. BitVM2 (Linus et al. 2024) reduced the dispute-resolution cost and admitted permissionless challenges. BitVM3 (Linus et al. 2025), employing garbled circuits (Yao 1986), reduced the on-chain block-space requirement for an assertion transaction to approximately 56 kB and for the disapproval transaction to approximately 200 bytes — roughly three orders of magnitude improvement over BitVM1 — and shifted the bulk of the verification off-chain into a cryptographically committed computation graph.

The consequence relevant to the U-Mentalism programme is that Bitcoin's base layer can today serve as the settlement layer for Turing-complete computation without any hard or soft fork. The smart-contract dimension of the proof-of-recollection proposal (§10.3) and of any code-as-law deployment (Lessig 1999, 2006; Wright & De Filippi 2015; De Filippi & Wright 2018) is therefore technically feasible on the existing Bitcoin substrate as a layer-2 protocol whose disputes anchor at L1.

F.2. Proof-of-Computable-Image as instantiation of proof-of-recollection

Section 10.3 proposed the redirection of proof-of-work into proof-of-recollection: the costly mining operation becomes the canonicalisation and ordinal insertion of a frame into a persistent record. We now describe a concrete realisation, *Proof-of-Computable-Image* (PoCI), in which the recollection content is precisely the output of a computable image scheme in the sense of Definition 2.1.

Scheme registry. Fix a computable enumeration $\{f_k\}_{k\in\mathbb{N}}$ of admissible image schemes drawn from the schemes of §§3 and 10. Admissibility is restricted to schemes with explicit primitive-recursive resource bounds $T_k(n)$, so that $\mathsf{Halt}_{\text{img}}$ does not obstruct the protocol: the Mandelbrot scheme of Example 3.5, the ray-tracer of Appendix B, neural-inference schemes of Proposition 10.3 with fixed weights, and the procedural schemes of §10.1 all qualify. The registry is fixed by protocol consensus and updated by formal proposal, as with any consensus-bound parameter.

Block structure. Each block header specifies, in addition to the conventional fields, a tuple (k, x_0, y_0, n_0) selecting a scheme and its evaluation parameters. The selection is not made freely by the miner but is derived deterministically from the previous block hash via a verifiable delay function or a commit-reveal protocol, removing miner control over scheme choice. This prevents the degeneration of the work into a SHA-256-equivalent search by selection of cheap schemes.

Mining and verification. The miner computes the frame $F_k(x_0, y_0, n_0)$ at the specified parameters using f_k and seeks a nonce ν such that $H(F_k \,\|\, \nu) < \theta$ for the current difficulty target θ, with H a conventional cryptographic hash function. Verification is structured asymmetrically by attaching to the block a zero-knowledge succinct proof certifying that F_k is indeed the output of f_k on (x_0, y_0, n_0). The frame itself is retained either on a data-availability layer or via Merkle commitment in the block with off-chain frame storage. The verifier checks the zk proof and the hash inequality; the verifier does not re-execute f_k.

Cost and asymmetry. The mining cost is dominated by frame computation plus zk-proof generation; the verification cost is dominated by zk-proof verification, which is succinct (polylogarithmic in the computation size). The conventional proof-of-work asymmetry is preserved, with the energy now expended on computation that produces a recoverable artefact (the frame) rather than a discarded hash.

F.3. Photonic alignment

The selection of computable image schemes as the work content is not arbitrary. Sections 9.4 and 12.4 established that the colour-native universal machine of §1.5 is structurally aligned with a photonic substrate: each operation processes 24-bit colour symbols natively, with a constant-factor speedup of approximately 24× at the symbol level versus binary-emulated computation, and energy advantages of approximately two to three orders of magnitude versus current CMOS

(Appendix D.4). A PoCI protocol in which the dominant computation is over Σ_{RGB} therefore advantages photonic mining hardware structurally — not by protocol decree but by the alignment of the work content with the substrate. The capital expenditure of competitive mining is, on this construction, channeled into photonic infrastructure that has applications beyond consensus: optical computing for rendering, scientific simulation, and the colour-native universal computation of the U-Mentalism programme more generally. The useful-proof-of-work criterion that has eluded most earlier proposals — Primecoin (King 2013) and proof-of-learning (Jia et al. 2021) in particular — is, in PoCI, satisfied at the level of the substrate rather than at the level of the artefact alone: the work itself trains the manufacturing and operational base for the engineering programme that the foundational paper makes mathematically precise.

F.4. Code-as-law via image-conditioned contracts

The smart-contract layer made feasible by BitVM3 admits contracts whose execution conditions are oracular data derived from computable image schemes. The liminal case is the bare structure: a transfer of value v on the chain conditioned on a predicate P evaluated over the output of a registered scheme f_k at protocol-specified parameters — that is, the transfer executes if and only if $P(F_k(x_0, y_0, n_0))$ holds. The predicate is deterministic in the scheme output; the scheme output is verifiable on Bitcoin L1 via BitVM3 fraud proofs or via zk-proofs analogous to those of §F.2; no trusted human oracle is required, and no off-chain consensus mechanism is invoked. From this minimal structure — value conditioned on a computable predicate over a computable image scheme — the full space of programmable smart contracts on Bitcoin opens, with schemes drawn from the registry of §F.2 and predicates drawn from decidable propositional logic freely composable.

This is the Lex Cryptographia of Wright & De Filippi (2015, 2018) in a form bounded by the present paper's foundational results. The bounds are not incidental: by Theorem 11.1, a contract whose conditions referenced first-order validity in general would not be in general executable, since $\mathsf{Halt}_{\mathrm{img}}$ is undecidable; by the restriction to primitive-recursively-bounded image schemes in §F.2, any contract expressible in the proposed framework is executable. The visual *Entscheidungsproblem* hypothesised in §11.5 marks the boundary of this implementational claim: a contract whose conditions involved deciding whether some visual content is generated by some scheme would, under the working hypothesis of §11.5, also be undecidable. The decidable contractual layer is the layer in which the schemes are given by name and the conditions concern their evaluation at specified parameters.

Code-as-law, on this construction, inherits exactly the boundary that Turing established in 1936 and that the present paper extends to the visual domain: the executable is the procedurally specifiable; the inexecutable is the higher-order question about whether some procedural specification of a given image exists. The boundary is precise, and it is the present paper's foundational result on the visual side that makes the inheritance precise.

F.5. Non-fungibility, declarative registration, and the dual fungibility of canonicalisation work

A natural reading of the PoCI tokens of §F.2 is in the vocabulary of non-fungible tokens (NFTs), but the reading requires a distinction of senses that the standard NFT literature does not generally make precise. The distinction is consequential for both the philosophy and the protocol economics, and is worth recording explicitly.

Two senses of fungibility. At the content level, each PoCI token is strictly non-fungible: the $24n^2$ bits of the frame, together with its ordinal position in the recollection (§§1.5, 10.3), identify it uniquely by construction. Token T_N associated with frame F_N at recollection ordinal N is not interchangeable with token T_M associated with frame F_M at ordinal $M \neq N$ — the bits differ, the ordinal differs, the identity differs. At the accounting level of *canonicalisation work*, however, every token in the issuance is interchangeable with every other: each represents one block reward's worth of accomplished canonicalisation effort, at the protocol-specified difficulty, and the work-unit it certifies is uniform across tokens. The same token is simultaneously non-fungible (as registered object) and fungible (as work-unit). The duality is not a contradiction; it tracks two distinct economic operations performed by the same protocol artefact.

The Ordinals analogue. The closest deployed Bitcoin-layer construction exhibiting the same dual structure is the Ordinals inscription mechanism (Bitcoin BIPs and the inscription-protocol community 2023–2025): an inscription's content is non-fungible (the data is what it is), while the satoshi on which it sits is part of the fungible base supply. PoCI, in the realisation register of §§F.1–F.4, is most naturally implemented as an Ordinals-style inscription whose content is the registered image-frame (or its Merkle commitment, where data availability is delegated) and whose carrier sat participates in the fungible token economy. The protocol-economic distinction is therefore not invented by PoCI; PoCI inherits it.

The distinctive U-Mentalism twist: declarative versus constitutive registration. A standard NFT contract *constitutes* the uniqueness of the token: the smart contract decides what is unique and what is not, and the artificial scarcity that the contract enforces is the source of the asset's distinctiveness. One could have minted differently; the contractual choice not to is what licenses the scarcity. PoCI tokens, by contrast, *register* a uniqueness that is mathematically pre-given. Every computable image-string at resolution n already occupies its ordinal slot in $\Sigma_{\mathrm{RGB}}^{n\times n}$ as a matter of arithmetic; there are exactly 256^{3n^2} such slots, and each is identified by its ordinal independently of any protocol decision. What PoCI canonicalises is *which slot is operationally indexed when*, and *by whom the indexing is performed*; the slot itself is not created by the protocol but recognised by it. In the vocabulary of legal philosophy, PoCI is a declarative rather than a constitutive registration — a recording of a pre-existing identity rather than an institution of a new one. This is, in the register of Lessig's *code is law* (Lessig 1999, 2006), the form of code-as-law least vulnerable to the standard critique of artificial scarcity, since the underlying uniqueness is not artificial.

NFT-films and the hierarchical structure. The natural extension to films is hierarchical and follows the cellular structure of §1.5. A film consisting of J frames is itself ordinally indexed at the film level — ordinal $\omega\cdot\alpha+\beta$ in the natural product order on (film-index, frame-index), or simply within a film-ordinal hierarchy on ω^2 when frame counts are unbounded — and each constituent frame retains its own frame-ordinal independently. The implementational consequence is that NFT-films admit three realisations on the Ordinals substrate: (i) a single film-level inscription whose content is the whole film (frame-ordinals encoded in the inscription metadata); (ii) a family of frame-level inscriptions whose linkage is recorded by a separate aggregating contract; (iii) a hybrid in which the film-level inscription is the canonical reference and frame-level inscriptions are derived on demand. The mathematical non-fungibility is two-tiered — the film as film, the frames as frames — and the protocol-economic accounting can be either single-token-per-film or one-token-per-frame, with the corresponding fungibility-as-work-unit applying at the chosen level.

Why the duality matters. The two senses of fungibility, kept distinct, allow PoCI to inherit the economic legibility of fungible-base-supply protocols (the work-unit is a uniform accounting object) while preserving the asset-level distinctiveness on which any visual marketplace must depend. The fungibility-as-value intuition that immediately presents itself — if every canonicalisation costs the same, the tokens are surely fungible — is correct at the accounting level and surfaces a genuine structural feature of the protocol, namely that the marketplace of canonicalised images is funded by a uniform work-unit currency and not by per-image bespoke pricing of the canonicalisation effort itself. What the marketplace prices is the asset; what the protocol prices is the work; the two prices are independent. The U-Mentalism programme commits to making this independence transparent at the level of the U.M. Assembly Language (§12.9), where the rights and duties of canonicalisation and the rights and duties of asset-disposition are separately specifiable.

F.6. Engineering open points

The proposal raises engineering questions that the present paper records without resolving. *Determinism*: frame computation must be deterministic at the bit level across implementations; floating-point operations must be replaced by exact-arithmetic or fixed-point equivalents, which the colour-native Σ_{RGB} framework of §1.5 already supports natively. *Scheme economics*: the ordinal-insertion fee structure of §10.3 must be designed to align miner incentives with scheme novelty — coverage maps over scheme-space — rather than with mere scheme cheapness. *zk-proof overhead*: succinct proof generation for image-scheme computation requires either specialised circuits or general-purpose zk-VMs, whose overhead may currently be substantial; reductions in this overhead are an active research area. *Data availability*: frame storage requires either dedicated data-availability layers or selective inscription mechanisms (ordinals-style), with the choice trading off cost against retrievability guarantees.

None of these is a foundational obstruction; each is an engineering parameter. The point of this appendix is to establish that the conceptual proposal of §10.3 has, in the existing 2025–2026 Bitcoin infrastructure, a concrete realisation path that respects the foundational boundaries of the present paper and the cryptographic security model of the underlying chain.

F.7. A critical note on attack-surface and the neuro-somatic-authentication horizon

The present appendix has, throughout, recorded the proposal in its constructive register. A symmetric critical register deserves explicit registration, in keeping with the methodological note of §10.5 and the wider U-Mentalism programme commitment to maintaining the distinction between formal capabilities and the social claims (§10.5, in fine). Five observations are worth making.

First, the relation between supercomputation depth and consensus security. The U.M. recollection-and-collection layer, when implemented as proof-of-recollection / proof-of-collection over a Bitcoin-class consensus chain (§§F.2–F.3), inscribes the recollection history into the chain's accumulated work-history. The integration with the supercomputation register of §§12.10–12.11 has a direct security consequence: the deeper the supercomputation stack of canonicalised image-and-film history, the deeper the consensus chain's accumulated state, and accordingly the larger the attack-surface for adversaries with the resources to mount Byzantine majority compromises (51%-style attacks and their consensus-layer relatives). The conventional security argument for proof-of-work chains — that re-mining a chain from a sufficient historical depth is computationally infeasible — weakens, in absolute terms, as the supercomputational resources available to the largest network participants grow; the relative security, between the largest miner and the median miner, may be preserved or eroded depending on how the supercomputational substrate distributes across the network. The U-Mentalism programme records this as a structural risk inherited from any deep consensus chain, accentuated rather than created by the proof-of-recollection / proof-of-collection extension, and applicable to attacks of all categories (not only the canonical 51%-rewrite; also long-range attacks on stake-based chains, selfish-mining strategies, eclipse attacks at the network layer, and front-running attacks at the mempool layer).

Second, the relation between economics and ethics in chain design. Blockchain technology is, the U-Mentalism programme observes, a register in which economic and ethical questions are inseparable in a manner that the standard economic literature on the topic does not always make explicit (Wright & De Filippi 2015 on *Lex Cryptographia* is the partial exception; the wider promotional literature is the rule). The choice of consensus algorithm, the issuance schedule, the fee structure, and the governance protocol are simultaneously economic decisions (about who is rewarded for what) and ethical decisions (about what kinds of participation are valorised, what kinds of harms are made costly, what kinds of obligations are made enforceable). The U.M. Assembly Language of §12.9, as the operative instrument of the charter of rights and duties of cybernetic anthropology, is the natural register in which the entanglement between economics and ethics in chain design becomes specifiable rather than implicit. The U-Mentalism programme registers this as a research direction adjacent to the present paper, with the proper development reserved for the planned Assembly Language and, more broadly, for the wider U-Mentalism research on ethics, neuroscience, and cybernetics that the introduction announces.

Third, the neuro-somatic-authentication horizon. The author recognises the question of a genetic-and-living signature — the binding of cryptographic identity to the unique genetic identity of a living body, as the deepest authentication mechanism the contemporary infrastructure does not provide — as an authentication frontier of the U-Mentalism programme. The recording is made not by way of guarding against the horizon but of keeping it present as the ultimate frontier against which the pressing questions of the present register find their long view: the operative questions of cryptographic-key infrastructure that precede it — key custody, recovery, succession, rotation, revocation, in-life impersonation, post-mortem transfer — retain their full operative weight in their own terms, and the horizon's role is to set them in their place rather than to displace them. A structural observation deserves to be added at this register: even were the signature half of the equation resolved — the private key provably bound to a body's genetic identity by some successor mechanism — the recognition half would remain open as a verification problem case by case. The computational verification of a genetic-and-living signature by every system that must rely on it is, at every individual act of verification, a fresh open question, and remains so even when formal verifiability is assured at the protocol level. The signature and the recognition are constitutively distinct halves of the same authentication equation, and the recognition half admits no final closure of the kind the signature half admits.

Fourth, consensus adaptation under supercomputational hash power. The integration of supercomputation-class resources (§§12.10–12.11) with the proof-of-recollection / proof-of-collection layer over the Bitcoin chain has direct implications for the chain's consensus operation. Cryptographic problem-resolution times — the wall-clock duration of finding a nonce satisfying the proof-of-work target — decrease as the network's aggregate hash power grows; Bitcoin's difficulty-adjustment algorithm, applied every 2016 blocks (approximately every two weeks), re-targets the per-block difficulty to maintain the ten-minute average block time. Under U.M.-class hash power, two pressures converge on the consensus regime. First, the adjustment lags hash-rate growth by up to a fortnight, during which faster-than-target block times produce confirmation acceleration: a transaction can clear the standard six-confirmation threshold in correspondingly less wall-clock time, with the consequence that Bitcoin approaches the latency profile of effective transactional money rather than slow-settlement store-of-value. The implication for monetary theory in practice is direct: a chain whose confirmation latency converges toward the latency profile of card-network settlement becomes integrable into the payment systems where instant settlement is the operational requirement. Second, the steady-state regime under U.M.-class power is one in which the difficulty target is correspondingly higher and the per-machine probability of finding a block correspondingly lower, with the centralisation risks recorded under the First observation above acting on the equilibrium that the adjustment converges toward. The transition is a parameter shift within the existing difficulty algorithm rather than a protocol change.

Fifth, the absence of a soft-fork proposal. The U-Mentalism programme does not propose a soft fork of the Bitcoin protocol to incorporate the colour-native register, even though the architectural integration of supercomputation with proof-of-recollection / proof-of-collection might suggest such a path. The reason is structural: U.M.-class computational power will be brought to bear on Bitcoin mining regardless of whether U.M. itself participates in the protocol layer, because the substrate-level capabilities of §12.10 are technology-class capabilities that any actor with substrate access can deploy. A soft fork integrating U.M. into the protocol would add one path of adaptation; the alternative — U.M. as an external system using Bitcoin's existing protocol as its persistence anchor — preserves Bitcoin's mature security and social architecture untouched while obtaining the persistence guarantee that U.M. recollection requires. Bitcoin will have to adapt to U.M.-class computational power in any event, from whichever angle the power is brought to bear; the programme's position is that one path of adaptation is preferable to two in the consensus-architecture register. The preference is suspended only when the integrity of Bitcoin's proof-of-work would otherwise be compromised — if U.M.-class hash power created an asymmetry severe enough to threaten the chain's resistance to majority-compromise attacks. In that contingency the appropriate intervention is a defensive soft-fork: a backward-compatible tightening of the protocol rules (in the BIP-66, BIP-141, BIP-341 lineage of historical soft-forks) that neutralises the U.M.-specific asymmetry by adding a validation criterion the existing proof-of-work alone does not impose — a memory-hard verification term (in the Equihash, Cuckoo Cycle, Argon2 line) or a recollection-anchored term requiring a Merkle proof of recollection-layer state, levelling the playing field across U.M. and non-U.M. substrates without modifying the underlying SHA-256 proof-of-work itself. The soft-fork would be defensive and provisional, with the modifications re-integrating into mainline Bitcoin operation through the standard protocol-upgrade trajectory once the substrate integration had stabilised and the consensus regime had converged under the modified rule. The contingency-cycle (defensive soft-fork → re-integration) is structurally distinct from the fork-and-fragment trajectory the principal case rejects, but is not without risk of its own: soft-fork activation requires majority-miner support that U.M.-class miners may not provide, the re-integration step requires a subsequent protocol modification with its own adoption risk, and the defensive criterion may itself become a vector for protocol-governance fragmentation. The cycle is recorded here as one the programme is prepared to enter only when the diagnostic of compromise is unambiguous, not as a recommendation in the absence of it. The adaptation should occur within Bitcoin's existing protocol envelope rather than through a protocol modification whose adoption risk and consensus-fragmentation risk would compound the integration risk that any new substrate technology already carries.

G Intermediate engineering point: binary TM with photonic sensorisation

This appendix records, for completeness, an intermediate engineering proposal between the conventional CMOS-Turing-machine and the colour-native photonic universal machine of §§1.5, 7, and 9.4 of the present paper. The proposal is not the engineering target of the U-Mentalism programme — that target is the RGB colour-native machine of §1.5 and its EEM-extended successor of §9.2 — but it represents a constructible intermediate point, useful as a benchmark and as a transition platform.

G.1. Construction

Let M be a binary Turing machine whose tape consists of consecutive frames of an RGB film, and whose head is realised by three parallel photodiodes situated to read the three colour channels of the head pixel independently. Each photodiode produces an 8-bit quantised intensity reading; the three readings together produce a 24-bit binary string per head position. The machine operates as follows:

(i) *Read.* The three photodiodes sample the head pixel synchronously and produce 24 bits of input.

(ii) *Decide.* A digital decision circuit on $24 + |Q|$ bits ($|Q|$ being the binary encoding length of the machine state) selects the next operation from the transition table.

(iii) *Write.* An optical modulator — three independent channels for R, G, B — writes the next 24-bit symbol back to the head pixel.

(iv) *Advance.* A mechanical or beam-steering mechanism translates the head to the adjacent pixel position.

The architecture is conventional-binary in its decision and arithmetic logic, but photonic in its input/output interface and in its memory medium (the RGB film itself).

G.2. Cost estimates under current components

The dominant costs under 2025 component performance:

Sensor bandwidth. CMOS imagers achieve $\sim$ 10 Gpixel/s readout per channel at single-pixel resolution; with 3-channel parallelism, 3×10^{10} bits/s of raw sensor throughput. Decision logic implemented in 7nm CMOS at $\sim 10^9$ instructions/s on a $1\,\mathrm{mm}^2$ die, with three orders of magnitude head-room available via ASIC dedication.

Modulator speed. Thin-film lithium niobate (TFLN) modulators achieve $\sim$ 100 GHz switching; spatial-light modulators (LCoS) achieve $\sim$ 1 MHz at higher channel counts. The throughput is sensor-limited at the photodiode read, not modulator-limited at write.

Energy per pixel-step. A complete read-decide-write cycle dissipates approximately $\sim 10^{-13}$ J ($\sim$ 100 fJ) in current technology, dominated by the modulator's switching energy. Compare to the colour-native photonic gate of §12.4, estimated at $\sim 10^{-15}$ J per gate, two orders of magnitude lower at the substrate level.

Head-translation speed. Beam-steering by acousto-optic deflectors achieves $\sim 10^6$ pixel transitions per second; mechanical translation of a physical film is slower by 2–3 orders.

G.3. Plausibility and trade-offs

The intermediate machine is constructible today with off-the-shelf components. The key trade-off versus the colour-native machine:

Versus colour-native U.M. 'C'. The intermediate machine treats the RGB colour as input/output representation only, with internal computation in binary. The advantage of Σ_{RGB} as the native computational alphabet — the 24× symbol-level speedup of §12.4 — is forfeited. The arithmetic and decision logic, being binary, inherit the von Neumann separation of §12.8 and its associated energy gap, with the photonic gain limited to the I/O interface.

Versus CMOS-only. The photonic I/O dissolves the bandwidth bottleneck between the sensor array and the processor, which is a substantial advantage for image-intensive workloads. The intermediate machine is, in this respect, attractive as a transition platform for the U-Mentalism engineering programme: it serves as a development substrate for the colour-native algorithms while the full colour-native photonic infrastructure of §§12.4, 12.8 is being matured.

The intermediate point is not the U-Mentalism engineering target. It is recorded here because it represents a constructible step within current technology, with the appropriate qualifications about which features of the full U.M. 'C' programme it does and does not realise.

H Implementation avenues and obstacles for the colour-native universal machine

This appendix surveys, without claim to completeness, the principal engineering avenues and the principal obstacles confronting the implementation of the colour-native universal machine of

§§1.5, 7, and 12.4. The intent is to record the territory openly so that subsequent work in the U-Mentalism programme can engage it at appropriate level of detail.

H.1. Substrate avenues

Silicon photonics. The mature integrated-photonic platform, with foundry processes available at 100–200 mm wafer scale. Strengths: established fabrication, CMOS-compatible, scalable to thousands of interconnected components. Limitations: silicon is centrosymmetric (no second-order nonlinearity), limiting native nonlinear processing; loss is moderate ($\sim$ 1 dB/cm); the operating wavelength range is centred on the 1300–1550 nm telecommunications bands, requiring wavelength conversion for visible-band operation.

Thin-film lithium niobate (TFLN). A rapidly maturing platform combining low loss ($\sim$ 0.1 dB/cm), strong second-order nonlinearity (electro-optic), and high-speed modulation ($\sim$ 100 GHz). Strengths: high modulation bandwidth, broad transparency window covering the visible band, available on bonded silicon wafers. Limitations: foundry availability is less mature than silicon photonics; component library is more limited; yields at wafer scale are improving but not yet at silicon levels.

Atomic and molecular systems. Cold-atom platforms and rare-earth-doped solids realise photon storage and processing at the single-photon level with very long coherence times ($\sim$ minutes for cold atoms via electromagnetically induced transparency). Strengths: quantum-level fidelity, exact-arithmetic deterministic operations under cryogenic conditions. Limitations: cryogenic operation, low component density, individual-element addressing currently limited.

Hybrid platforms. The most likely engineering target combines silicon photonics for routing, TFLN for modulation, and atomic systems for memory, integrated via wafer-level bonding. The complexity of such hybrids is non-trivial but is the structural correlate of the colour-native machine's heterogeneity of operations: transport, modulation, memory.

H.2. The memory-of-light obstacle

The structural challenge for any photonic universal machine is the persistence of computational state. Photons propagate at the speed of light in their medium and do not stand still without an active storage mechanism. Current storage options:

Optical resonators. High-Q ring resonators store energy at the order of μs to ms; the longest-lived demonstrated optical resonators reach $\sim$ s but only at cryogenic temperatures with specialised geometries.

Bragg-grating delay lines. Fixed-length delay lines with $\sim$ ns to μs delay; useful for short-term buffering but not for arbitrary memory.

Atomic-cloud storage. EIT-based storage in cold atoms achieves seconds to minutes of coherent photon storage, but requires cryogenic infrastructure and is presently limited to small numbers of stored photons.

Persistent classical memory. For non-coherent storage, traditional photographic film, optical disc, or phase-change materials provide arbitrary storage times at the cost of slow read/write.

The implication for the U-Mentalism programme is that the engineering implementation of a fully photonic colour-native machine, at the colour-native register of §1.5, will combine fast optical processing with classical or atomic-mediated memory, with the boundary between the two adjusted to the throughput requirements of the recollection register of §§10.3, 10.6, 12.8. This is not a foundational obstruction; it is an architectural parameter.

H.3. The quantum non-locality challenge

For the quantum extension of the colour-native machine — KLM-style schemes (§9.3) employing entangled photons — the architecture inherits the standard challenges of photonic quantum computation:

Photon loss and decoherence. Even at the lowest loss rates available, multi-photon entangled states suffer exponential degradation with the number of photons; current platforms achieve $\sim$ 10–100 entangled photons, with proposals (boson sampling, photonic measurement-based quantum computation) reaching higher numbers under heralding architectures.

Non-deterministic gates. The KLM scheme realises universal quantum computation with linear optics plus measurement, but at the cost of non-deterministic gate operations corrected by feed-forward measurement. Engineering the feed-forward at the speed of photon propagation is a long-standing challenge.

Synthesis of colour with quantum coherence. The colour-native synthesis of §1.5, when realised on a quantum substrate, must preserve the coherence of the photon population across the three RGB channels. Spectral entanglement between channels — a non-trivial state — appears to be both necessary for the colour-native interpretation and a source of architectural complexity. The interplay of spectral entanglement and spatial entanglement is, in this register, the principal open problem for the quantum extension.

H.4. The colour-synthesis non-commutativity

Section 9.3 established that the operational structure $(\Sigma_{\mathrm{RGB}}^{n\times n}, \circ_{\mathrm{crypt}})$ is non-commutative (Proposition 9.3.2). In the photonic platform, the corresponding physical structure is the composition of optical operations: phase shifters, beam splitters, polarisation rotations, and the specifically colour-mixing operations (additive or subtractive synthesis through nonlinear or interference processes). These operations do not commute, and the compactness of the engineered gate set is bounded by the per-gate loss budget. Each additional gate adds ~ 0.1 dB of loss in current TFLN, ~ 1 dB in current silicon photonics; the total optical budget for a computation is the product of the number of gates and the per-gate loss. The implication: optimisation of the gate set — which non-commutative operations are needed; what is the minimal set; what is the per-gate cost — is the architectural correlate of the algebraic structure of §9.3. The choice of gate set is, in this register, the engineering analogue of the polynomial basis selection in the algebraic register of §9.3.

H.5. Summary

The implementation of the U-Mentalism colour-native universal machine is engineering-feasible at substrate level (silicon photonics, TFLN, hybrid platforms exist and are maturing), confronts a structural memory-of-light challenge that hybrid architectures already address, and presents a quantum-non-locality challenge in its quantum extension that connects directly to the Hermann-Bell-MIP* lineage of §9.3. The colour-synthesis non-commutativity of §9.3.2 manifests at the engineering level as the architectural correlate of an algebraic property of the substrate. None of these is a foundational obstruction; each is a parameter of an engineering programme whose target is the colour-native machine and whose successor is its EEM-extended (Type-N, $N \to \infty$, $k \to \infty$) realisation.

bc1q6eqt7sl8sgfggqfu3guhasta0zyq5tyn6m394v